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**Department of
Mechanical and Aerospace Engineering**

2025/2026

Final Year Design Project Report

Project Title: **Robotic ultrasonic inspection system
development for key engineering
components and infrastructure**

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1 General Section

By All Group Members

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1.1 Introduction

Natural gas pipelines, a typical one of which is shown in Fig. 1.1, form critical energy infrastructure that requires regular non-destructive testing (NDT) to detect cracks and prevent catastrophic failures [1]. Traditional manual ultrasonic inspection is labor-intensive, inconsistent and often unsafe in confined spaces, while commercially available robotic inspection systems are typically prohibitively expensive or too bulky for widespread adoption by smaller operators [2]. The present project therefore seeks to develop a compact, affordable and remotely operated robotic platform that performs automated ultrasonic inspection from inside the pipe. The architecture is designed to minimize couplant dependency, with a longer-term goal of achieving fully contactless operation. By combining a concentric three-legged mechanical architecture, low-cost stepper-motor actuation, a carefully selected angle-beam shear-wave transducer and machine-learning-based signal classification, the team aims to create a practical inspection tool whose total component cost is an order of magnitude lower than current industrial offerings. The resulting platform is intended not only as a functioning laboratory prototype but also as a modular, scalable base that can be further commercialized into a value-oriented solution suitable for routine maintenance of aging pipeline networks.



Fig. 1.1 A typical natural gas pipeline infrastructure [3]

1.2 Objectives of the Project

The project's overarching objective is to design, build and validate a low-cost robotic system capable of autonomous ultrasonic crack detection inside natural-gas transmission pipes. This involves developing a self-centering mechanical platform that maintains concentricity and enables precise transducer positioning, conducting a systematic evaluation of multiple ultrasonic transducer configurations to identify the most suitable option for single-sided robotic inspection, and integrating a signal-processing and machine-learning pipeline that reliably classifies crack versus intact surfaces. A further objective is to demonstrate an end-to-end automated inspection sequence under realistic pipe conditions, validating the full system from locomotion to defect classification. Secondary goals are to maintain total build cost below USD 1500, to deliver a modular hardware and software architecture that can be reproduced with standard fabrication tools, and to document clear pathways for future commercialization and field deployment.

1.3 Teamwork Assignment, Milestones and Deliverables

The project team consisted of four members under the supervision of Prof. Fan SHI as well as PhD student YI, Zhi. The work was organized into several interconnected modules: Pipeline

specimen preparation, Ultrasonic transducer evaluation, Robot mechanical design, hardware and software development, Signal processing, and System integration. Although each member led a distinct sub-area, the practical execution of experiments, prototype assembly, and system-level validation was inherently collaborative. Table 1.1 summarizes the task distribution across all members. A tick (✓) indicates substantial contribution, a dot (●) indicates minor contribution, and a blank indicates (almost) no contribution to that part.

Table 1.1 Task assignments across the four group members

Task	ZENG Minyu (Leader)	CHEUNG Ka Yip	XU You Qi	LAU Tsz Kit
Literature review and market research	✓	●	●	●
Pipeline specimen procurement and preparation	●	✓		✓
Defect design and fabrication (laser-cut and abrasive-ground cracks)	●	✓		✓
Contact ultrasonic transducer testing (twin-crystal, pulse-echo, 0.25–2.5 MHz)		✓		✓
Air-coupled ultrasonic transducer testing (through-transmission, pitch-catch, 420 kHz–1 MHz)		✓		✓
EMAT transducer evaluation and testing		✓		✓
Angle-beam shear-wave probe (5 MHz) testing and validation	✓	✓		✓
Robot concept exploration and architecture selection (three design iterations)	✓		✓	
Front-end sensing module mechanical design (V1 → V2)	✓			
Driving module mechanical design			✓	
Control hardware design and circuit assembly (Arduino Mega, CNC Shield, power system)	✓			
Embedded software development (motor control, PS2 interface, automation sequence)	✓		●	
Signal processing pipeline			✓	
Machine learning classification pipeline (SVM training, cross-validation, permutation testing)			✓	
Data visualization (polar-coordinate rendering of inspection results)	●		✓	
Robot assembly and subsystem verification	✓	●	✓	✓
System integration and automation sequencing	✓		●	✓
Pipe validation testing (clamping, rotation, locomotion, automation demo)	✓	●		✓
Project management, procurement, and budget administration	✓	●		
Mid-term progress report drafting	✓	✓	✓	✓
Final report drafting	✓	✓	✓	✓
Final presentation preparation	✓	✓	✓	✓

The project timeline spanned from September 2025 to April 2026. Table 1.2 lists the major phases, key activities, and deliverables for each period. The first semester (September–December 2025) focused on concept exploration, initial prototype development, and preliminary ultrasonic experiments, culminating in the mid-term progress report. The second semester (January–April 2026) delivered a ground-up redesign of the robot, systematic transducer evaluation, the complete signal-processing and machine-learning pipeline, and full system integration with an end-to-end automated inspection demonstration.

Table 1.2. Team milestones and deliverables

Period	Phase	Key Activities	Deliverables
2025.09–10	Concept exploration	Literature review; robot architecture concept evaluation; pipe material specification and procurement; transducer configuration planning	Literature review, concept sketches, pipe specimen
2025.11	Detailed design	V1 sensing module CAD; V1 driving module CAD; initial defect fabrication; preliminary contact UT experiments	V1 CAD model, preliminary contact UT data
2025.12	V1 prototyping and mid-term reporting	V1 prototype manufacture and assembly; in-pipe motion and DoF verification; air-coupled UT experiments; mid-term progress report	V1 prototype, mid-term progress report, air-coupled UT data
2026.01	V2 redesign and alternative UT exploration	V2 sensing module CAD redesign; EMAT transducer study and preliminary experiments; refined crack fabrication	V2 CAD model, EMAT preliminary data, refined crack specimens
2026.02–03	Procurement, assembly, and hardware/software build	Parts machining and 3D printing; V2 robot assembly; control platform selection; circuit assembly; embedded software development; signal processing pipeline development	V2 prototype, complete hardware system, embedded software, Python signal processing pipeline
2026.04 early	System integration	Module connection and integration testing; drive-wheel resonance diagnosis and resolution; wheel configuration revision	Optimized software, integrated robot
2026.04 mid	Transducer evaluation and ML classification	Comparative testing of five transducer configurations; SVM classifier training and validation	Transducer comparison dataset, trained ML classifier
2026.04 mid–late	Pipe validation and demonstration	In-pipe mechanical and motion validation; automated 25-point inspection sequence; data pipeline and visualization verification	Test data, automation demo, visualization outputs
2026.04 late	Reporting	Final report drafting; presentation preparation	Final report, final presentation

Most milestones were met within the overall schedule. The most significant schedule disruption occurred when the EMAT transducer, which had been returned to the HKUST(GZ) campus for asset inventory during the Chinese New Year period, could not be retrieved for several months. The robot was originally designed with the EMAT probe as the intended non-contact inspection sensor, and the V2 mechanical architecture was originally built on top of it. The unavailability of the EMAT transducer delayed the planned system integration, and the limited EMAT dataset collected before the probe was relocated proved insufficient for robust statistical evaluation. In response, the team switched to the angle-beam shear-wave probe as an alternative. Other adjustments—including the replacement of steel ball casters with rubber wheels (identified during integration, Section D) and the cancellation of the IMU-based closed-loop angular feedback (Section B)—were resolved within their respective work packages without affecting the critical path.

1.4 Summary of Achievements in Mid Progress Report

In the mid-progress report, the team's progress throughout the concept exploration, initial design, and first prototype fabrication stages was recorded. The following significant accomplishments had been achieved during that phase:

A full-scale pipe specimen was procured: two 500 mm segments of API 5L X65 PSL2 carbon steel pipe with an outer diameter of 512 mm, inner diameter of 468 mm and wall thickness of 22 mm. Ten initial artificial defects were fabricated by laser cutting, though these were subsequently replaced by five more representative V-shaped surface cracks produced by abrasive grinding for the final experiments, as detailed in Section 2.

Preliminary ultrasonic experiments showed that contact transducers at 1 MHz and above produced clear back-wall echoes suitable for wall-thickness measurement with errors under 4%. Air-coupled transducers, however, suffered severe signal attenuation at the air-steel interface, demonstrating impractical sensitivity for reliable crack detection.

Three architectural concepts were evaluated: a wall-adherent cart, a base cart with sweeping arm, and a concentric three-legged design. The concentric architecture was selected for its inherent self-centering capability. A first-generation prototype (V1) was built with a CAM-follower mechanism for synchronized leg extension, a frontal scanning arm with three degrees of freedom (360° rotation, vertical lift, manual angle adjustment), and a rear driving module with independent stepper motors on each leg. The robot was controlled wirelessly via a PS2 gamepad.

The V1 prototype was constructed and tested inside the pipe specimen. All mechanical degrees of freedom—sweeping arm rotation, arm lift, and support leg diameter adjustment—functioned correctly. The robot successfully maintained its position and moved forward and backward under remote control, validating the concentric three-legged architecture as a viable concept.

Despite these successes, critical shortcomings were identified: oversized artificial defects, unreliable air-coupled transducer signals, and excessive weight and backlash in the sensing module. These issues directed the second semester's work.

1.5 Main Results and Achievements

The final system successfully met or exceeded the majority of the design specifications established at the outset of the project. The following subsections summarize the principal results across the mechanical, sensing, signal-processing and system-integration domains.

Mechanical Platform. The V2 front-end module, constructed from lightweight aluminum plates and a carbon-fiber-reinforced PETG rotary disk, achieved reliable concentric clamping across the full 440–510 mm pipe-diameter range. The cross-roller bearing and internal ring-gear drive delivered 360° continuous rotation with less than 0.2° resolution and zero perceptible backlash. A lead-screw-driven linear slide provided 62 mm of radial probe travel with 0.01 mm per-step precision. The rear driving module was redesigned with rubber wheels and an independent three-legged suspension, enabling smooth forward and backward locomotion at approximately 2–3 cm/s inside the pipe. Total material cost remained below the USD 1,500 target, satisfying the affordability objective.

Ultrasonic Transducer Selection. Comparative trials of five transducer configurations (contact P-wave, contact T-wave, air-coupled, EMAT and angle-beam) confirmed that a 45° angle-beam shear-wave probe at 5 MHz provided the best detection performance. It achieved 100 % detection of both inner-wall and outer-wall artificial cracks with high signal-to-noise ratio, whereas contact probes showed inconsistent results on outer-wall defects, air-coupled transducers suffered severe attenuation, and EMAT exhibited sensitivity only to the deepest cracks. The angle-beam configuration also best satisfied the practical constraints of robotic deployment: single-side internal access, manageable couplant requirements, and compatibility with the robot’s front-end payload.

Signal Processing and Machine Learning. A comprehensive data processing pipeline was developed, extracting 25 physics-informed features spanning time-domain statistics (peak amplitudes, energy ratios, crest factor, zero crossing rate) and frequency-domain spectral descriptors (dominant frequency drop, spectral centroid shift, high-frequency ratios). Cohen’s d was used to evaluate the discriminative power of each feature. A linear support vector machine trained on principal-component-analysis-reduced features achieved a cross-validated macro F1-score of 95.6% on the angle-beam dataset, statistically validated by a permutation test with 1 000 iterations ($p = 0.001$). This represented a substantial improvement over single-feature threshold-based classification, which reached only 80 % macro F1-score.

System Integration and Automation. End-to-end automation sequences demonstrated repeatable 25-point grid inspection of a known crack location inside the test pipe. The robot traversed the pipe autonomously under pre-programmed C++ control, positioned the angle-beam transducer at each grid point, acquired ultrasonic signals, and all recorded measurements were correctly classified by the ML pipeline as either crack or normal surface. An interactive 3D point-cloud visualization tool was developed using Plotly, mapping classification results to spatial coordinates along the pipe interior to enable intuitive defect localization for field inspectors.

Limitations. Several constraints should be acknowledged. The training data were collected from artificially machined cracks in a controlled laboratory environment, and generalization to naturally occurring weld defects in operational pipelines remains unverified. The absence of real-time angular feedback for the transducer and the manual pre-programming of motion parameters limit the system’s autonomy. Future work will focus on adding closed-loop sensing, exploring contactless transducers, and validating the platform under field conditions.

1.6 Conclusions

This project has delivered a fully functional, low-cost robotic ultrasonic inspection platform that addresses the gap between prohibitively expensive commercial crawlers and unreliable manual methods. The concentric mechanical architecture, optimized angle-beam sensing, and machine-learning signal classification together enable automated crack detection with high reliability at a fraction of current market prices. While certain limitations—most notably the absence of real-time angular feedback and the need for field validation on naturally occurring defects—remain to be addressed, the modular design, open documentation and repeatable fabrication processes provide a solid foundation for subsequent commercialization. Future work will focus on adding closed-loop sensing, migrating to an industrial-grade controller, and exploring contactless sensors without couplant needs, thereby transforming the present laboratory prototype into a deployable, value-oriented inspection solution for the energy sector.

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Final Report

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2 Individual Section A: Experiment Setup and Ultrasonic Techniques

By CHEUNG Ka Yip

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2.1 Introduction
2.2 Theoretical Background
2.3 Experimental Methodology
2.4 Results and discussion
2.5 Summary of achievements

2.1 Introduction

The long-term objective of this project is to develop a robotic platform capable of performing automated ultrasonic inspection inside natural gas pipelines without the use of liquid couplant. Non-destructive ultrasonic testing (UT) is the most critical part of achieving this target.

The laboratory houses a range of ultrasonic transducers spanning multiple coupling mechanisms, wave modes, and operating frequencies. These include contact piezoelectric probes generating longitudinal or shear waves, air-coupled transducers that eliminate the need for liquid couplant, electromagnetic acoustic transducers (EMATs) that excite waves directly in the pipe wall via Lorentz force and magnetostriction, and angle-beam probes that introduce shear waves at oblique incidence for enhanced sensitivity to vertically oriented cracks, and so on. Each transducer type has distinct practical constraints, for instance, couplant dependency, surface preparation requirements, single-side versus two-side access, and tolerance to pipe curvature and surface roughness. All these constraints directly affect the transducer's suitability for robotic deployment.

During the first semester, contact transducers and air-coupled transducers were tested at several frequencies to establish basic feasibility. The contact transducers produced clear multi-echo signals, confirming that thickness measurement and flaw detection in the target pipe material were achievable. The air-coupled transducers demonstrated non-contact through-transmission but suffered severe signal attenuation at the air-steel interface, raising questions about their practical sensitivity and reliability for industrial usage. These preliminary findings motivated us for a more systematic investigation encompassing additional transducer types, a wider range of frequencies, and controlled artificial defects whose geometries were more representative and manufacturable.

The present section documents that investigation and tries to answer the following questions:

1. To characterize the detection performance of five distinct ultrasonic transducer configurations - contact longitudinal-wave (P-wave), contact shear-wave (T-wave), air-coupled, EMAT, and angle-beam - against surface-breaking cracks of known geometry on a carbon steel pipe specimen representative of in-service gas pipelines.
2. To quantify how the location of a crack (inner wall versus outer wall) and its depth affect the signal amplitude, time-of-flight, and signal-to-noise ratio obtained by each configuration.
3. To identify the transducer configuration that best satisfies the practical constraints of robotic deployment: single-side internal access, minimal or zero couplant dependency, and sufficient tolerance to the surface condition expected inside an operational pipe.

The section is organized as follows. Section 2.2 reviews the theoretical background of ultrasonic wave propagation, the operating principles of each transducer type, establishing the physical basis for interpreting the experimental results. Section 2.3 describes the pipe specimen, the fabrication and dimension of the artificial defects, the probe configurations tested, and the data acquisition and signal processing protocols. Section 2.4 presents the experimental results, organized by transducer type, with quantitative metrics including time-of-flight, peak amplitude, and so on, and compares the transducer configurations against a common set of performance criteria and discusses the implications for integration with the robot's front-end

sensing module (Section B). Section 2.5 summarizes the principal findings and recommends a transducer configuration and operating frequency for the integrated system.

2.2 Theoretical Background

2.2.1 Fundamentals of Ultrasonic Wave Propagation

An ultrasonic wave is a mechanical elastic wave propagating through a solid at a frequency above the audible range (>20 kHz) [4]. In nondestructive testing, the wave is introduced into the test piece by a transducer, and the echoes returning from internal features like material boundaries, defects, or the far wall carry information about the condition of the material [1].

Sound velocity: In an isotropic elastic solid, two independent body wave modes can propagate [5]. The longitudinal wave (P-wave), in which particles oscillate parallel to the propagation direction, travels at

$$c_L = \sqrt{\frac{E(1-\nu)}{\rho(1+\nu)(1-2\nu)}} \quad (2.1)$$

The shear wave (T-wave or S-wave), in which particles oscillate perpendicular to the propagation direction, travels at

$$c_S = \sqrt{\frac{G}{\rho}} = \sqrt{\frac{E}{2\rho(1+\nu)}} \quad (2.2)$$

where E is Young's modulus, G the shear modulus, ν Poisson's ratio, and ρ the density. For carbon steel at room temperature ($E \approx 205$ GPa, $\nu \approx 0.29$, $\rho \approx 7850$ kg/m³), these evaluate to $c_L \approx 5900$ m/s and $c_S \approx 3200$ m/s [6].

Acoustic impedance: When an ultrasonic wave encounters an interface between two media, the fraction of energy reflected versus transmitted is governed by the acoustic impedance of each medium, defined as

$$Z = \rho c \quad (2.3)$$

Table 2.1 lists the acoustic impedance of materials relevant to this study. The impedance of carbon steel is approximately five orders of magnitude larger than that of air. Any gas-filled discontinuity, such as a crack, a void, or a delamination, therefore, reflects nearly all incident ultrasonic energy, which is the fundamental reason ultrasonic testing is sensitive to crack-like defects.

Table 2.1. Acoustic impedance of relevant materials at room temperature

Material	Density ρ (kg/m ³)	Sound velocity (m/s)	Acoustic impedance Z (MRayl)
Carbon steel	7,850	5,900 (L) / 3,200 (S)	~46.4 (L)
PMMA (Perspex)	1,180	2,730 (L)	~3.2
Glycerin	1,260	1,920 (L)	~2.4
Water	1,000	1,480 (L)	~1.5
Air	1.2	343 (L)	~0.0004

Normal-incidence reflection: For a wave striking an interface at normal incidence, the intensity reflection coefficient is

$$R = \left(\frac{Z_2 - Z_1}{Z_2 + Z_1} \right)^2 \quad (2.4)$$

The intensity transmission coefficient is $T = 1 - R$. At a steel–air interface, $R \approx 1.0$ meaning virtually total reflection. At a steel–water interface, $R \approx 0.88$, meaning only 12% of the incident energy enters the water; this illustrates why a coupling medium is essential for contact transducers and why the choice of couplant affects signal strength.

Oblique incidence and Snell's law: When a wave strikes an interface at an oblique angle, both reflection and refraction occur, and at a solid–solid interface the refracted wave undergoes mode conversion — a single incident longitudinal wave can produce both a refracted longitudinal wave and a refracted shear wave in the second medium. The angles satisfy the generalized Snell's law:

$$\frac{\sin \theta_{L1}}{c_{L1}} = \frac{\sin \theta_{L2}}{c_{L2}} = \frac{\sin \theta_{S2}}{c_{S2}} \quad (2.5)$$

where θ_{L1} is the incident angle in medium 1 (typically a Perspex wedge), θ_{L2} and θ_{S2} are the refraction angles of the longitudinal and shear waves in medium 2 (steel), and c_{L1}, c_{L2}, c_{S2} are the corresponding wave speeds. Two critical incident angles follow directly from this relationship.

The first critical angle θ_{c1} is the incident angle at which the refracted longitudinal wave reaches 90° and ceases to propagate into the bulk of the steel:

$$\theta_{c1} = \arcsin \left(\frac{c_{L1}}{c_{L2}} \right) \quad (2.6)$$

For a Perspex wedge on carbon steel ($c_{L1} \approx 2730\text{m/s}$, $c_{L2} \approx 5900\text{m/s}$): $\theta_{c1} \approx 27.6^\circ$ deg. Above this angle, only a shear wave enters the steel, which is the condition required for pure shear-wave angle-beam inspection.

The second critical angle θ_{c2} is the incident angle at which the refracted shear wave reaches 90° :

$$\theta_{c2} = \arcsin \left(\frac{c_{L1}}{c_{S2}} \right) \quad (2.7)$$

For Perspex on steel: $\theta_{c2} \approx 58.5^\circ$. Above this angle, all body waves are totally reflected, and a Rayleigh surface wave is generated instead. The practical working range for shear-wave angle-beam probes is therefore 27.6° to 58.5° , corresponding to refracted shear-wave angles in steel of 0° – 90° . Commercially, probes with refracted angles of 45° , 57° are standard [7]. Fig. 2.1 illustrates the general Snell's law.

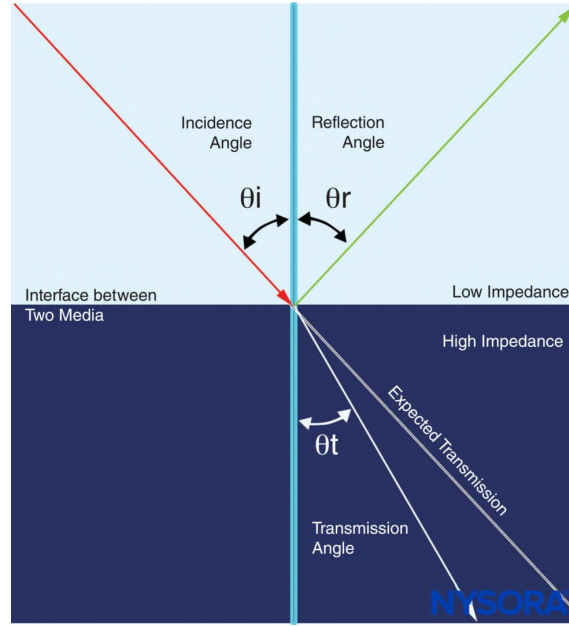


Fig. 2.1 the general Snell's law [8]

Attenuation: As an ultrasonic wave propagates, its amplitude decays with distance due to absorption (conversion of elastic energy to heat) and scattering at grain boundaries. The amplitude after propagating a distance x follows:

$$A(x) = A_0 e^{-\alpha x} \quad (2.8)$$

where α is the attenuation coefficient. Attenuation increases with frequency because scattering losses scale with the ratio of grain size to wavelength. For carbon steel at frequencies below 3 MHz, the longitudinal-wave attenuation is less than 0.01 dB/mm (10 dB/m). The shear-wave attenuation is higher: approximately 10 dB/m at 3 MHz and 40–60 dB/m at 4 MHz. In practice, this frequency dependence shows a trade-off: higher inspection frequencies improve defect resolution (shorter wavelength) but reduce penetration depth (higher attenuation).

2.2.2 Ultrasonic Wave Modes for Pipeline Inspection

Four typical ultrasonic wave modes are relevant to pipeline inspection. Their distinguishing feature is the relationship between particle motion and wave propagation direction, which determines how each mode interacts with crack-like defects.

Longitudinal wave (P-wave): Particles oscillate parallel to the propagation direction, producing alternating regions of compression and tension. This mode propagates in solids, liquids, and gases. In carbon steel, $c_L \approx 5900$ m/s. Because the particle motion is parallel to the wave direction, a longitudinal wave at normal incidence reflects most strongly from defects oriented perpendicular to the beam — i.e., laminations, inclusions parallel to the surface, and the back wall itself. It is the primary mode for thickness measurement and for detecting defects that lie parallel to the scanning surface. For surface-breaking cracks oriented perpendicular to the surface, however, a normally incident longitudinal wave produces only a weak, scattered reflection, because the crack face is nearly parallel to the particle motion. Fig. 2.2 illustrates the longitudinal-wave particle motion.

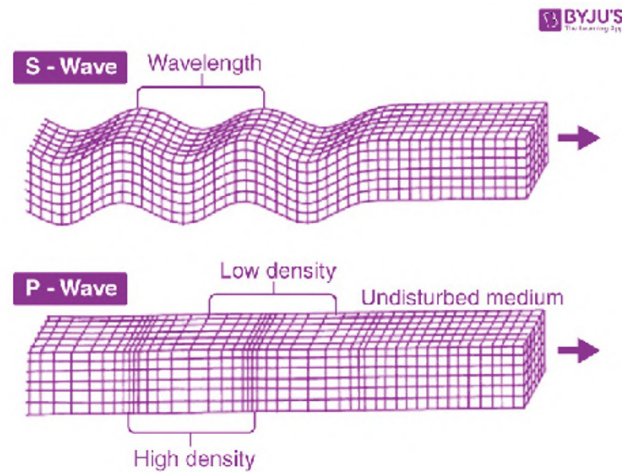


Fig. 2.2 S-wave and P-wave propagation in material [9]

Shear wave (S-wave): Particles oscillate perpendicular to the propagation direction. This mode propagates only in solids, because liquids and gases cannot sustain shear stress. In carbon steel, $c_s \approx 3200$ m/s. For a given frequency, the shear wavelength is approximately 55% of the longitudinal wavelength (0.64 mm versus 1.18 mm at 5 MHz), giving shear waves inherently finer spatial resolution [10]. More importantly, when a shear wave strikes a surface-breaking crack at an oblique angle, its particle motion is nearly perpendicular to the crack face, producing a strong, unambiguous reflection. This makes shear waves, particularly when introduced at oblique incidence via an angle-beam probe, the preferred mode for detecting vertically oriented cracks such as fatigue cracks, stress-corrosion cracks, and lack-of-fusion in welds. Fig. 2.2 also illustrates the shear-wave propagation.

Surface wave (Rayleigh wave): Particles near the free surface execute an elliptical trajectory in the plane containing the propagation direction and the surface normal. The wave energy is confined to within approximately one wavelength of the surface, and the velocity is $c_R \approx 0.87 \sim 0.95c_s$ (~ 2800 – 3000 m/s in steel) [11]. Rayleigh waves are generated when the incident angle exceeds the second critical angle. They are sensitive to surface-breaking defects but cannot detect sub-surface flaws, and they are strongly attenuated by surface roughness and liquid loading on the pipe wall. They were not used as a primary detection mode in this study.

Guided wave (Lamb wave): In structures where the wall thickness is comparable to the ultrasonic wavelength, such as pipe walls at frequencies below a few hundred kHz, the wave is constrained by both the inner and outer surfaces and propagates as a guided mode [12]. Guided waves are strongly dispersive: the phase and group velocities vary with frequency, and multiple modes can coexist at a single frequency. The $T(0,1)$ torsional mode is non-dispersive (constant group velocity ≈ 3200 m/s in steel) and propagates long distances with full cross-sectional coverage, making it the dominant mode for long-range pipeline screening.

2.2.3 Transducer Technologies

The five transducer configurations evaluated in this study exploit these wave modes in different combinations.

Contact P-wave probe: A contact longitudinal-wave probe uses a piezoelectric crystal, which is typically lead zirconate titanate (PZT) or X-cut quartz, that expands and contracts in the thickness direction when driven by a high-voltage electrical pulse. The crystal is bonded to a

heavy damping backing block that absorbs rearward radiation and shortens the pulse duration, improving axial resolution. A thin wear plate protects the crystal face, and a quarter-wavelength matching layer is often placed between the crystal and the wear plate to reduce the acoustic impedance mismatch between the crystal ($Z \approx 30 - 35 \text{ MRayl}$) and steel ($Z \approx 46 \text{ MRayl}$), thereby increasing transmitted energy. The probe is pressed against the pipe surface with a liquid or gel couplant to exclude air from the interface. Because the crystal vibration is parallel to the probe axis, a longitudinal wave is launched into the steel at normal incidence. The schematic diagram of P-wave probe is shown in Fig. 2.3. The contact P-wave probe is the standard tool for wall-thickness measurement and for detecting defects that lie parallel to the scanning surface.

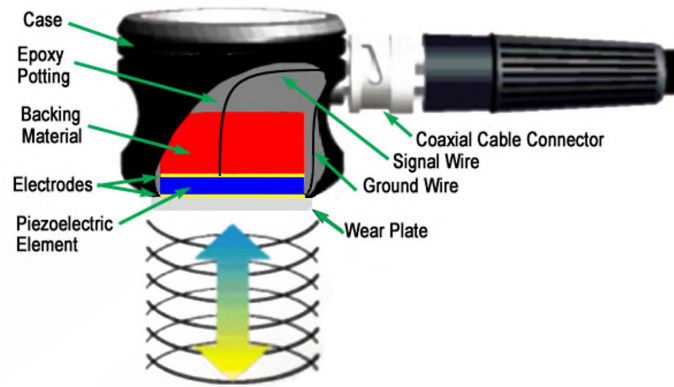


Fig. 2.3 Schematic of a typical P-wave transducer [13]

Contact T-wave probe: A normal-incidence shear-wave probe uses a specially cut piezoelectric crystal (Y-cut quartz, or a shear-polarized PZT element) that undergoes shear deformation under an applied electric field rather than thickness expansion. The crystal oscillates parallel to the contact face. When coupled to the steel surface, this in-plane shear motion is transmitted directly into the test piece, generating a shear wave that propagates normal to the surface with particle motion parallel to the contact plane. Because liquids and conventional gels cannot sustain shear stress, the probe requires a high-viscosity, resin-like shear-wave couplant with a non-zero shear modulus, applied under firm mechanical pressure. The internal construction is somehow similar to the P-wave probe. The normal-incidence T-wave probe offers finer spatial resolution than its P-wave counterpart because, at the same frequency, the shear wavelength in steel is approximately half the longitudinal wavelength (0.64 mm versus 1.18 mm at 5 MHz), enabling it to resolve smaller defects.

Air-coupled transducer: An air-coupled ultrasonic transducer contains a piezoelectric element bonded to a low-impedance matching layer designed specifically to radiate into air. The generated longitudinal wave crosses an air gap, strikes the pipe surface, and a small fraction enters the steel. At every air-steel interface, however, the acoustic impedance mismatch is approximately five orders of magnitude (Table 2.1): over 99.99 % of the incident energy is reflected. The received signal is therefore extremely weak. For this reason, air-coupled transducers are almost operated in through-transmission mode rather than in pulse-echo mode. Their principal advantage is that no liquid couplant is needed, making them the only truly non-contact piezoelectric method. In practice, air-coupled inspection is limited to thin, low-attenuation components; for carbon steel pipes of representative wall thickness, signal-to-noise ratio is the dominant challenge.

EMAT: An electromagnetic acoustic transducer generates and detects ultrasound directly in the surface layer of a conductive test piece, without a piezoelectric element or a coupling medium. A radio-frequency coil is placed against, or with a small clearance from, the pipe surface. When driven by a high-current burst, the coil induces eddy currents in the electromagnetic skin depth of the steel. A permanent magnet provides a static bias magnetic field normal to the surface. The eddy currents interact with this field through the Lorentz force,

$$f = J \times B_0 \quad (2.9)$$

producing an alternating body force at the ultrasonic frequency. In ferromagnetic carbon steel, magnetostriction — the dimensional change of the material in response to an applied magnetic field — provides an additional, and in many configurations' dominant, generation mechanism. The force orientation and the wave mode are determined by the coil geometry. A racetrack or spiral coil beneath a normal-field magnet produces shear forces parallel to the surface, launching a shear wave that propagates into the pipe wall at normal incidence (Fig. 2.4). At lower frequencies on thin-walled pipes, the same transducer geometry can excite guided shear-horizontal or torsional modes for long-range inspection. EMATs require no couplant and tolerate a lift-off gap of up to a few millimeters, allowing operation on rough, coated, or hot surfaces. Their main limitation is transduction efficiency, and high-power excitation and signal averaging are essential to compensate.

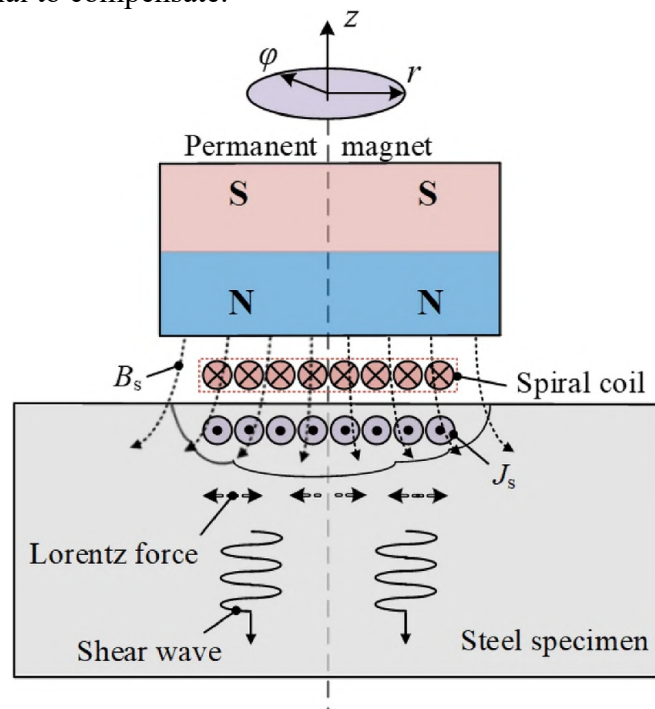


Fig. 2.4 Schematic of EMAT generating shear wave [14]

Angle-beam probe: An angle-beam probe consists of a longitudinal-wave piezoelectric crystal bonded to a Perspex wedge cut at a precise angle. The crystal generates a longitudinal wave that travels through the wedge and strikes the wedge–steel interface at the wedge angle. Through Snell's law and mode conversion at the interface (Section 2.2.1), a shear wave is refracted into the steel. The wedge angle is designed so that the incident angle falls between the first and second critical angles (27.6° – 58.5° for Perspex on steel), ensuring that only a shear wave enters the steel while the longitudinal component is totally reflected. Once inside the pipe wall, the shear wave follows a zigzag path, as shown in Fig. 2.5, reflecting alternately from the inner and outer surfaces. This skip technique allows a single probe to inspect both surfaces

across an extended lateral range. When the shear wave encounters a surface-breaking crack, its particle motion producing a strong reflection. A couplant is required between the wedge and the pipe surface.

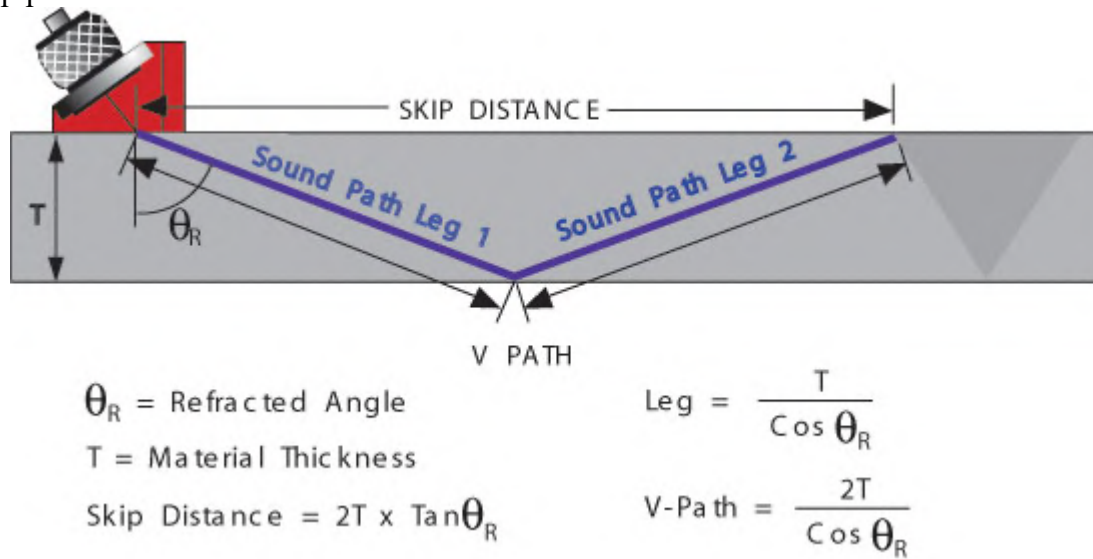


Fig. 2.5 Schematic of angle-beam transducer wave path [13]

2.3 Experimental Methodology

2.3.1 Pipe Specimen and Artificial Defects

The test pipe was an API 5L X65 PSL2 carbon steel pipe of outer diameter 512 mm, inner diameter 468 mm, and wall thickness 22 mm, cut into two 500 mm segments for handling. The material is a high-strength low-alloy (HSLA) steel with a specified minimum yield strength of 65 ksi (~450 MPa), representative of the grade used in high-pressure natural gas transmission pipelines.

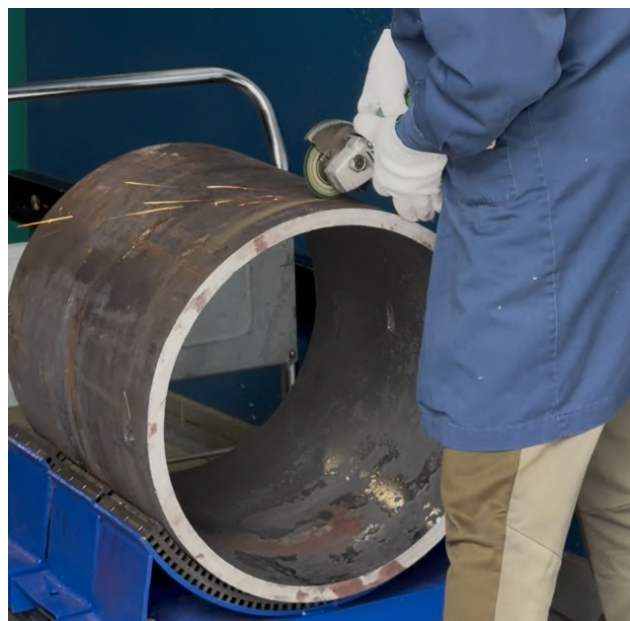
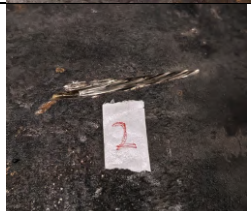
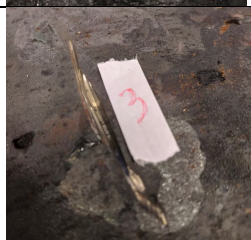
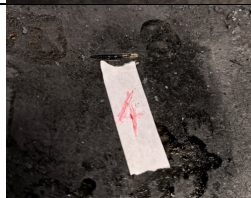
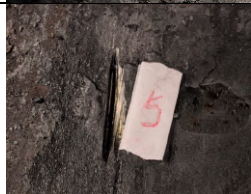


Fig. 2.6 Pipe specimen and defect manufacturing

Five artificial surface cracks were fabricated on the pipe using an abrasive grinding wheel. The cracks are V-shaped grooves cut with an abrasive grinding wheel, consistent with the cross-section of a thin cutting disc. Fig. 2.6 shows the overall pipe specimen and how the final defects were produced by abrasive grinding. Three cracks (C1–C3) were placed on the outer surface and two (C4–C5) on the inner surface, at randomly distributed axial positions around the pipe circumference. C3 was oriented parallel to the pipe axis (axial); all other cracks were oriented perpendicular to the pipe axis (circumferential).

The cracks were produced in three stages. Several through holes were cut by plasma during the pipe manufacturing but soon proven to be meaningless for its naked-eye visibility. The second set of five shallow cracks was cut, but preliminary ultrasonic testing with contact transducers showed that the defect echoes were too weak for reliable detection. The cracks were therefore deepened in situ using the same grinding method. Table 2.2 lists the dimensions of each crack before and after deepening; the final dimensions are the ones against which all subsequent probe evaluations were performed.

Table 2.2. Artificial crack dimensions

Crack	Location	Orientation	Initial: depth × length × width (mm)	Final: depth × length × width (mm)	Photo
C1	Outer wall	Circumferential	1.88 × 3.16 × 26	4.05 × 50.21 × 6.16	
C2	Outer wall	Circumferential	1.37 × 2.73 × 28	6.23 × 6.75 × 66.37	
C3	Outer wall	Axial	1.88 × 3.16 × 26	5.59 × 7.06 × 72.83	
C4	Inner wall	Circumferential	2.08 × 4.12 × 46.74	1.63 × 3.42 × 30.94	
C5	Inner wall	Circumferential	0.80 × 5.31 × 40.30	4.35 × 9.13 × 50	

The detection of C4 and C5 required the probe to be placed against the inner wall, directly over the crack (inside to inside). For C1–C3, the probe was placed on the inner wall, and the ultrasonic beam was directed outward through the full wall thickness toward the external crack (inside to outside). In all contact-probe measurements, the pipe segment was laid horizontally on a support stand with the open ends accessible, and the probe was manipulated by hand from inside the bore.

2.3.2 Probe Configuration and Test Matrix

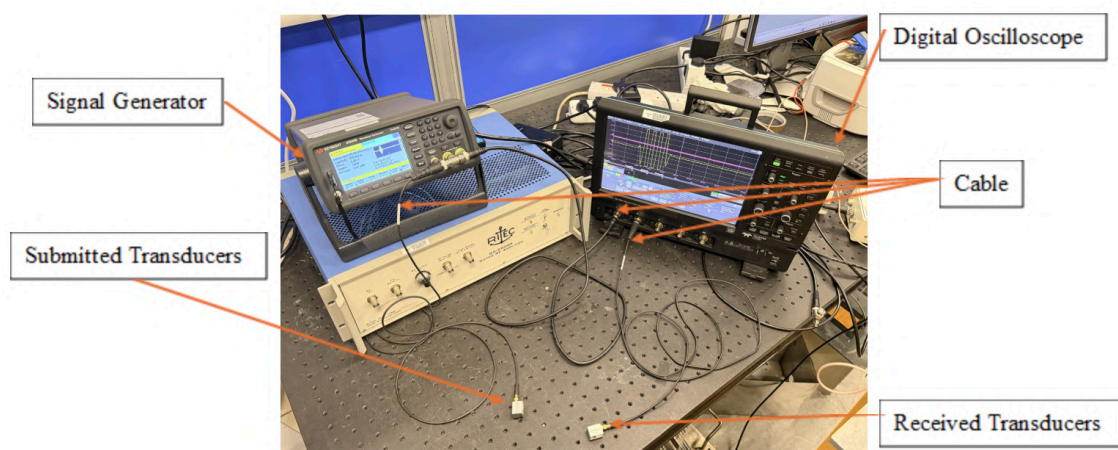


Fig. 2.7 A common experiment setup

Measurement modes: Two distinct ultrasonic inspection modes were employed. In pulse-echo mode, a single transducer served as both transmitter and receiver: the excitation pulse was

routed to the transducer, and after a short electronic recovery interval the same transducer detected the returning echoes. This mode requires access from only one side of the pipe — a critical advantage for robotic deployment — and was used for all contact P-wave, contact T-wave, EMAT, and single-probe angle-beam measurements. In through-transmission (pitch-catch) mode, separate transmitting and receiving transducers were placed on opposite sides — or, in the 8 cm parallel configuration, at a fixed lateral separation on the same side — of the pipe wall. The transmitter launched a continuous or tone-burst signal and the receiver captured the energy that traversed the specimen. This mode was used for the air-coupled transducer pair and for the coupled angle-beam probe pair. Fig. 2.8 illustrates the two measurement methods in practice.



Fig. 2.8 Illustration of through-transmission (left) and pulse-echo mode (right)

Couplants: Three coupling strategies were used, corresponding to the requirements of each transducer type. The contact P-wave probe and angle-beam probes used a water-based ultrasonic gel couplant, applied as a thin layer between the probe face and the pipe surface. The contact T-wave probe required a glycerin-based couplant. Unlike water-based gels, glycerin possesses a non-negligible shear modulus and can transmit the in-plane shear motion of the T-wave crystal into the pipe wall; a standard liquid couplant would allow the crystal to slip without transferring shear stress into the steel [4]. Two types of couplants are shown in Fig. 2.9. Both gel and glycerin couplants were applied by hand in quantities sufficient to form a continuous film with no visible air bubbles. The EMAT and air-coupled transducers required no couplant.



Fig. 2.9 Water-based (left) and glycerin-based (right) couplant

Contact P-wave probe: Frequencies of 0.25, 0.5, 1, 2.5, and 5 MHz were first evaluated on intact pipe sections to characterize the frequency-dependent signal quality and to identify the frequency offering the best trade-off between spatial resolution and penetration. For crack detection, 5 MHz was used exclusively because its shorter wavelength (~ 1.18 mm in steel)

provides finer defect resolution. The probe was pressed against the inner pipe wall by hand with water-based gel couplant.

Contact T-wave probe: The T-wave campaign was conducted in two stages. The first, on the initially shallow cracks, used both C1 and F1. After the cracks were deepened, a second campaign was carried out.



Fig. 2.10 Air-coupled: through-wall (left) parallel (right) configuration

Air-coupled transducer: The transducers were clamped in a mechanical fixture that maintained coaxial alignment across the pipe wall. Two measurement geometries were tested (Fig. 2.10). In the through-wall configuration, the transmitter and receiver were placed on opposite pipe surfaces; the beam crossed two air–steel interfaces, incurring extreme transmission loss. In the 8 cm parallel configuration, both transducers were placed on the inner wall, 8 cm apart, so that the beam propagated along the pipe wall.

EMAT: The probe was placed directly against the inner wall with a lift-off gap under 1 mm. The received signal was amplified, filtered by a 256-tap FIR digital filter with an autocorrelation noise-suppression algorithm, and averaged on-board over 64 successive pulse-echo cycles before transmission to the PC. The manufacturer's EMABOX software-controlled acquisition and exported two-column Time(μ s)–Amplitude CSV files. One normal measurement taken from the outer wall (outside to inside) was included as a direction reference.

Angle-beam probe: Two angle-beam configurations were evaluated. In the first, a single 45° shear-wave probe (Perspex wedge, refracted angle 45° in steel; Doppler A5P6 $\times$ 6A45 OLE404, 5 MHz) was hand-positioned on the inner wall; the zigzag skip path provided natural coverage of both pipe surfaces. In the second, a pitch-catch pair — one 45° probe transmitting and one 57° probe receiving (Perspex wedge, refracted angle 57° ; Doppler TF1-L50-IHC, 5 MHz) — was manually aligned along the same axial line on the inner surface.



Fig. 2.11 Angle beam transducer measurement

Table 2.3 summarizes all probe configurations and dataset sizes. Repeated acquisitions at the same measurement position denote independent trials at constant settings.

Table 2.3 Summary of probe configurations and dataset sizes

Probe configuration	Centre freq. (MHz)	Mode	Direction	Normal records	Crack records
Contact P-wave, multi-frequency	0.25, 0.5, 1, 2.5, 5	Pulse-echo (C1, C2)	Inside → outside	19	—
Contact P-wave, 5 MHz crack detection	5	Pulse-echo (C1)	Inside → outside, inside → inside	5	5
Contact T-wave, small crack	5	Pulse-echo (C1, F1)	Inside → outside, inside → inside, outside → inside	18	10
Contact T-wave, big crack	5	Pulse-echo (C2, F1)	Inside → outside, inside → inside	38	50
Air-coupled, through-wall + 8 cm	0.42, 1	Through-transmission (C1, C2, F1)	Through-wall, parallel	10+	4+
EMAT	5	Pulse-echo (board ADC)	Inside → outside, inside → inside, outside → inside	5	5
Angle-beam 45°, first dataset	5	Pulse-echo (C1)	Inside → outside, inside → inside	10	25
Angle-beam 45° + 57°, coupled	5	Pitch-catch (C1, C2)	Inside → outside, inside → inside	10	12
Angle-beam 45°, second dataset (extended normal)	5	Pulse-echo (C1)	Inside → outside, inside → inside	25	9
Angle-beam, robot validation	5	Pulse-echo (C1)	Inside → outside, inside → inside	22	3

2.4 Results and discussion

2.4.1 P-wave Contact Transducer

Fig. 2.12 displays the results from the P-wave pulse-echo transducer with 0.25 MHz and 0.5 MHz, respectively. For the pulse-echo longitudinal-wave measurement, the first back-wall echo is not clearly resolved because it arrives within the transducer ring-down period ($10\mu\text{s}$) and overlaps with the transmitted pulse ($t=2d/v=0.044/5900=7.46\mu\text{s}$). As a result, the early reflected signal is mixed with the excitation waveform, so the absence of a distinct first echo is mainly caused by limited temporal resolution (dead zone) rather than the absence of reflection itself. Therefore, these two frequencies were not considered to inspect the pipeline.

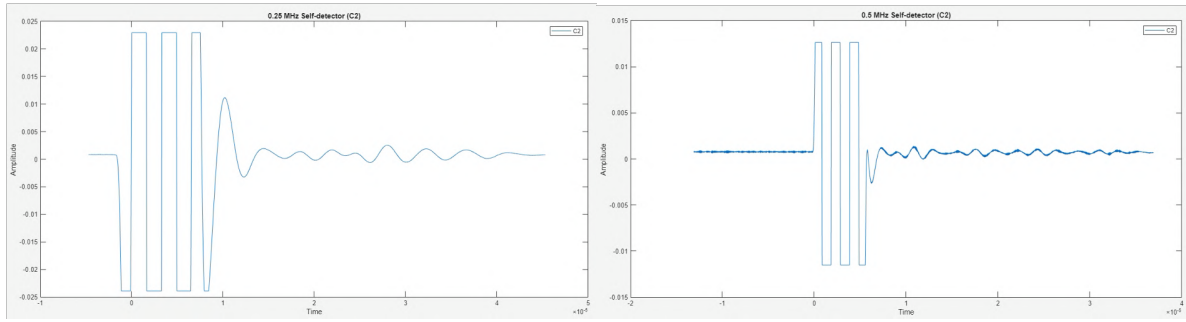


Fig. 2.12 Normal surface signals of a pulse-echo P-wave contact transducer, with frequencies of 0.25 MHz (left) and 0.5 MHz (right)

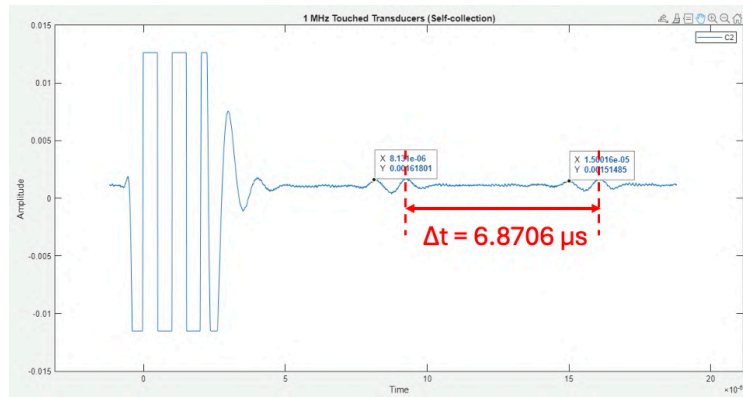


Fig. 2.13 Normal surface signal of a 1MHz pulse-echo P-wave contact transducer

Fig. 2.13 shows the received signal of 1MHz pulse-echo P-wave contact transducer. The first wave packet received at $8.131\mu\text{s}$ (amplitude $\approx 1.618 \times 10^{-3}$) finished one-round reflection; the second echo at $15.006\mu\text{s}$ (amplitude $\approx 1.515 \times 10^{-3}$) finished two-round trips. The time-of-flight between the two echoes is $6.8706\mu\text{s}$. Given the wave travelling speed of 5900m/s , the round-trip propagation distance is calculated to be 4.054cm . The expected round-trip distance is based on twice the nominal pipe-wall thickness 4.4cm (pipe thickness is 22mm), yielding a relative error of 7.86% . The result agrees reasonably well with the theoretical prediction.

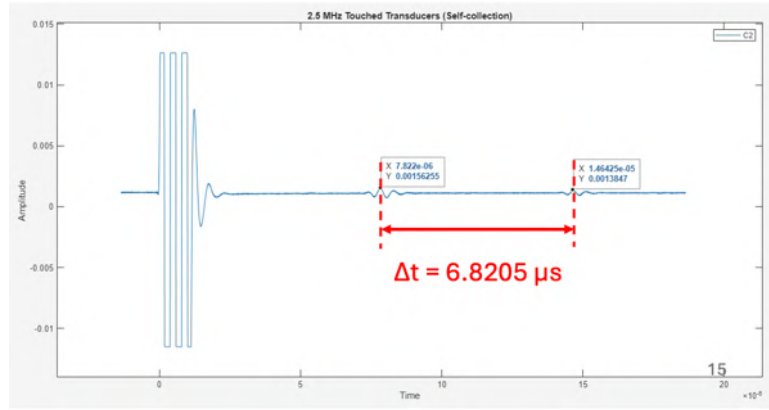


Fig. 2.14 Normal surface signal of a 2.5MHz pulse-echo P-wave contact transducer

Fig. 2.14 shows the received signal of 2.5MHz pulse-echo P-wave contact transducer. The first wave packet received at $7.822 \mu s$ (amplitude $\approx 1.562 \times 10^{-3}$) finished one-round reflection; the second echo at $14.643 \mu s$ (amplitude $\approx 1.385 \times 10^{-3}$) finished two-round trips. The time-of-flight between the two echoes is $6.8205 \mu s$. After the theoretical calculation, the round-trip propagation distance is evaluated to be 4.024 cm , yielding a relative error of 8.55% . The result also agrees reasonably well with the theoretical prediction.

Eventually, 1MHz or higher frequencies are suitable to inspect the test pipeline, and the transducers at these frequencies would be utilized in future investigation.

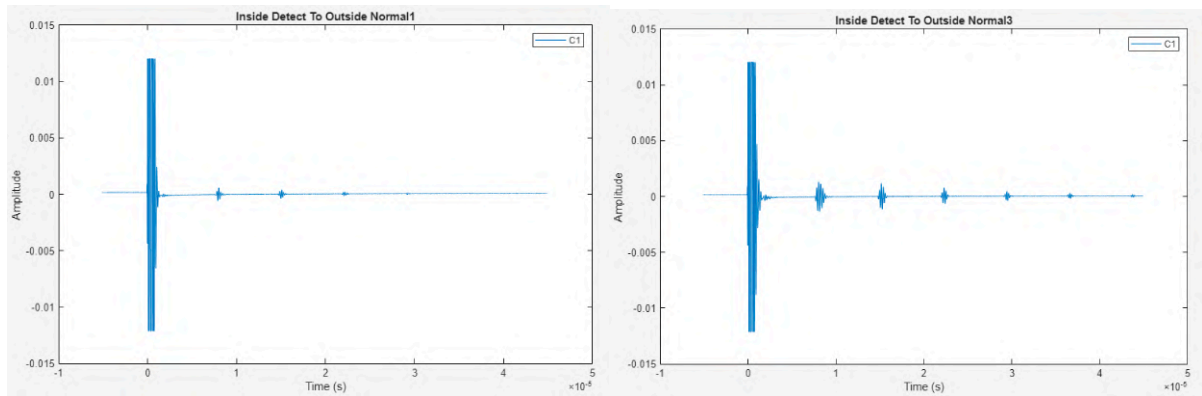


Fig. 2.15 Normal surface signals of a 5MHz pulse-echo P-wave contact transducer

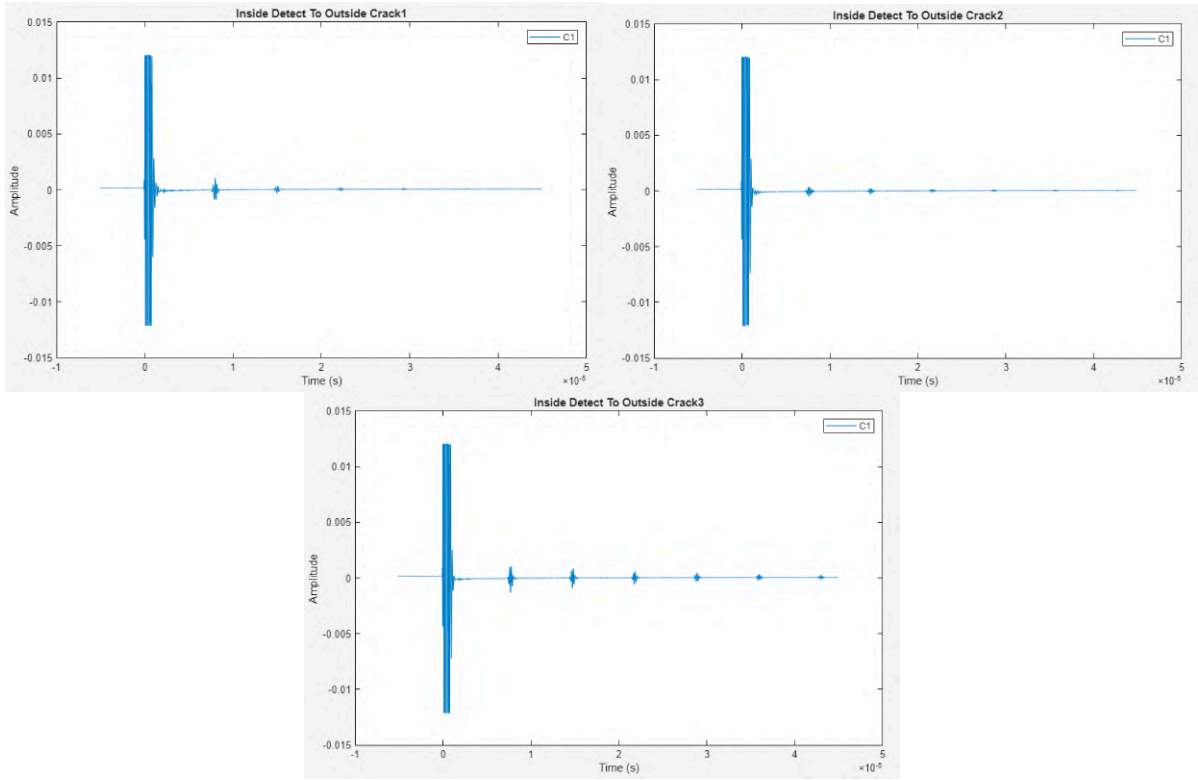


Fig. 2.16 Outer cracks signals of a 5MHz pulse-echo P-wave contact transducer

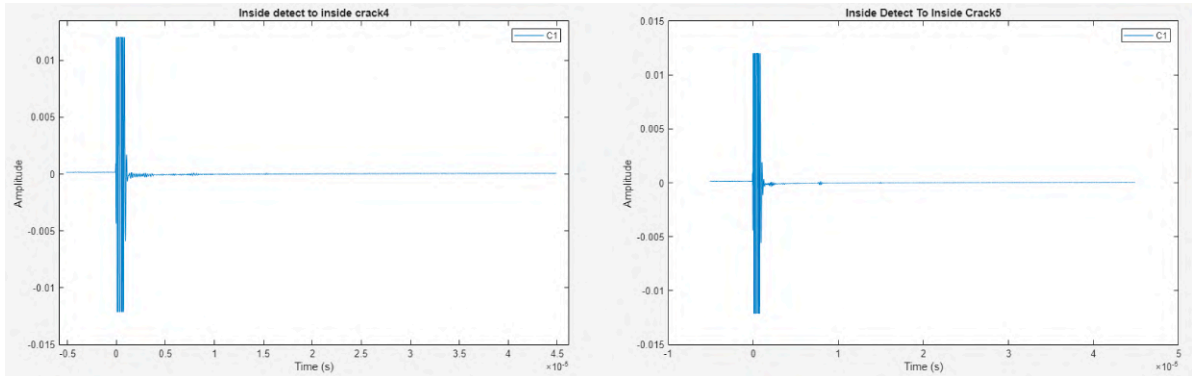


Fig.2.17 Inner cracks signals of a 5MHz pulse-echo P-wave contact transducer

Fig. 2.15 (normal surface signal), 2.16 (outer cracks) and 2.17 (inner cracks) display the recorded signals of the P-wave pulse-echo transducer at 5MHz. During these tests, the transducer was attached to the inner surface. While the arrival times of all signals correctly indicate the pipe thickness, no apparent disparity on time-of-flight occurs between the normal ones and defects (the wave path was estimated to be shorter for cracks thus returns earlier) - this is because the dimension of defects is much smaller than the pipe thickness itself. A clear and significant amplitude of attenuation was observed for the signals corresponding to inner cracks 4 and 5, very likely because the defects worsen the contact between transducer and wall surface. In contrast, the echoes from outer cracks 1, 2, and 3 exhibited no measurable difference in amplitude or waveform when compared to the response from an intact pipe surface. As a result, it is not possible to reliably identify the presence of outer cracks based on the pulse-echo measurements alone.

To summarize, during this P-wave contact transducer experiment, the outer crack identification cannot be achieved, while inner crack identification can be verified.

2.4.2 T-wave Contact Transducer

For the contact probe measurements, a total of 44 samples were acquired after the cracks were deepened, comprising 19 normal specimens and 25 cracked specimens. The three most prominent signals from each cracked specimen are presented below (Fig. 2.18 right and Figs. 2.19, 2.20) and compared with the corresponding signals from the normal specimens (Fig. 2.18 left). All signals were acquired using the same 5 MHz standard contact transducer in pulse-echo mode.

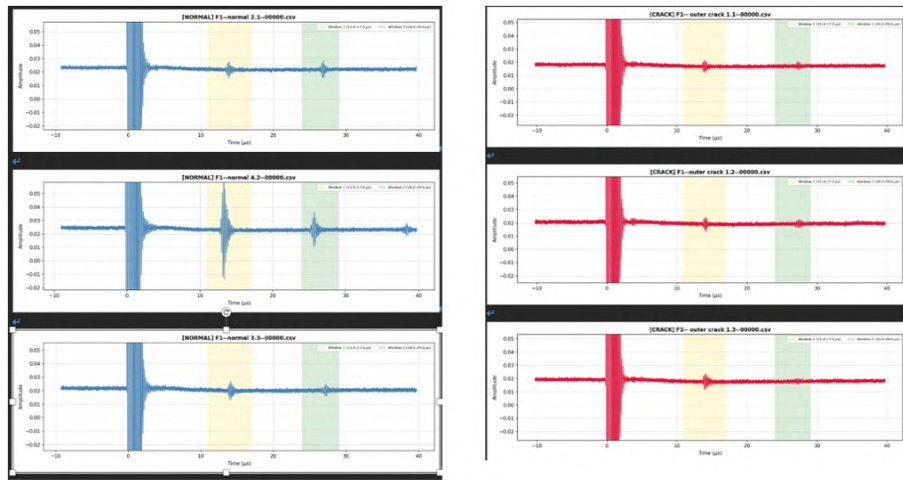


Fig. 2.18 Normal surface (left) and crack 1 (right) signals of a 5MHz pulse-echo T-wave contact transducer

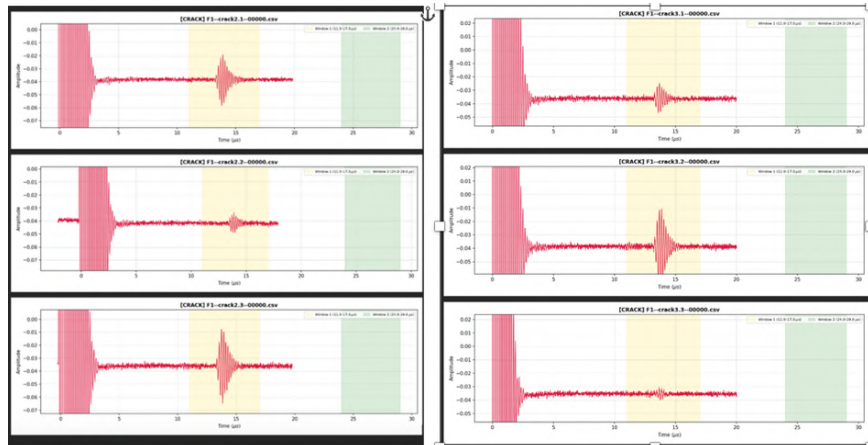


Fig. 2.19 crack 2 (left) and crack 3 (right) signals of a 5MHz pulse-echo T-wave contact transducer

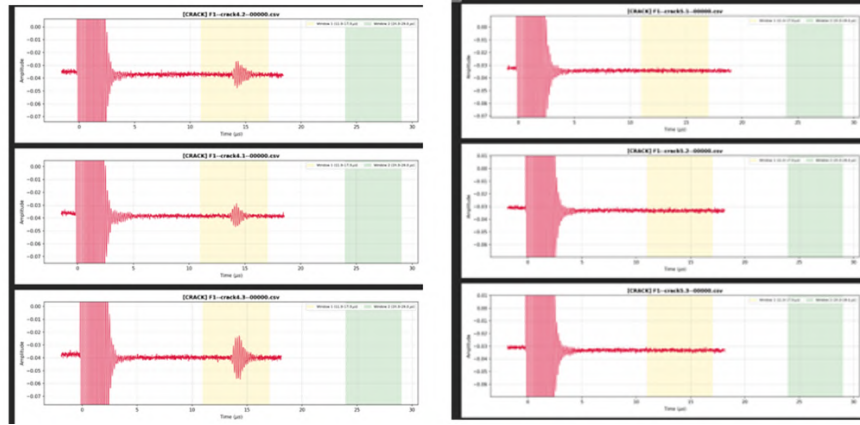


Fig. 2.20 crack 4 (left) and crack 5 (right) signals of a 5MHz pulse-echo T-wave contact transducer

Table 2.4 Comparison of First Echo (Backwall Echo) Time and Amplitude

Specimen	Repeats	Averaged first echo time(μ s)	Time shift vs normal (μ s)	Averaged first echo amplitude	Amplitude reduction vs Normal mean (%)
Normal	12	26.8	—	0.024	—
Crack 1	5	26.5	−0.3	0.019	20.8
Crack 2	5	26.4	−0.4	0.015	37.5
Crack 3	5	26.3	−0.5	0.017	29.2
Crack 4	5	26.7	−0.1	0.020	16.7
Crack 5	5	26.2	−0.6	0.008	66.7

The standard contact T-wave probe produced some measurable differences between normal and cracked conditions, including a small advance in first-echo arrival time (0.1–0.6 μ s earlier than the normal mean of 26.8 μ s) and amplitude reductions ranging from 16.7 % to 66.7 %. However, the time difference was minimal to determine a threshold between normal and defect signal, similar to the P-wave result, given that pipe's local thickness variation may also introduce error. Moreover, the amplitude drops were inconsistent across cracks, and the normal baseline exhibited high variability (standard deviation ± 0.006). These fluctuations were mainly caused by manual glycerin couplant application, inconsistent pressing force, and progressive surface rust on the carbon steel pipe.

In conclusion, the standard contact T-wave probe is not reliable for this project. Its performance depends too heavily on operator technique and pipe surface condition, making it unsuitable for automated robotic pipeline inspection.

2.4.3 Air-coupled Transducer

Fig. 2.21 displays the recorded signal of the air-coupled transducer at 420 kHz and 1 MHz with through-transmission technique, respectively. At this moment, the abrasive defects were not yet provided, thus only normal cases will be discussed.

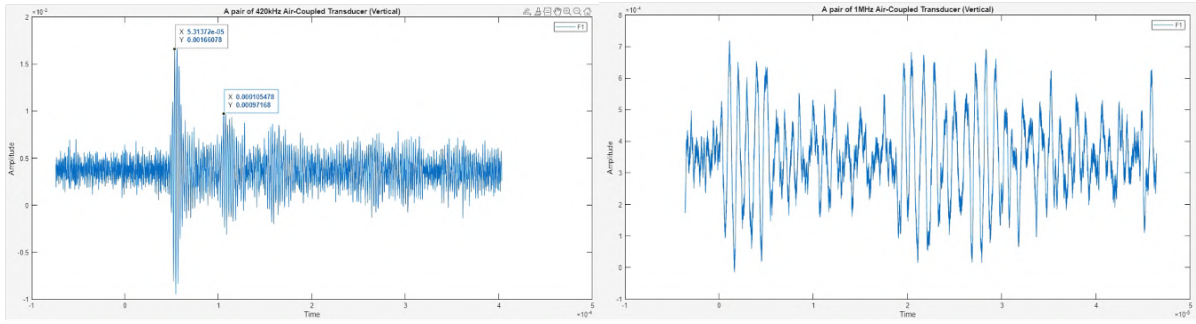


Fig. 2.21 Normal surface signals of through-transmission air-coupled transducers, with frequencies of 420 kHz (left) and 1 MHz (right)

The 420 kHz waveform exhibits two distinguishable arrivals: the first at 53.14 μs (amplitude 1.66×10^{-3}) and the second arrival at 105.48 μs (amplitude 0.97×10^{-3}), yielding a time difference of 52.34 μs . This time discrepancy is not related to the thickness of the steel wall; they are instead dominated by acoustic path in the air between pipe and receiver, as shown in Fig. 2.22. This makes sense as 99% of signal tends to reflect instead of transmitting due to extreme acoustic-impedance mismatch at the air–steel interface, as discussed in theory.

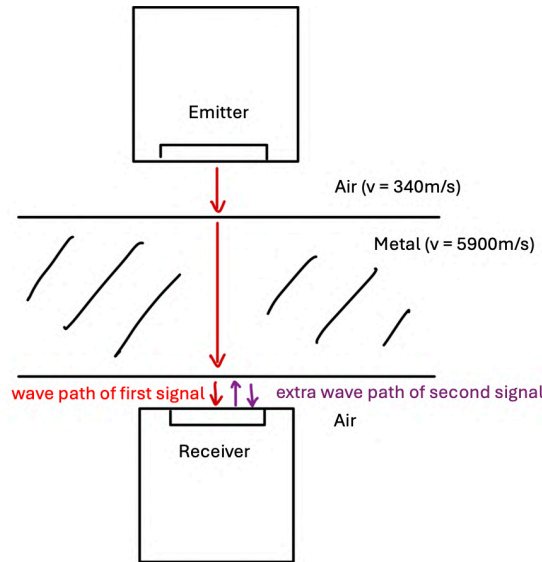


Fig. 2.22 Schematic of the 1st and 2nd signals' wave paths

Another observation is the low signal-to-noise ratio (SNR) at 1MHz compared with 420kHz, because acoustic attenuation in air increases strongly with frequency, causing much greater transmission loss at 1MHz. Across the entire acquisition window, the amplitude fluctuates between 0s and 7 μs and between 20 μs and 30 μs , with no discernible pulse or wave packet rising above the background noise.

From the robot design perspective, such through-transmission configuration is not acceptable as the robot can only travel inside the pipe. The pitch-catch configuration (meaning two transducers are both placed at the same side) might be viable, but the guide wave it produces requires further investigation. This is a path for future exploration.

2.4.4 EMAT

Figs. 2.23, 2.24 and 2.25 display the signals of 5MHz EMAT inspecting normal surfaces, inner crack and outer crack, respectively.

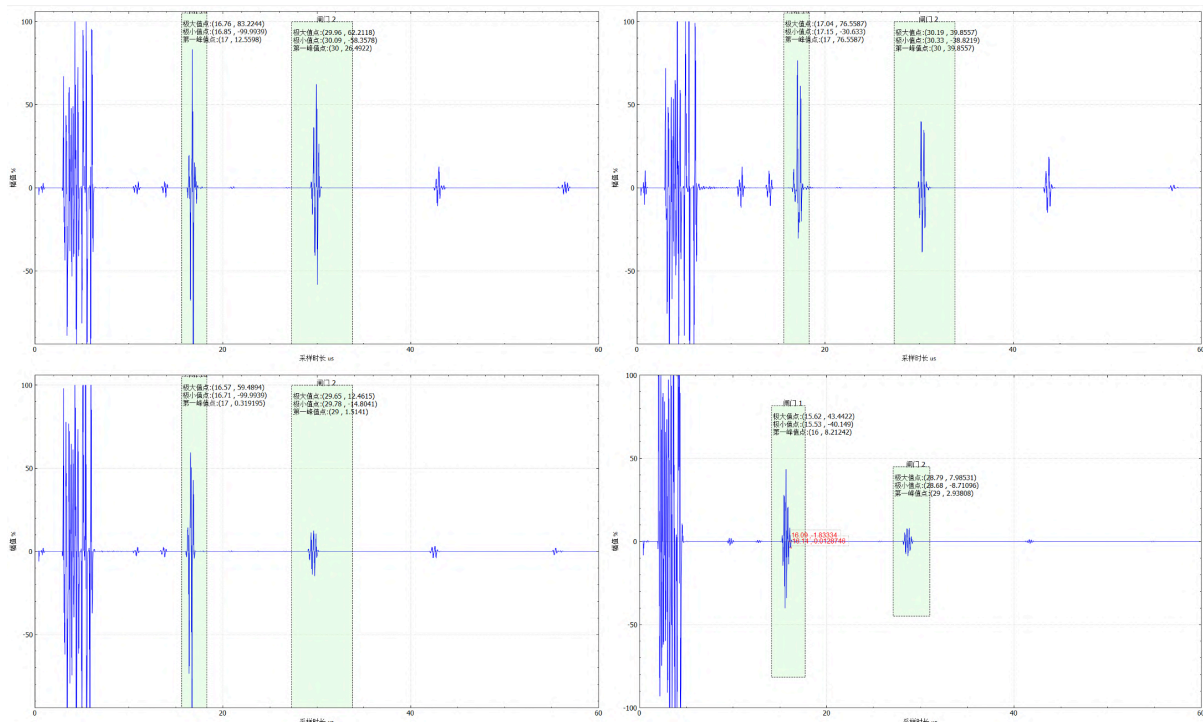


Fig. 2.23 Normal surface signals of EMAT at frequency of 5MHz

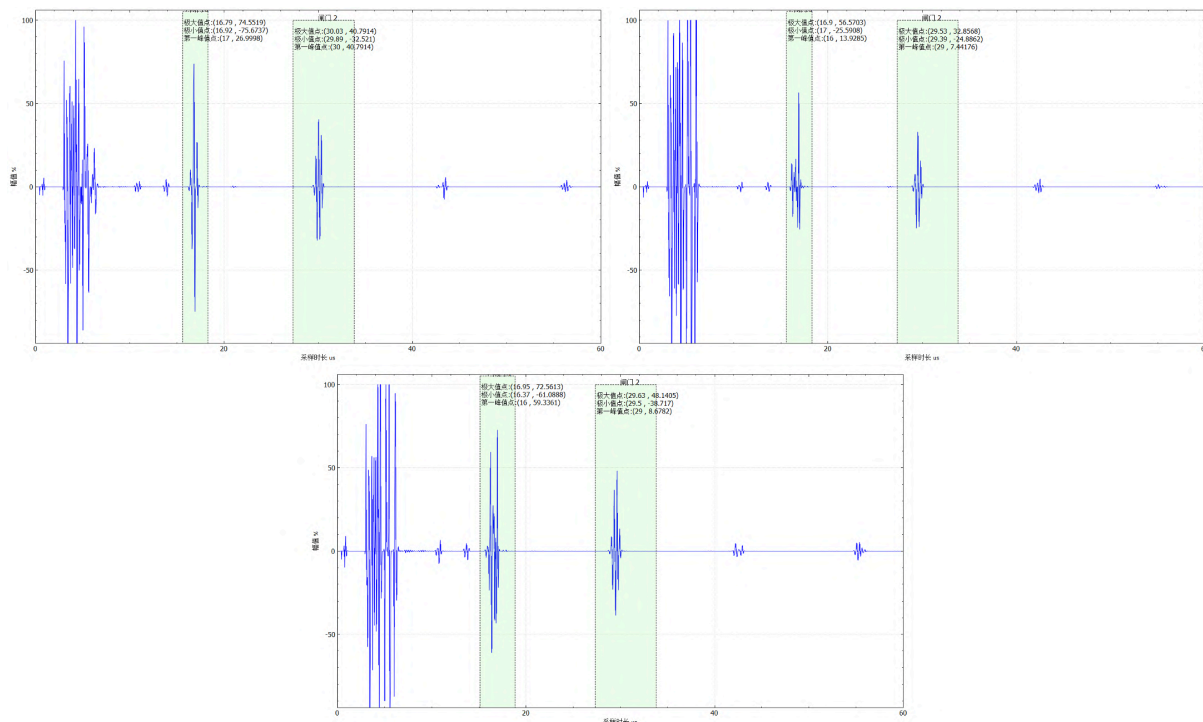


Fig. 2.24 Crack 1 (left top), crack 2 (right top) and crack 3 (bottom) signals of 5MHz EMAT

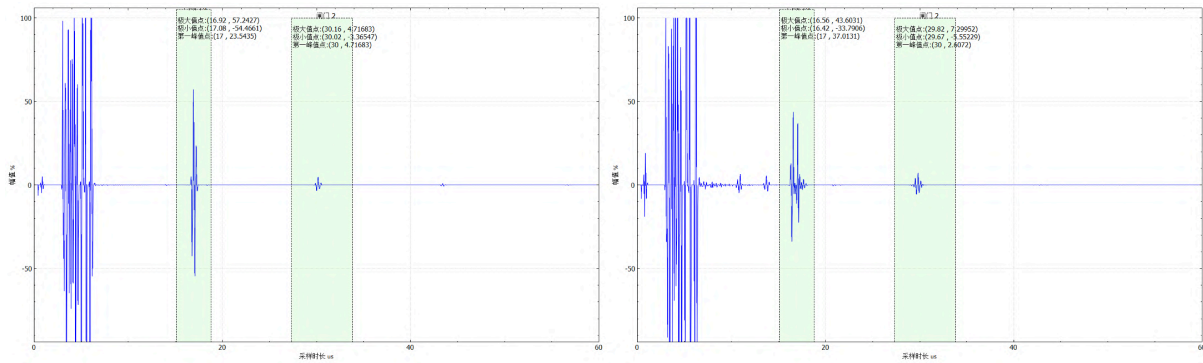


Fig. 2.25 Crack 4 (left) and crack 5 (right) signals of 5MHz EMAT

The table below summarizes the maximum echo amplitudes recorded in Gate 1 and Gate 2 using the EMAT probe. All amplitudes are expressed as a percentage of full scale (%). Amplitude reduction is calculated relative to the mean Gate 2 amplitude of the five normal surface measurements. A total of ten datasets were collected, consisting of five normal points and five crack points.

Table 2.5 Peak Amplitude Comparison in Gate 1 and Gate 2 for Normal Surface and Surface Crack Conditions (EMAT Probe)

Specimen	Type / Location-Orientation	Gate 1 Amp (%)	Gate 1 Time (μs)	Gate 2 Amp (%)	Gate 2 Time (μs)	Reduction in Gate 2 vs Normal Mean (%)
Normal 1	Normal surface	83.22	16.78	62.21	29.96	—
Normal 2	Normal surface	76.56	17.04	39.86	30.19	—
Normal 3	Normal surface	59.49	16.57	12.46	29.65	—
Normal 4	Normal surface	24.37	17.11	58.24	29.94	—
Normal 5	Normal surface	43.44	15.62	7.99	28.79	—
Normal Mean	—	57.42	—	36.15	—	—
Std Dev	—	23.12	—	23.85	—	—
Crack 1	Outer wall / Circumferential	74.55	16.79	40.79	30.03	-12.8 (increase)
Crack 2	Outer wall / Circumferential	56.57	16.90	32.86	29.53	9.1
Crack 3	Outer wall / Axial	72.56	16.95	48.14	29.63	-33.2 (increase)
Crack 4	Inner wall / Circumferential	57.24	16.92	47.17	30.16	-30.5 (increase)
Crack 5	Inner wall / Circumferential	43.03	16.50	7.30	29.82	79.8

The EMAT probe was tested on five normal surface conditions and five surface crack specimens. The normal measurements exhibited substantial variability in Gate 2 amplitude (mean = 36.15 %, standard deviation = 23.85 %), which is characteristic of EMAT systems and primarily attributable to small changes in lift-off distance and surface conditions. Despite this variation, Crack 5 (deep inner-wall circumferential crack) produced a clear amplitude reduction of 79.8%. The pattern is unclear for the remaining cracks, indicating that the current EMAT configuration has limited sensitivity to shallower or less severe surface cracks under the experimental conditions used. The result implies that the EMAT probe might have the potential

to detect certain surface-breaking cracks through amplitude attenuation caused by scattering and mode conversion [15]; however, the overall signal strength and repeatability remain lower than those typically achieved with the angle beam probe (next part). The data provide a useful reference for future robotic integration, although further optimization of excitation parameters and lift-off control is recommended to improve signal stability and defect detectability.

2.4.5 Angle Beam Probe

The 5 MHz 45° angle-beam shear-wave probe was evaluated in pulse-echo mode, the oblique shear-wave beam propagates through the wall via a zigzag skip path. A total of 85 individual records were collected across cracked and defect-free regions via robotic and manual methods.

As shown in Fig. 2.26, the round-trip acoustic path length for one full skip — from the inner wall to the outer wall and back — is determined by the refracted angle $\theta = 45^\circ$ and the wall thickness $t = 22$ mm. The half beam path (one-way through the thickness) is given by $t / \cos \theta$:

$$\text{One - way path} = \frac{t}{\cos(45^\circ)} = \frac{22}{1/\sqrt{2}} = 31.11\text{mm} \quad (2.10)$$

$$\text{Round - trip path} = 2 \times 31.11 = 62.22\text{mm} \quad (2.11)$$

With the shear wave velocity of 3200 m/s in carbon steel, the round-trip time of flight in the steel alone is:

$$\Delta t = \frac{(62.22 \times 10^{-3})}{3200} = 19.44\mu\text{s} \quad (2.12)$$

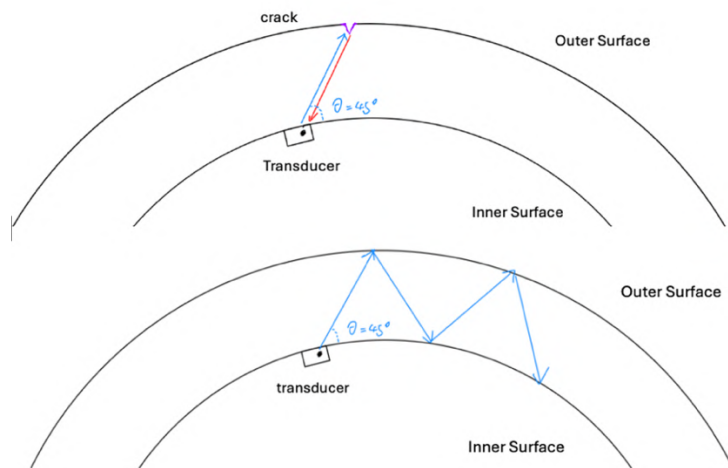


Fig. 2.26 Schematic of wave paths encountering normal (lower) and surface defect (upper)

In addition to the steel propagation time, the ultrasonic pulse must traverse the Perspex wedge twice (forward and return). The wedge transit time depends on the wedge geometry and the longitudinal wave velocity in Perspex (2730 m/s). For the compact wedge used in this probe, the round-trip wedge delay is estimated at 3–6 μs . The total expected time of flight for the first echo when encountering the surface defect is therefore approximately 22–25 μs . The

other case is that when the surface is intact, the wave will bounce up-and-down and propagate farther away, and there will be no echo expected.

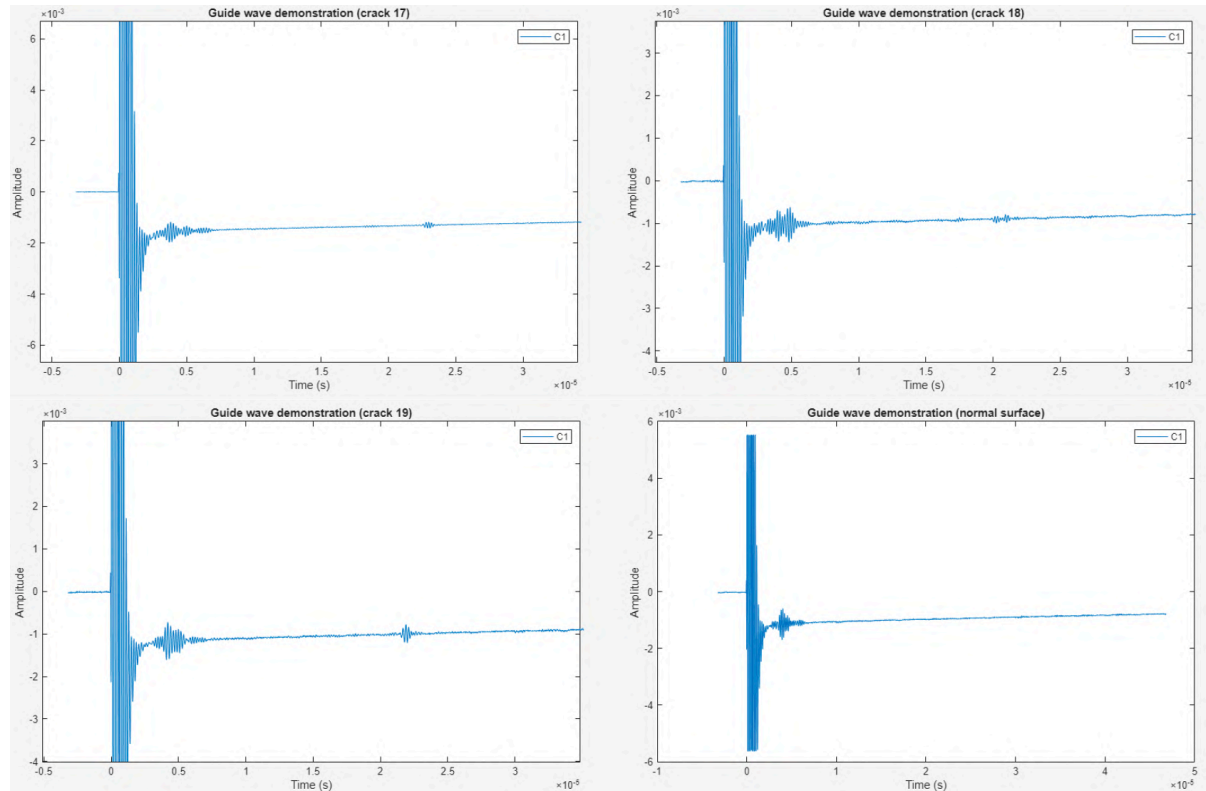


Fig. 2.27 Representative signals acquired with angle-beam probe: normal sample (bottom-right), defect (all others)

This theoretical estimation is verified by experiment results. The signals obtained on the target cracks exhibit a clear, well-defined reflection echo arriving at a time-of-flight between 20 μ s and 25 μ s, which are consistent with the calculation. In contrast, the signals recorded on intact (normal) pipe surfaces lack any comparable event in the same time window; the waveforms remain essentially flat and noise-limited, with no feature that could be misinterpreted as a crack-reflected pulse.

The complete alignment of the measured crack echoes with the theoretical time-of-flight—and their absence on defect-free surfaces—confirms that the 45° shear-wave angle-beam technique, even when deployed robotically, is capable of reliably detecting and discriminating the target crack in the pipeline.

2.4.6 Influence of Defect Geometry on Detectability

Crack depth and location exerted markedly different effects on detectability across the probe types. Deeper cracks (final depth > 4 mm, e.g., C2, C3, C5) generally produced larger amplitude reductions or clearer defect echoes, while shallower cracks (C1, C4) were more challenging to detect. Outer-wall cracks (C1–C3) were particularly difficult for normal-

incidence probes (P-wave and T-wave) because the wave path is nearly parallel to the crack face, resulting in weak scattering. Inner-wall cracks (C4–C5) were more readily detected by contact probes owing to direct proximity to the transducer. The EMAT probe showed strong sensitivity only to the deepest inner-wall crack (C5, 79.8 % reduction) but performed poorly on shallower or outer-wall defects. The contact P-wave probe was effective solely for inner-wall cracks, while the contact T-wave probe exhibited moderate but inconsistent performance that deteriorated further with surface rust and shallow defects.

The 45° angle-beam probe was least sensitive to these geometric factors. Its oblique shear-wave path and zigzag skip technique enabled strong reflections from cracks of any depth or location, with consistent echoes appearing in the predicted 20–25 μ s window regardless of whether the crack was on the inner or outer wall.

2.4.7 Implications for Robotic Integration

Table 2.6 Comparison of the four ultrasonic probes for carbon steel pipeline inspection

Probe Type	Wave Mode	Coupling Method	Key Characteristics	Most Suitable Crack Types	Limitations in Pipeline Inspection
Air-Coupled (ACU)	Longitudinal (also accept more advanced modes)	Non-contact (air)	High attenuation due to impedance mismatch; low SNR	Surface/near-surface cracks on rough or coated surfaces	Severe energy loss; limited penetration in thick walls
Standard Contact	Longitudinal/Share	Contact (gel/glycerin couplant)	Strong signal, normal incidence;	Inner-wall cracks, planar defects parallel to surface	Requires couplant; weak for vertical surface cracks
EMAT	Shear / Guided waves	Non-contact (electromagnetic)	Generates waves directly in material; no couplant	Surface/near-surface cracks, coated or high-temp pipes	Lower energy coupling; limited data volume in your tests; equipment reliability issues
Angle Beam (Shear Wave)	Shear (Transverse)	Contact (wedge + couplant)	Angled incidence (45°/60°/70°)	Surface-breaking cracks , fatigue cracks, SCC, weld defects	Requires good surface preparation; couplant needed

Several practical constraints must be considered for integration with the tunnel inspection robot: couplant dependency, probe size and weight, single-side accessibility, tolerance to surface roughness/rust, and ease of robotic mounting/manipulation.

The contact P-wave and T-wave probes require liquid/glycerin couplant and consistent manual pressing force, rendering them impractical for automated deployment. Air-coupled transducers eliminate couplant but suffer unacceptable SNR. EMAT offers true non-contact capability and good lift-off tolerance but suffers from high variability and limited sensitivity.

The 45° angle-beam shear-wave probe (5 MHz) is a suitable configuration for robotic integration. Although it requires water-based gel couplant (which can be automated via a small dispensing system), it has demonstrated reliable performance in both manual and early robotic scanning trials. Its compact wedge design is compatible with the robot's front-end payload integration, and the clear separation of defect echoes from the initial pulse facilitates straightforward signal-processing algorithms for autonomous defect detection.

Recommendation: Adopt the 45° angle-beam shear-wave probe as the primary NDT sensor for the tunnel inspection robot. Supplementary use of EMAT may be explored for future non-contact upgrades once lift-off control and excitation parameters are optimized. This choice maximizes detection reliability while satisfying the core robotic requirements of minimal operator intervention and single-sided access inside operational natural gas pipelines.

2.5 Summary of achievements

This section addresses the research questions and summarizes the principal findings of the ultrasonic transducer evaluation.

Response to Research Questions

1. Detection performance of the five transducer configurations the five probes were systematically tested against five artificial surface-breaking cracks of known geometry on an API 5L X65 carbon steel pipe (22 mm wall thickness). The 45° angle-beam shear-wave probe achieved 100 % detection of both inner- and outer-wall cracks with distinct echoes in the predicted 20–25 μ s window and high SNR. The contact P-wave probe reliably detected only inner-wall cracks, while the contact T-wave probe produced inconsistent amplitude reductions with high operator-dependent variability. The EMAT probe showed limited sensitivity (clear detection only for the deepest inner-wall crack C5), and the air-coupled transducer yielded noise-dominated signals with negligible defect information.
2. Influence of crack location and depth Crack depth and wall location significantly affected detectability. Deeper cracks (>4 mm, e.g., C2, C3, C5) generally produced larger amplitude reductions or clearer defect echoes. Outer-wall cracks (C1–C3) were poorly detected by normal-incidence probes (P-wave and T-wave) because the wave path runs nearly parallel to the crack face, resulting in weak scattering. Inner-wall cracks (C4–C5) were more readily identified by contact probes due to proximity. The 45° angle-beam probe was least sensitive to these geometric factors; its oblique shear-

wave zigzag path generated strong, consistent reflections from cracks of any depth or location.

3. Suitability for robotic deployment the 45° angle-beam shear-wave probe best satisfies the practical constraints of robotic internal pipeline inspection: single-side access, acceptable water-based gel couplant (amenable to automated dispensing) and demonstrated compatibility with robotic scanning. Contact probes suffer from couplant dependency and manual pressing variability, while air coupled and EMAT configurations exhibit insufficient SNR or repeatability under the test conditions.

Several limitations should be acknowledged. Only five artificial cracks were fabricated, limiting the statistical robustness of the results. All testing was conducted in a controlled laboratory environment on a clean (though progressively rusted) pipe segment; real operational pipelines will present additional challenges such as heavier corrosion, internal coatings, residual product, curvature effects, and dynamic robot motion. Equipment constraints (e.g., unavailability of EMAT) further reduced data completeness for some configurations. Finally, manual probe placement was used for most measurements, whereas full robotic deployment will introduce additional variables in probe alignment and contact pressure.

There are suggestions for next-stage robotic integration. For the subsequent integration phase (Section B), the 45° angle-beam probe should be selected as the primary NDT sensor. Recommended actions include: (1) design and fabrication of a compact robotic mounting fixture with automated gel dispensing and constant-force contact mechanism; (2) development of a precise probe alignment and constant-pressure mounting mechanism on the robot chassis to maintain optimal contact angle and coupling quality during dynamic movement inside the pipeline; (3) validation trials on the robot platform under simulated pipeline conditions (rust, curvature, vibration); and (4) optional parallel investigation of an optimized EMAT as a future non-contact backup once lift-off control is improved. These steps will ensure reliable, automated surface-crack detection inside operational natural gas pipelines, advancing the overall project objective.

MECH 4900 Final Year Design Project Final Report 2025/2026

3 Individual Section B: Robot Structural, Hardware and Software Design

By ZENG Minyu

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3.1 Introduction

The target of the inspection robot is to carry different types of ultrasonic transducers to the inner wall of natural gas pipeline and position them to detect types of defects. Inspection quality depends on the robot reaching the scheduled position and maintaining a precise distance between the transducer and pipe surface during inspection. This demands three interdependent engineering layers: a mechanical structure that holds concentricity with the pipe and provides the necessary degrees of freedom (DoFs) for sensor positioning; a hardware system that powers actuators and reads operator inputs; and embedded software that generates precise motion timing, coordinates multi-axis movements, and automates the inspection sequence. A failure in any layer degrades the entire system.

This section documents my work on all three layers for the robot's front-end sensing module. It covers the mechanical redesign from the first-generation prototype (V1) to the second-generation version (V2), the control circuit built around an Arduino Mega 2560, and the full embedded software stack — from low-level stepper motor pulse generation to automated inspection sequences. Ultrasonic techniques, signal processing, and system integration are addressed in Sections A, C, and D respectively.

The section is organized as follows. Section 3.2 defines the design requirements and the iterative prototyping methodology. Section 3.3 presents the schedule, milestones, and deliverables throughout the project. Section 3.4 will first review the V1 prototype and then trace the mechanical evolution from V1 to V2 across the framework, support mechanism, wheel carriers, rotating scanning assembly, and scanning arm, concluding with a summary comparison. Section 3.5 details the hardware architecture: control platform selection, dual power supply, and motor driver configuration. Section 3.6 covers the software system, emphasizing the timer-interrupt-driven motion architecture and the diagnosis of a mid-frequency resonance problem encountered during integration. Section 3.7 presents experimental results from single-DoF tests to system-integration pipe trials. Section 3.8 summarizes the achievements and identifies directions for future work.

3.2 Design Requirements and Methodology

The front-end sensing module must satisfy the detailed specification listed in Table 3.1. These requirements originated from the project's overall objective and were unchanged between V1 and V2.

My objective was to deliver a front-end module that meets Table 3.1 across all three engineering layers identified in Section 3.1: a mechanical structure satisfying the kinematic and dynamic requirements, a control circuit that powers six stepper motors and reacts to input signals and sensor states, and embedded software that generates precise multi-axis motion and automates the inspection workflow. The module was developed through two full prototype cycles — V1 and V2 — with the latter informed directly by the former's test results.

The design methodology followed an iterative cycle of concept sketches, CAD modelling, manufacture, separate testing, integration testing, and design revision. Within each prototype cycle, subsystem design proceeded from the rear of the module forward: the self-centering support mechanism was designed first, establishing the frame geometry; the rotating scanning assembly followed, its bearing placement and disk diameter constraining the scanning arm; the scanning arm and probe interface came last, fitted within the envelope defined upstream. This

back-to-front, constraint-driven ordering prevented rework when a design decision in one subsystem affected another.

Table 3.1 Design specifications for the front-end sensing module

Parameter	Requirement	Rationale
Pipe inner diameter range	440–510 mm	Robot should fit into pipe with different diameters in the prescribed range
Concentricity	Module center must coincide with pipe centerline	Any eccentricity introduces significant transducer-to-wall distance variance, degrading measurement consistency
Transducer-to-wall distance	Constant, precise and adjustable via radial DoF	Different transducer types require different stand-off distances; contact transducers require zero gap
Rotational coverage	360° continuous sweep and discrete point standby	Full-circumference inspection without reorienting the robot
No. of DoFs (sensing)	≥ 2	Circumferential rotation + radial extension; a third DoF (incidence angle) was present in V1 but eliminated in V2 after single-probe pulse-echo operation proved sufficient
Transducer mounting	Interchangeable, different diameter probes	Enables rapid switching between transducer types
Weight	< 6 kg (sensing module)	Must be transportable by a single person and must not overload the rear driving module
Structural material	Lightweight, rigid, manufacturable with accessible tools	Informed by V1's experience with acrylic: 2D-only fabrication forced multi-layer stacking and limited design freedom

Kinematic parameters were determined in SolidWorks by sketching each mechanism in its limiting positions in 2D and exploiting the software's built-in parametric constraint solver for rapid validation. For example, linkage geometry was tuned until robot's most typical posture stays in the best mechanical environment (in terms of force and velocity propagation), and all components' geometries are optimized for potential failures like buckling. The design is inspired by published industrial pipeline inspection robots [16][17][18] and classical mechanism of machinery [19]. The transition from V1 to V2 reflected a deliberate shift in design philosophy: from rapid proof-of-concept fabrication to a production-grade prototype, which will be elaborated in Section 3.4.

The hardware and software followed three primary principles: conservative design — minimizing complexity and modification for robustness and isolating risk across modules (e.g., a relay separates the control circuit from power circuit, so the system boots inert); layered modular architecture — each software module (motion, input, display, automation) was developed, tested, and integrated bottom-up with clear interfaces, enabling rapid iteration without destabilizing earlier layers; and root-cause resolution — when issues surfaced, the

underlying mechanism was identified before a fix was applied, rather than accumulating patches atop an unexamined foundation.

3.3 Project Schedule and Milestones

Table 3.2 summarizes the personal milestones from October 2025 to April 2026. The first semester focused on concept exploration and a rapid proof-of-concept prototype (V1); the second semester delivered a ground-up redesign (V2) and a complete embedded control system.

All milestones were met within the team's overall schedule. No blocking delays were passed to other members.

Table 3.2. Personal timeline and milestones

Period	Phase	Key activities	Deliverables
2025.10	Concept exploration	Literature survey on pipeline inspection robots; Concept 1 and Concept 2 considered and rejected; Concept 3 selected	Concept sketches, literature review
2025.11	V1 detailed design	Full CAD of V1 sensing module: CAM-follower support mechanism, 3-DoF scanning arm	V1 CAD model
2025.12	V1 prototyping & testing	V1 manufactured, assembled, and tested inside the pipe. All DoFs functional; robot successfully moved. Mid-term progress report submitted	V1 prototype, mid-term report
2026.01	V2 mechanical design	Complete redesign of front-end module in SolidWorks: hexagonal aluminum frame, self-centering 4-bar linkage, cross-roller bearing + PETG-CF rotary disk, U-channel scanning arm. Design time ~3 weeks	V2 CAD model
2026.02–03	Procurement & assembly	Parts outsourced for machining, metal, and 3D printing. Delayed ~2 weeks by Chinese New Year factory closures. Assembly completed early March	V2 prototype
2026.03	Control platform selection	ESP32 (3.3 V logic incompatible with A4988) → Arduino UNO (insufficient GPIO) → UNO Q (immature toolchain) → Arduino Mega 2560 (selected). WiFi control plan downgraded to USB serial for robustness	Hardware platform decision
2026.03	Hardware build & software	Circuit assembled. Software developed bottom-up: single-motor → multi-motor interrupt-driven → PS2/OLED integration	HW, SW development
2026.04 early	System integration	Front and rear modules connected via universal joint. Mid-frequency resonance discovered on drive-wheel axis during first pipe trial; IMU closed-loop control abandoned due to I2C signal degradation over long cable; Automation sequence → Python data pipeline	Software updates
2026.04 mid	Resonance resolution	Root cause diagnosed (330 Hz main-loop frequency locked motor speed to resonance band). Timer1 interrupt + 1/8 micro stepping deployed. Omni-ball casters replaced with rubber wheels (3D-printed adapters) after friction proved excessive	Optimized software, revised wheel configuration
2026.04 mid	Pipe validation	Five system-level tests completed: pipe clamping, cantilever touch-retract, full rotation, forward/backward motion, and automated inspection sequence demo	Test data, automation sequence demo recording
2026.04 late	System readiness	All subsystems verified; robot ready for integrated defect-detection demonstration	Defect-detection demonstration

3.4 Robot Structural Design: V1 to V2

3.4.1 Overall Configuration and Framework: Shared Heritage, Material Upgrade

The sensing module, in both its V1 and V2 design, adopts a multi-plate, multi-column frame architecture: two parallel plates separated and rigidly connected by six transverse aluminum cylindrical rods. This configuration provides torsional stiffness along the axial direction while leaving the interior volume open for the support mechanism. Exploded and assembled views of both prototypes are shown in Fig. 3.1 (V1) and Fig. 3.2 (V2).

V1 used four stacked acrylic plates of 8 mm thickness, fastened to the six rods with threaded standoffs. The plates carried all mechanical subsystems directly: the front face held the rotating scanning arm, the middle layers enclosed the CAM-follower support mechanism, and the rear plate interfaced with the driving module via a universal joint. While this arrangement proved functional — V1 stood, moved, and performed all DoF motions inside the pipe — the acrylic material imposed two fundamental constraints. First, acrylic is limited to 2D fabrication by laser cutting; complex three-dimensional features such as integrated bearing seats, stiffening ribs, or folded flanges could not be realized without bonding separate pieces. This forced a multi-layer stacking strategy where each functional layer required its own plate, driving up the module's thickness (194 mm) and weight (>5 kg). Second, acrylic's low specific strength meant that plates had to be thick enough to carry structural loads, further deteriorating the weight issue.

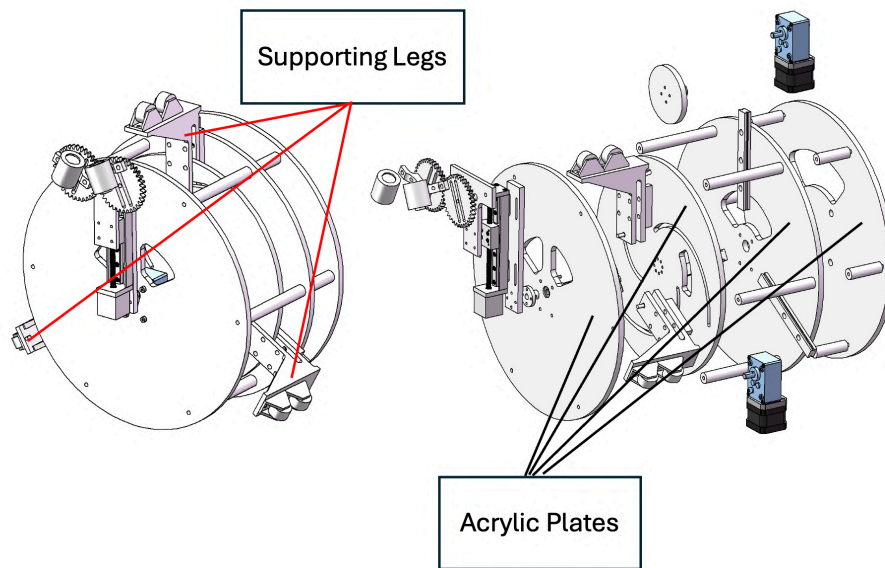


Fig. 3.1 Assembled view (left) and exploded view (right) of V1 prototype

V2, shown in Fig. 3.2, replaced the four 8-mm acrylic plates with two 2.5-mm aluminum alloy hexagonal end plates connected by the same six-rod architecture. The reduction from four plates to two was enabled by aluminum's three-dimensional formability: features that in V1 required separate plates — such as mounting flanges, clearance cutouts, and cable routing channels — could be integrated into a single bent or machined part. The end plates were cut into irregular hexagons, with three rectangular cutouts at alternating edges (120° spacing) providing physical clearance for the support legs when fully retracted, thereby reducing the module's minimum passable diameter. Curved slots and circular holes drilled into the plates

served as internal cable holes between the front sensing assembly and the rear hardware module, thus preventing mechanical interferences.

Although aluminum alloy (2.8g/cm^3) is approximately 2.3 times denser than acrylic (1.2g/cm^3), reduction in total number of plates and thickness resulted in a net reduction in frame mass. The switch to aluminum unlocked design freedoms that shaped every subsequent subsystem: the folded U-channel scanning arm, the wheel-carrier brackets, and the rear bracket for the universal joint all depended on metal's ability to carry load in three dimensions. The six-rod column architecture, inherited from V1, proved to be a sound structural choice — it provided torsional rigidity in both prototypes and simplified assembly by defining a clear alignment axis.

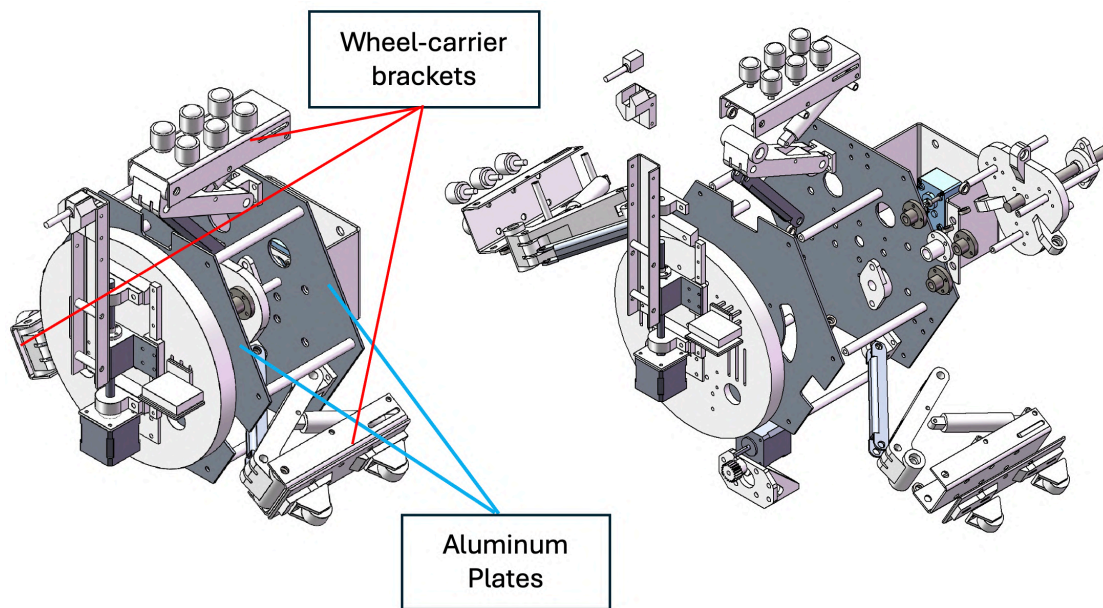


Fig. 3.2 Assembled view (left) and exploded view (right) of V2 prototype

3.4.2 V1 Sensing Module: Design and Lessons Learned

V1, as illustrated in Fig. 3.1, was a functional proof of concept for the concentric three-legged architecture. This section briefly documents its design and the problems that motivated the V2 redesign; detailed descriptions of V1's subsystems appear in the midterm progress report.

The V1 sensing module had an envelope diameter of 350 mm and a thickness of 194 mm. Fig. 3.3 shows the key subsystems in a front-view schematic: the CAM-follower support mechanism and the three support legs. The heart of the design was an internal CAM plate — a flat acrylic disk into which three identical Archimedean spiral slots were laser-cut at 120° spacing. A stepper motor with a 100:1 worm-gear reducer rotated the CAM plate about the module's central axis. Shoulder bolts riding in each slot converted the rotary motion into synchronized radial displacement of the three legs, each constrained to move along a dedicated linear guide rail. Because the spiral profile is monotonic, a single motor could drive all three legs outward or inward, and the worm-gear reducer provided self-locking to hold position without continuous power. The sweeping arm (in Fig. 3.1), mounted on the front motor, provided three degrees of freedom: 360° rotation, vertical lift via a ball-screw linear slide, and manual incidence-angle adjustment through a meshed gear pair. An adjustable clasp accommodated transducers of 28–35 mm diameter.

V1 was assembled and tested in Fall 2025 and all DoFs functioned; the robot successfully stood, moved forward and backward, and responded to PS2 wireless control. These results validated the concentric three-legged architecture as a viable concept for in-pipe inspection.

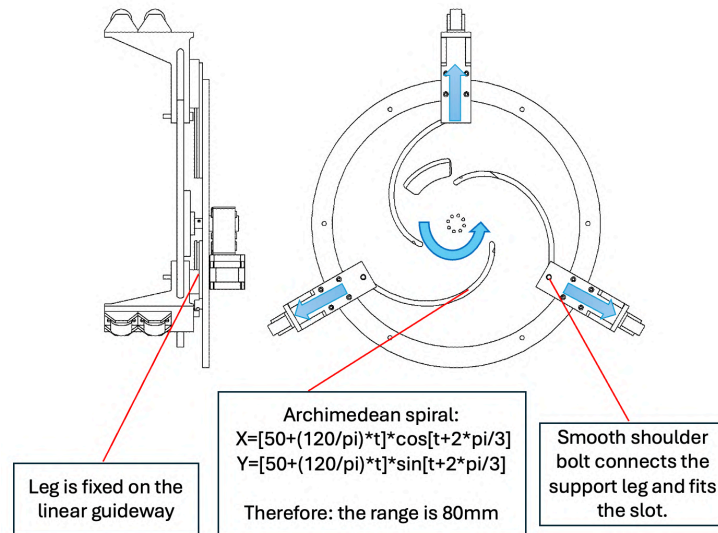


Fig. 3.3 The side-view (left) and front-view (right) of CAM-follower support mechanism and support legs

Lessons learned. Testing revealed five systemic weaknesses that could not be resolved within the acrylic-and-CAM design paradigm:

1. Conflated load paths. The three support legs transferred their full supporting and bending loads through the CAM plate directly onto the geared motor's output shaft (clearly seen in Fig.3.3 side view). The motor was required to bear the entire structural weight of the sensing module in addition to providing actuation torque, and this was a single-point failure risk.
2. Backlash in the CAM and coupling. The CAM slot, machined from acrylic by laser cutting, had inherent dimensional tolerances that produced backlash at the leg tips. The flange coupling connecting the geared motor to the CAM plate relied on a single set screw biting into the key slot on the motor shaft; the size mismatch between the small shaft and the large disk amplified the backlash to several degrees of free play.
3. Restrictive leg geometry. Each leg was an unbalanced cantilever fixed to the linear rail. The leg length was constrained by the inter-plate spacing, and any attempt to lengthen it for larger diameter coverage would produce unacceptable bending deflection at the tip.
4. Poor maintainability. Accessing any leg or the CAM-follower mechanism required disassembling the middle two plates, and they were the ones who need the most frequent repair.
5. Excessive weight. The module weighed over 5 kg, primarily due to the thick acrylic plates.

These five problems defined the design objectives for V2: separate structural and actuation load paths; eliminate backlash at every transmission interface; free the legs from inter-plate constraints and give them independent suspension; make every module independently accessible; and reduce weight. Sections 3.4.3 through 3.4.7 address how each of these objectives was met.

3.4.3 Support Mechanism: Self-Centering Four Bar Linkage

The CAM-follower was the most severely flawed subsystem in V1. It concentrated structural load and actuation torque onto a single motor shaft, introduced backlash at two series interfaces (CAM slot and flange coupling), and buried the mechanism between plates where it could not be accessed. The V2 redesign replaced it with a self-centering four-bar linkage, so named because its three synchronized linkages visually resemble a folding umbrella frame when extending and retracting driven by a central lead screw. Fig. 3.4 shows the overall assembly and Fig. 3.5 the kinematic diagram of a single leg.

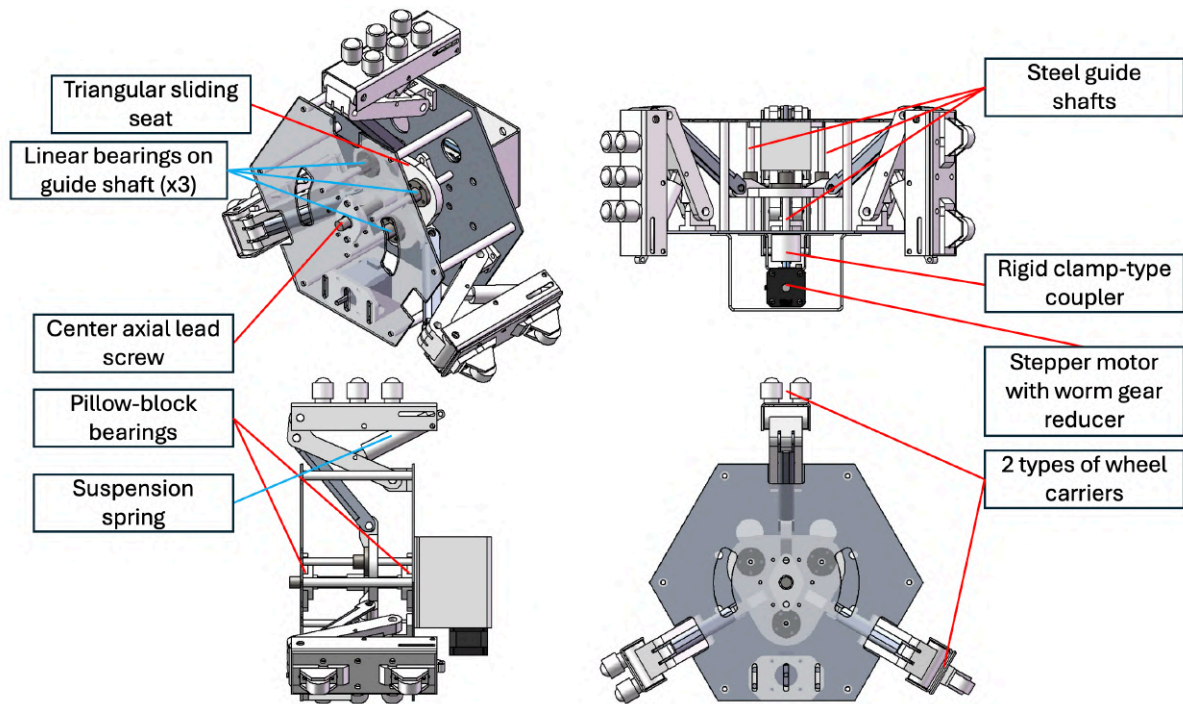


Fig 3.4 Assembly of self-centering mechanism: isometric view (upper left), bottom view (upper right), side view (lower left), front view (lower right)

A stepper motor with a 100:1 worm-gear reducer, which is the same model used in V1, drives a central axial lead screw (pitch 2 mm, stroke 50 mm) through a rigid clamp-type coupler. The lead screw is supported at both ends by pillow-block bearings mounted on the front and rear end plates, which constrain it to pure rotation. As the screw rotates, a triangular 3d-printed sliding seat, which is driven by a nut embedded in its center, moves linearly along the module axis. The seat is prevented from rotating by three hardened steel guide shafts that pass through it and are fixed to both end plates; these shafts also absorb radial load, ensuring the seat moves with strict concentricity relative to the module axis. The shafts get in contact with the linear bearings on the sliding seat thus allowing it to slide back and forth under minimum rolling friction.

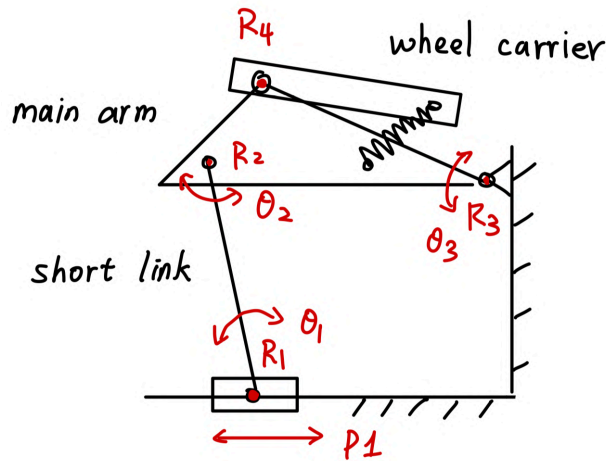


Fig. 3.5 Kinematic diagram of a single leg

Three identical four-bar linkages, spaced at 120° , connect the sliding seat to the wheel carriers. Each linkage comprises a short link (the coupler) and a long main arm (the rocker). Joints R1 (seat to short link) and R2 (short link to main arm, at mid-span) use pin shafts with embedded ball bearings to minimize friction. Joint R3 (main arm to rear end plate) is the fixed pivot; joint R4 (main arm tip to wheel carrier) is the output. The kinematic logic is straightforward: as the sliding seat moves toward the front plate, the short link pushes the main arm outward, rotating it about R3 and driving the wheel carrier radially outward. Reversing the screw retracts all three legs. Because the three linkages are constrained to a common sliding seat, their motion is mechanically synchronized — no independent calibration is needed. The 1:100 gearbox and fine-pitch lead screw together provide self-locking: axial forces on the legs cannot back-drive the screw, so the mechanism holds position without continuous power.

Fig. 3.6 illustrates the most fundamental structural improvement: the separation of actuation and load-bearing paths. In V1, the three legs channeled all gravitational and bending loads through the CAM plate into the motor shaft. In V2, the three guide shafts carry the entire structural weight of the sensing module; the lead screw experiences only the axial component of the linkage forces required to hold or adjust the leg position. The motor is consequently freed from structural duty and only responsible for motion.

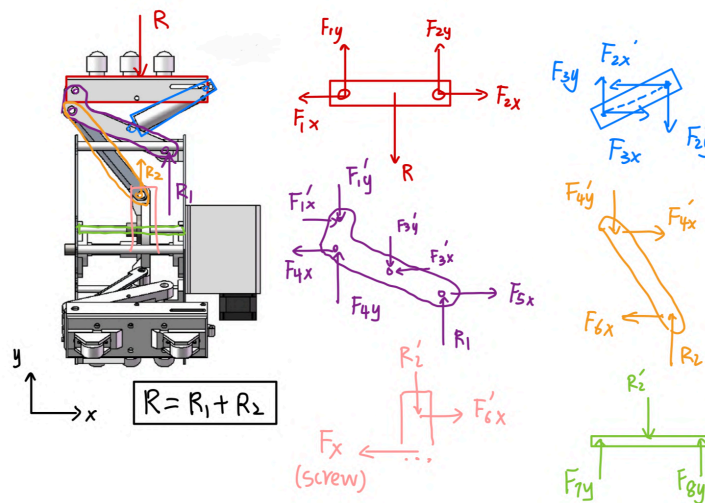


Fig. 3.6 Structural static analysis

These changes addressed the backlash that plagued V1. The CAM slot — a sliding-contact interface with millimeter-scale tolerances — was eliminated entirely; the four-bar linkage uses rolling-element bearings at all joints (R1–R4), which have negligible clearance. The flange coupling, whose single set screw had been the dominant source of free play in V1, was replaced by a clamp-type coupling that grips the shaft circumference uniformly.

3.4.4 Wheel Carrier and Suspension

V1's legs were cantilevers with one side fixed on linear rails, their length constrained by the inter-plate spacing (Fig. 3.3). The V2 wheel carrier, shown in Fig. 3.7, is an independent assembly that decouples leg geometry from the frame. Each carrier is fabricated from an aluminum plate bent into a U-channel; the main arm (36 mm wide) passes through the interior of the U and pivots at the front via joint R4. Two short aluminum columns brace the U-walls internally, forming a local torsion box that resists twisting and inward deformation. This dual-pivot arrangement, with R4 at the front, a suspension spring at the rear, replaces V1's single-side mounting and allows the leg to be substantially longer without bending.

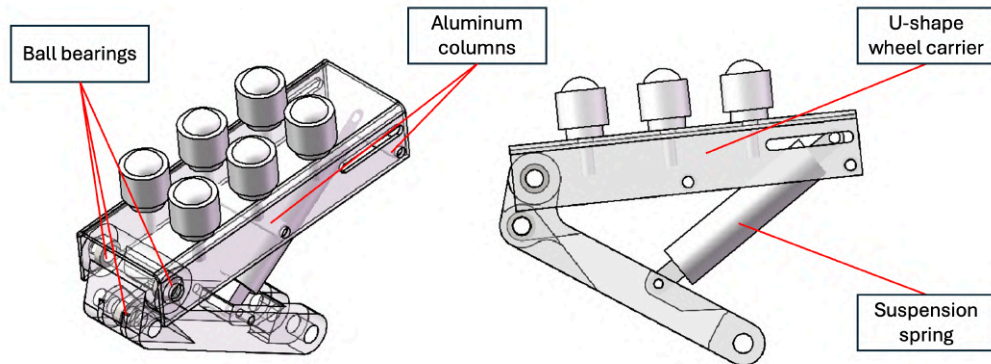


Fig. 3.7 The wheel carrier and suspension assembly

The suspension spring connecting the main arm to the rear of the carrier provides passive compliance. In its free state, the carrier tilts rearward. When the robot enters the pipe and the linkage extends, the rear wheel contacts the wall first, compressing the spring; further extension brings the front wheel into contact. Once both wheels are seated, the spring preload maintains consistent wall pressure while absorbing local irregularities (estimated compliance of approximately ± 5 mm).

The wheel configuration underwent one design change during integration. The initial V2 assembly used six steel omni-directional ball casters per carrier (18 total), intended to give the front module rotational freedom inside the pipe. This freedom proved unnecessary, and the steel balls generated excessive rolling resistance that the rear driving module could not overcome. The omni-ball casters were replaced by two rubber wheels per carrier, mounted via 3D-printed adapters, reusing V1's rubber wheels. This change, documented in Section D, reduced rolling resistance sufficiently even under spring compression for reliable forward motion while retaining vibration damping from the rubber compliance.

3.4.5 Rotating Disk Assembly

V1's acrylic scanning arm was driven directly by a geared motor through a flange coupling (Fig. 3.1). The coupling's single set screw, which has been identified in Section 3.4.2 as the

dominant backlash source, connected a small-diameter motor shaft to the arm with no intermediate support, leaving the arm cantilevered and prone to angular wobble. V2 introduced a disk to consolidate the entire drivetrain with the assembly shown in Fig. 3.8: an assembly view and a side exploded view revealing the layered construction.

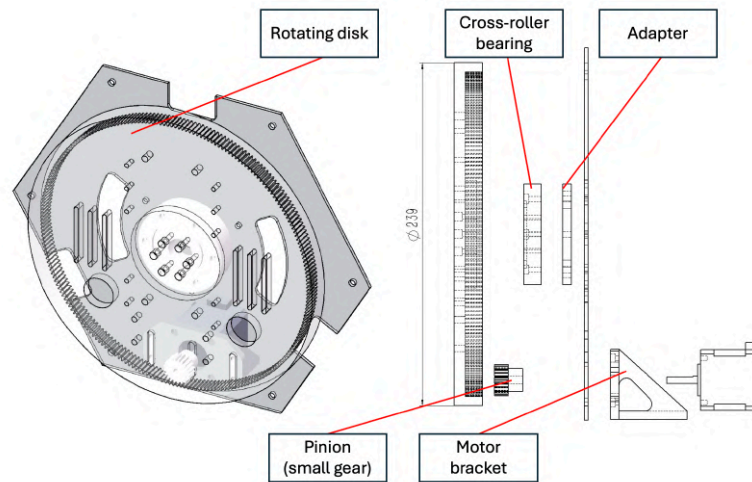


Fig. 3.8 Assembly view (left) and side exploded view of rotating disk

The core of the assembly is a high-precision cross-roller bearing [20] mounted at the center of the front-end plate via an adapter (which is used to create some clearance for screw tightening). Unlike a conventional deep-groove bearing, a cross-roller bearing supports radial, axial, and overturning-moment loads simultaneously, which is a requirement driven by the scanning arm's overhung mass and load. The bearing's outer ring bolts to the front plate; its inner ring carries a rotary disk, 3D-printed from carbon-fiber-reinforced PETG (PETG-CF) [21], whose inner circumference integrates a full internal ring gear (module 1, 224 teeth). A stepper motor, mounted on the front plate via a 3D-printed bracket, drives a 20-tooth pinion that meshes with the internal gear. The resulting reduction ratio of 11.2:1 gives an angular resolution of 0.16° per full step, and the corresponding 2240 steps for a disk's full revolution.

Backlash in the rotary axis is controlled by two design choices. First, the flange coupling was eliminated entirely by using the pinion to drive the ring gear. The internal engagement mechanism makes good use of the large diameter of disk to split function into highly concentric rotation and eccentric driving. Second, the motor bracket is secured to the front plate through four screws passing through slotted holes. Loosening the screws allows the bracket and therefore the pinion to be slid radially toward or away from the ring gear, adjusting the mesh clearance. Once the desired engagement is achieved, the screws are torqued to lock the position. This simple manual adjustment proved sufficient to eliminate perceptible backlash in the rotary axis during operation.

3.4.6 Scanning Arm and Transducer Interface

The introduction of rotating disk (Section 3.4.5) fundamentally changed the design space available for the scanning arm. In V1, the arm was directly coupled to the geared motor shaft, inheriting all the backlash and instability. In V2, the arm mounts on top of the rotating disk — a large, rigid platform whose rotation is already handled by the cross-roller bearing. The separation of functions that the arm only deals with radial positioning allowed each subsystem to be optimized independently.

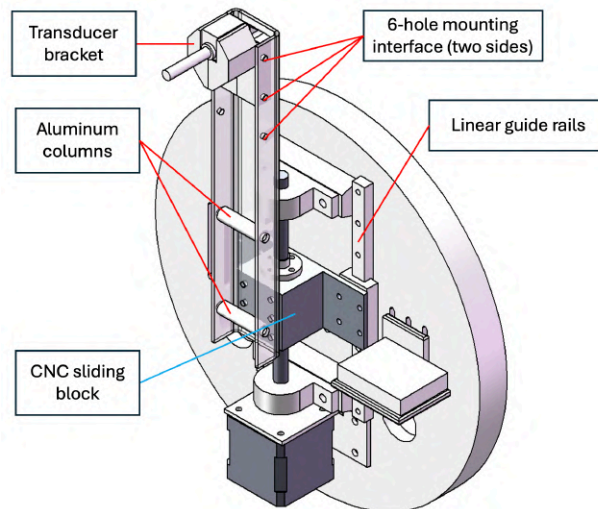


Fig. 3.9 Scanning arm assembly

The arm itself, shown in Fig. 3.9, is fabricated from aluminum sheet bent into a U-channel cross-section. It provides high bending stiffness with minimal mass, and like the wheel carriers (Section 3.4.4), two short aluminum columns brace the U-walls internally to resist twisting and inward deformation.

The radial DoF is achieved by a pair of miniature linear guide rails mounted on the disk face. A custom CNC-machined slide block, driven by an independent stepper motor through a lead screw (pitch 2 mm, direct-drive, no reducer), moves the arm along the rails. This assembly gives a displacement resolution of 0.01 mm per step and a radial travel range of 0 to 62 mm. When the arm is operated at a pulse interval of 6000 μ s, the corresponding linear speed is 1.67 mm/s, ensuring smooth, controllable contact with the pipe wall during positioning.

V1's third degree of freedom (a meshed gear pair for manual incidence-angle adjustment) was eliminated in V2. Experimental evaluation during the first semester demonstrated that single-probe pulse-echo operation, provided sufficient signal quality for defect detection without the complexity of angle adjustment. This decision eliminated one mechanical joint, one manual calibration step, and a potential source of positioning error.

The arm tip carries a six-hole precision mounting interface. Different transducer brackets, for instance, contact, air-coupled, or angle-beam probes can attach to this interface via matching bolt patterns. The angle-beam shear-wave probe used in the final defect-detection demonstration (Section C and D) was mounted through this interface.

3.4.7 Conclusion of Robot Structural Design

A common upgrading principle runs through the subsystems described above. V1 accumulated functions onto shared components: the motor shaft bore structural weight, the CAM plate served as both bearing surface and motion guide, the sweeping arm both rotated and retracted. Each function was achieved, but the system was highly coupled and limited for any improvement. V2 systematically undid these couplings. The 3 steel guide shafts carry weight; the lead screw only delivers torque. The cross-roller bearing handles smooth rotation and the internal gear determines the precise angle. The suspension springs absorb unevenness on the terrain while the linkages ensure concentricity. Each component was given exactly one responsibility and a clean interface to its neighbors. This brings the benefits that one subsystem

can be designed, tested, and upgraded in isolation, and makes V2 a rational prototype compared with the concept-checking V1. (Fig. 3.10 is an actual comparison of two versions of prototype) Sections 3.5 and 3.6 extend this idea into the electronic and software domains.

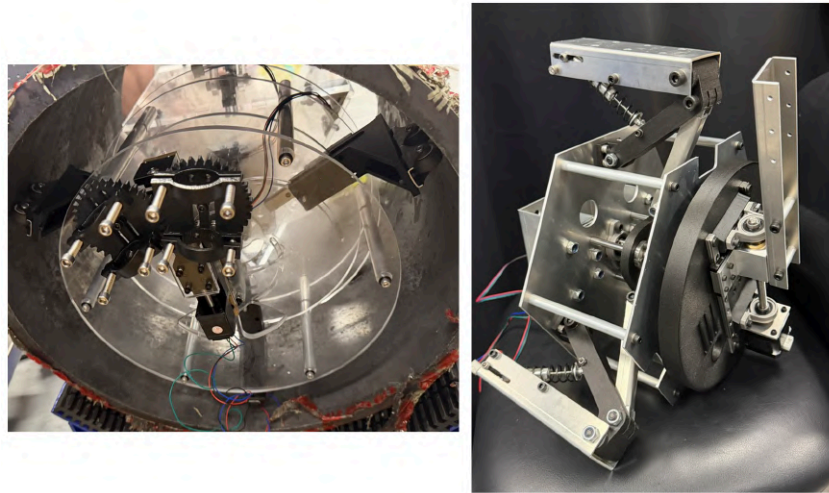


Fig. 3.10 Comparison of two prototypes

3.5 Robot Hardware Design

3.5.1 Control Platform Selection

Selecting the microcontroller for the robot needs to meet four constraints: the number of general-purpose I/O pins needed to drive six stepper motors and interface with peripherals; compatibility with the 5 V logic level expected by the A4988 stepper drivers [22]; maturity of the software toolchain (libraries); and the communication for remote operation. Four platforms were evaluated empirically. Each was programmed and tested with the actual motor hardware before the final choice was made.

Firstly, Espressif ESP32 was attractive for its integrated Wi-Fi and Bluetooth, which would allow the robot to be controlled through local area network (LAN) without a wired connection. A functional network-control firmware was written and validated using the ESP32's onboard LED. However, when connected to the A4988 driver, the ESP32's 3.3 V logic outputs failed to reliably trigger the driver. The same code transferred to an Arduino UNO drove the motor without issue, confirming that the problem was hardware-level incompatibility rather than a software defect. The ESP32 was abandoned.

The Arduino UNO was the natural fallback: 5 V native logic, vast community support and pin-compatible with the CNC Shield v3 that carries the A4988 modules makes it a rational choice for fast prototyping. However, once the CNC Shield was seated, the UNO's remaining GPIO pins were insufficient for the PS2 receiver (4 pins), OLED display (2 pins for I2C), and relay (1 pin) required by the system. The Arduino UNO Q, a recently released dual-core board combining a Linux-capable microprocessor with a real-time microcontroller, was also briefly considered for its ability to run a local web server natively. But its software ecosystem proved too immature: the Qualcomm SoC and new MCU lacked stable compiler support and community libraries, and its GPIO amount was identical to the standard UNO.

The Arduino Mega 2560 [23] satisfied all constraints. Its 54 digital I/O pins accommodated the CNC Shield, PS2, OLED, relay, and reserve pins for future expansion. It operates at the 5 V logic level required by the A4988. The mature Arduino ecosystem reduced development risk.

The trade-off was the loss of integrated wireless capability. An ESP8266 MQTT module was briefly evaluated to support IoT, but the complexity of deploying a message broker was judged disproportionate to the benefit. The final configuration adopted a USB serial link (115200 baud) to a host PC running Python scripts for data logging, while a PS2 wireless gamepad provided local real-time control.

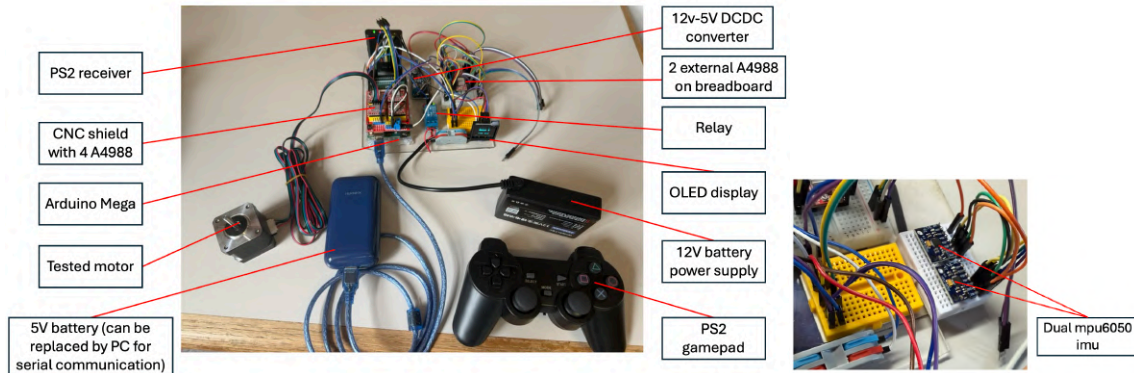


Fig. 3.11 Complete hardware system overview

The complete hardware system, photographed in Fig. 3.11, integrates the Mega, CNC Shield, six A4988 drivers (four on the shield, two on an external breadboard to accommodate the three drive-wheel motors sharing a single step/direction signal), dual power supplies, relay, OLED display. The auxiliary view in Fig. 3.11 is the dual IMU (MPU 6050), which were expected to measure the rotation angle but eventually discarded for which reasons will be discussed in section D.

3.5.2 System Architecture

Fig. 3.12 shows the system-level connection diagram. The Arduino Mega sits at the center, receiving commands from the PS2 controller over SPI and exchanging data with the host PC over USB serial. It outputs step and direction signals to the CNC Shield, which distributes them to the six A4988 stepper drivers. The enable pin (D8) is shared across all axes, allowing a single logic signal to disable all motors simultaneously. An SSD1306 128×64 OLED display, connected via hardware I2C (SDA=D20, SCL=D21), provides real-time feedback on motor positions, relay status, and system alerts.

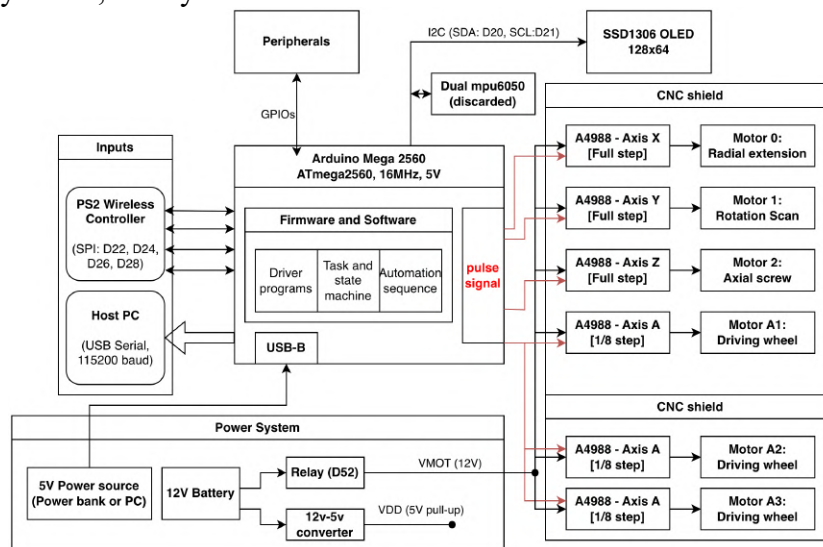


Fig. 3.12 Hardware architecture diagram

Power is supplied by two independent batteries. A 5 V power bank connects to the Mega's USB-B port and powers the microcontroller. A 12 V Li-Po battery (approximately 6 A maximum discharge) powers the CNC shield and motors. This 12 V rail passes through a normally open relay controlled by Mega pin D52. The relay's default-open state ensures that the system boots inert: no motor can receive power until the operator explicitly closes the relay via the PS2 START button. All motor motion commands in firmware are gated behind the relay status check, providing a hardware-level safety interlock independent of software state. From the relay, the 12 V rail branches to the VMOT pins of all six A4988 drivers. A separate branch feeds a 12 V-to-5 V DC-DC buck converter, whose regulated output supplies the VDD logic pins of the A4988 modules, keeping the driver logic supply isolated from the Arduino's own 5 V rail.

Microstepping was applied selectively across the four axes. The drive-wheel axis (Motor A) was configured for 1/8 microstepping by jumpering the M0 & M1 pins on its three A4988 drivers, yielding 1600 microsteps per revolution. This reduced the step-angle discontinuity identified as a contributor to mid-frequency resonance (Section 3.6.5). The remaining three axes, radial extension (Motor X), disk rotation (Motor Y), and axial linkage (Motor Z) — were operated in full-step mode. All step counts in firmware are expressed in raw driver pulses; the microstep ratio for Motor A is applied downstream in the Python coordinate-conversion script.

The initial hardware design included two MPU6050 inertial measurement units, mounted on the rotary disk at I2C [24] addresses 0x68 and 0x69 (see Section D), intended to provide closed-loop angular position feedback for the scanning disk. The MPU6050 integrates a 3-axis gyroscope and a 3-axis accelerometer with an on-chip Digital Motion Processor (DMP) that performs sensor fusion by combining the gyroscope's short-term accuracy with the accelerometer's long-term gravity reference, and outputs quaternions or Euler angles directly, offloading the computation from the main microcontroller. Of the three Euler angles, roll and pitch are gravity-referenced and remain stable over time; yaw, however, has no absolute reference and accumulates unbounded drift through integration of gyroscope bias, making it unsuitable for long-term angle tracking without a magnetometer. The dual-sensor arrangement was designed to improve reliability through redundancy: their roll-angle readings were averaged to produce the disk's angular position. The sensors were calibrated at startup by recording and subtracting the initial roll offset. During benchtop testing with short I2C cables, both sensors returned stable, consistent roll-angle measurements. However, when the front module was connected to the rear hardware compartment, the cable run of approximately one meter, and the I2C signals experienced severe degradation due to the combination of cable capacitance, voltage drop over the long wiring, and electromagnetic interference from the 12 V motor circuits. The I2C specification [25] recommends a maximum bus capacitance of 400 pF, a limit readily exceeded by a 1-metre unshielded ribbon cable. The industrial-level robust alternative considered, an RS485-protocol inclinometer, was physically too large (approximately palm-sized) to mount on the rotary disk. The closed-loop IMU approach was therefore abandoned in favor of open-loop step counting, with the disk angle derived from the gear reduction ratio (2240 steps per revolution). This limitation is discussed further in Section D.

A complete pin assignment table for the Arduino Mega, including CNC Shield signal mapping and all peripheral connections, is provided in Appendix B.1 for reference and replicability.

3.6 Robot Software Design

3.6.1 Software Architecture

The software was developed in C++ using the PlatformIO [26] on the Arduino Mega 2560. Its architecture extends the separation-of-concerns principle: splitting program into 3 layers from top to bottom and dividing into multiple independent modules, and separating timing-critical signal generation from user-interaction logic into independent execution threads.

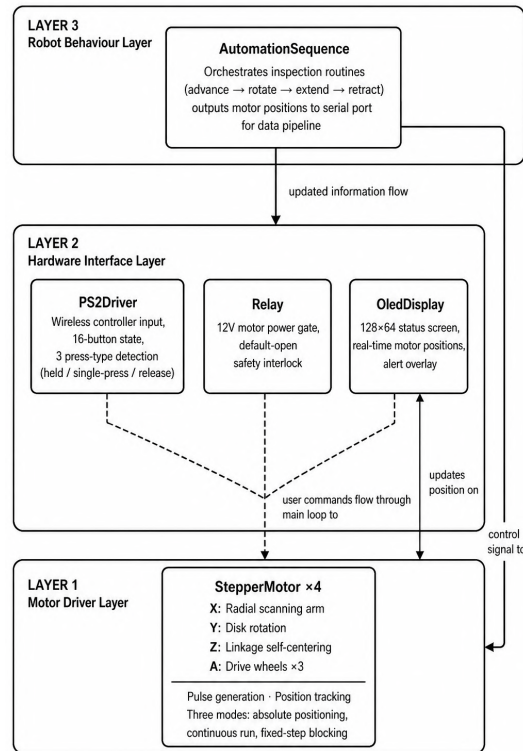


Fig. 3.13 Software architecture diagram

The five classes defined in program form a layered architecture, as shown in Fig. 3.13. At the foundation, **StepperMotor** encapsulates all pulse generation and position tracking for a single motor axis. It provides three motion modes (absolute positioning, continuous run, and fixed-step execution) so that higher layers can command motors without any knowledge of pin voltages or pulse widths. Building on this foundation, three interface classes bridge the software to the physical world: **PS2Driver** translates the wireless controller's 16-button state into the three press-type distinctions needed for correct operation (distinguishing a held button for sustained rotation from a single press for a prescribed step, for instance); **Relay** models the safety relay that gates motor power, ensuring no motion command can execute before the operator explicitly enables the 12 V rail; and **OledDisplay** renders real-time motor positions and alert messages so the operator can monitor system state without a connected PC. At the top, **AutomationSequence** stores a predefined list of motion steps - advance, rotate, extend the scanning arm, retract - and outputs the resulting motor positions to the serial port after each step. The control signals are passed to Layer1 at the right time to perform an expected robot behavior.

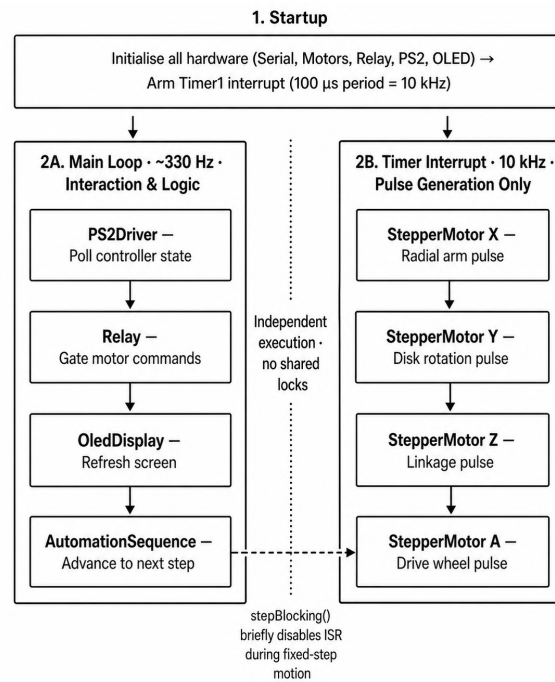


Fig. 3.14 System runtime execution diagram

The following are the program's sequential logic, and Fig. 3.14 illustrates how execution is divided across parallel threads. At power-up, an initialization sequence activates all hardware including serial port, four stepper motors, the safety relay, the PS2 wireless controller, and the OLED display, and then arms an onboard timer to fire every 100 microseconds. The startup will not pass if any hardware issue occurs. After self-checking, two threads run in parallel. The main loop, shown on the left of Fig. 3.14, cycles at approximately 330 Hz and handles all user-facing tasks: reading controller inputs, interpreting which button was pressed and in what manner, updating the on-screen display, and advancing the automation sequence when active. On the right, a timer interrupting service routine fires at 10 kHz with a single responsibility: generating the step pulses that move all four stepper motors. This split means that pulse timing, which requires microsecond-level consistency, is never disrupted by a slow display refresh or a lengthy serial transmission. The main loop decides what the robot should do; the interrupt decides precisely when each step pulse arrives at the motor driver. Section 3.6.5 demonstrates that this separation was not a design preference but a functional necessity: without it, the drive wheels could not move the robot.

Each class and thread carries a single responsibility and communicates through clean interfaces. The remainder of Section 3.6 explains how each module and layer was developed and integrated, moving from the StepperMotor foundation upward.

3.6.2 Stepper Motor: Interrupt-driven Pulse Generation

A stepper motor advances when its driver receives a rising edge on the STEP pin. When the robot's typical interval is 5500 µs per step, the motor demands roughly 180 pulses per second at precisely maintained spacing. An update function in main loop cannot deliver this: PS2, OLED, and serial communication each consume variable time, and the resulting jitter (typically ± 1500 µs around a 3ms cycle) makes the speed parameter meaningless. One may consider using a blocking function to generate a precise pulse, i.e. delay for certain time and let CPU

stay idle during the period, it works but the drawback is trivial: it causes a huge waste of resource and CPU will no longer be able to handle any other tasks like updating the display, serial, or even worse, driving several motors to move simultaneously. The solution is to offload pulse timing to a hardware peripheral that runs independently of program flow.

The ATmega2560's Timer1 is a 16-bit counter clocked directly by the system's 16 MHz quartz crystal [27]. On every clock cycle it increments; when it reaches a preloaded reference value, a hardware comparator sets a flag that forces the CPU to suspend the main loop, jump to a designated interrupt service routine, execute it, and resume the main loop at the exact point of interruption. Because the crystal oscillates on its own circuit and the match is detected by a dedicated comparator with no software polling, the interval between interrupts is immune to main-loop congestion. The TimerOne library [28] is used to configure Timer1 to fire every 100 μ s with zero long-term drift.

The three motor operating modes are built atop this mechanism. Continuous mode uses the ISR alone: when the operator holds $\blacktriangle$ or $\times$ on the gamepad, the main loop has preset the motor's direction and speed, and the interrupt service routine (ISR) generates pulses autonomously. Blocking mode, triggered by single-press R1/R2, is used for automated sequence steps. It disables the ISR globally with `cli()` function and produces pulses in a tight loop with `delayMicroseconds()`, re-enabling the ISR with `sei()` [29] upon completion — this prevents the two execution contexts from racing on the same STEP pin while guaranteeing that a prescribed number of steps completes before the next sequence action. Absolute-position mode (`moveTo()`) was retained for potential closed-loop use. The pseudo code for execution flow is shown below.

```

loop():                                // main program loop, ~330 Hz
  ps2.update()                          // poll controller
  handlePS2Input()                      // dispatch buttons to motor commands
  //      |
  //      | Suddenly Timer1 fires (100  $\mu$ s since last match)
  //      | CPU suspends loop, pushes registers to stack, jumps to:
  //      ↓
  motorISR():                           // interrupt service routine, 10 kHz
    motorX.updateDirectInInterrupt()
    motorY.updateDirectInInterrupt()
    motorZ.updateDirectInInterrupt()
    motorA.updateDirectInInterrupt()
  //      | CPU pops registers, resumes loop at interrupted instruction
  //      ↓
  oled.update()                         // refresh display
  sequence.update()                     // advance automation
  // ... loop continues

updateDirectInInterrupt():
  now = micros()
  if pulse is active:
    if now - pulseStart >= 200: // A4988 minimum pulse width met
      STEP_PIN = LOW           // end this step
      pulseActive = false
      position += direction
  else:

```

```

if now - lastStep >= interval:
    STEP_PIN = HIGH      // begin next step
    pulseStart = now
    pulseActive = true
    lastStep = now

```

The ISR's two-state logic, that one tick to raise the pin, subsequent ticks to hold until the pulse width expires, keeps each interrupt service under 100 ns, preserving ample time for main-loop tasks. The interval and direction variables are written only in the main loop (in response to operator input) and read only by the ISR (to determine when and in which sense to pulse), so the two threads share data without locks. This realization was developed in direct response to a disastrous case – the resonance band discussed in Section 3.6.5.

3.6.3 Control Interface

The operator controls the robot through a PS2 wireless gamepad and an SSD1306 128×64 OLED display, each encapsulated in its own class (PS2Driver and OledDisplay) that hides the underlying communication protocol from the rest of the program.

The PS2 controller transmits a 16-button state packet over SPI every frame. The PS2Driver class, built on the PS2X library [30], stores the current and previous bitmasks and derives three detection modes from their difference. `pressed()` returns true if a button stays down, `justPressed()` is triggered once on the falling edge when a button is just pressed, and `justReleased()` fires on the rising edge when a button is released. Table B.2 maps each button to its function, detection mode, and relay protection status.

The jog increment N step (triggered by pressing R1/R2) is axis-dependent and can be changed subjected to the target. This variable sizing is adaptive to the four axes' widely different motion resolutions. All motion commands pass through a relay gate in `handlePS2Input()`: if a motion button is pressed while the relay is open, the command is silently discarded, and the OLED shows an alert.

The OLED is driven by the U8g2 library [31] in page-buffer mode over hardware I2C. It maintains two display layers. During normal operation, the selected motor index (0–3) and its cumulative step count are shown, refreshed only when the state changes via a dirty-flag check that avoids redundant rendering. An alert layer superimposes when a command is rejected — displaying "Access Denied" for 1.5 s — and the normal view automatically restores afterward. A project-identifying boot screen is shown for the same duration at startup.

3.6.4 Automation Sequence and Python Data Pipeline

The AutomationSequence class executes a predefined list of motion steps and streams the resulting motor positions to the host PC for coordinate reconstruction.

A sequence is an ordered array of SequenceStep structs, each specifying a type, target motor, parameter, and speed. Six step types are defined: motor steps, motor angle (reserved), wait, arm extend, arm retract, and repeat. Two trigger modes are provided: pressing ■ calls `start()` to launch the whole sequence, with the main loop's `update()` advancing one command per frame automatically; pressing ↑ calls `executeOneStep()` for single-command debugging. Each frame, `update()` dispatches the current step by type: calling `stepBlocking()` for motor movements,

delay() for waits, or resetting the index for repeats, and then calls printMotorStates() to output the four-axis step counts to serial and advances the index.

On the host PC, “serial_logger.py”, using the PySerial library [32], detects the Arduino serial port, attaches an elapsed-time stamp to each line read, and saves the complete session log. “log_to_coords.py” parses the log and converts the raw step counts into cylindrical coordinates (r, θ , z) referenced to the pipe centerline. The radial coordinate is fixed at the pipe inner radius, $r = 234$ mm. The angular coordinate is derived from the calibrated disk revolution constant:

$$\theta = S_Y \times 360^\circ / 2240 \quad (3.1)$$

where S_Y is the cumulative step count of Motor Y and 2240 steps/rev. The axial coordinate is derived from the drive wheel geometry and microstepping configuration:

$$z = S_A \times \pi D / (n_\mu n_s) = S_A \times \pi \times 70 / (8 \times 200) = S_A \times 0.137 \text{ mm} \quad (3.2)$$

where S_A is the cumulative step count of Motor A, $D = 70$ mm is the wheel diameter, $n_\mu = 8$ is the 1/8 microstep ratio, and $n_s = 200$ steps/rev is the full-step count per motor revolution. The output CSV contains elapsed time, all three cylindrical coordinates in both degrees and radians, and the raw step counts for traceability, and serves as the handoff to You Qi's data visualization pipeline (Section C).

3.6.5 Mid-Frequency Resonance: Diagnosis and Solution

The most challenging software problem encountered during integration was a mid-frequency resonance. Its diagnosis exposed the fundamental timing flaw in the original main-loop update logic; its solution motivated both the Timer1 ISR architecture and the hybrid microstepping strategy.

During the first integrated pipe trial, Motor A, which drives the three wheels, vibrated severely when fitted with rubber wheels. (At this moment, the motor-updating task was placed in the main loop) The motor shook with negligible net displacement. Two observations ruled out simple torque overload. First, the same motor operated normally when bare, with no load. Second, the motor ran without issue when installed on the lead screw under several newtons of axial resistance. A rubber wheel weighing tens of grams caused loss of synchronization; a lead screw under load did not. The pattern pointed to a frequency-dependent mechanism rather than a strength problem.

To diagnose this issue, the first measurement was the main loop frequency. A counter inserted into loop() recorded approximately 330 iterations per second — a cycle time of roughly 3 ms. Since the original update() function issued at most one STEP pulse per iteration, the actual pulse interval must be multiples of the loop period. In other words, whether the speed parameter was set to 700 μ s or 3000 μ s, the motor received pulses at 3 ms (or occasionally 6 ms) intervals. The phenomenon is called false speed regulation.

The rubber wheel's unique vibration could be explained by the stepper motor's dynamics. A stepper motor behaves as a torsional spring–inertia system [33], [34]: the magnetic field between rotor and stator provides a position-dependent restoring torque, and the combined rotor–load moment of inertia determines the system's natural frequency, $\omega_n \sim 1/\sqrt{J}$. The bare

rotor has minimal inertia and a natural frequency well above 330 Hz — no resonance. The lead screw adds large inertia, pushing ω_n far below 330 Hz, while the metal sliding contact in the screw threads provides strong friction that damps any transient oscillation before it can accumulate energy over successive cycles. The rubber wheel encounters the worst case in the inertia-damping parameter space as it adds enough rotational inertia to pull ω_n into around 330 Hz, but mechanical damping is negligible. Each pulse arrives in phase with the growing oscillation, the amplitude escalates cycle by cycle, and the rotor eventually loses synchronization by falling back by one or more full steps and producing the observed stuttering motion.

A simple experiment confirmed the diagnosis. Gripping the motor housing by hand during operation immediately restored normal rotation. The hand introduced sufficient external damping to suppress the oscillation, proving that the root cause was low-damping resonance rather than a driver fault, wiring defect, or defective motor. A further confirmation came after the Timer1 ISR was deployed: when the ISR generated perfectly uniform pulses at intervals of 3000–3500 μs (corresponding to 285–333 Hz), the stutter also reappeared. This demonstrated that the trigger was pure frequency matching: even flawless pulse waveforms could not prevent resonance if the step rate itself fell within the system's resonant band.

Two complementary measures were applied, one in software and one in hardware. The software measure was the Timer1 ISR detailed in Section 3.6.2. By migrating pulse generation from the main loop to a 100 μs hardware timer interrupt, the speed parameter regained control over the pulse interval. Any speed outside the narrow 3000–3500 μs resonant band could be selected and would produce clean, vibration-free motion. The hardware measure was to configure Motor A's three A4988 drivers for 1/8 microstepping. In full-step mode, each pulse drives the rotor abruptly from one position to the next, during which the sudden acceleration is unfavorable. Microstepping divides each full step into eight smaller increments, with the A4988 modulating the coil currents sinusoidally to produce a smooth transition [35]. The impulsive energy delivered per increment is reduced, the speed increment resembles a continuous function.

After both measures were deployed, Motor A drove the robot smoothly at all speed settings outside the 3000–3500 μs band. The continuous-rotation mode became reliably usable, and speed control took effect. Three engineering conclusions can be drawn from this challenge. First, a microcontroller's main loop is not a clock: any signal requiring microsecond-level periodicity must be generated by a hardware timer peripheral, where a quartz crystal but not code path length governs the cadence. Second, a stepper motor's resonant behavior is governed not by load magnitude but by inertia and damping; a light, undamped load can be far more unstable than a heavy, damped one. Third, microstepping suppresses resonance by reducing the per-step excitation amplitude below the threshold that the mechanical system can accumulate into a growing oscillation [34], [35].

3.7 Results and Discussion

3.7.1 Design Review Against Specifications

Section 3.2 has defined eight design specifications for the front-end sensing module (Table 3.1). Table 3.3 reviews each against the delivered V2 system. Mechanical validation data supporting the assessments will be introduced in detail in Section D.

Table 3.3. Design specification compliance review

#	Specification	Requirement	Status	Evidence
1	Pipe diameter range	440–510 mm	Yes	Support mechanism covers full range; clamping completed within minutes
2	Concentricity	Module center = pipe centerline	Yes	Self-centering mechanism + spring suspension; front module stood independently at centerline
3	Transducer-to-wall distance	Constant, adjustable via radial DoF	Half	0–62 mm range, 0.01 mm/step, auto adjustment mechanism pending
4	Rotational coverage	360° continuous sweep	Yes	2240 steps/rev, smooth under scanning arm + probe load; cross-roller bearing stable at all angles
5	No. of DoFs (sensing)	≥ 2	Yes	2 DoFs delivered; third DoF intentionally eliminated
6	Transducer mounting	Interchangeable	Yes	Six-hole precision interface; contact, air-coupled, and angle-beam shear-wave probes can be mounted with carrier
7	Weight	< 6 kg (sensing module)	Half	Actual weight pending measurement
8	Structural material	Lightweight, rigid, manufacturable	Yes	Main body aluminum alloy sheet (bent/CNC) + PETG-CF 3D print + hardened steel shafts

3.7.2 Resolution of Mid-term Identified Problems

The V1 prototype exposed five systemic weaknesses. Each defined a design objective for V2. Table 3.4 summarizes the resolution.

Table 3.4. V1 problems and V2 solutions

V1 Problem	V2 Solution
Conflated load paths: motor shaft bore structural weight	Three steel guide shafts carry structural loads; lead screw handles axial torque only; clamp-type coupling replaces set-screw flange coupling
Backlash: CAM slot tolerance + flange coupling free play	Four-bar linkage with bearings replaces sliding CAM; internal ring gear engagement with slotted adjustment replaces flange coupling
Restrictive leg geometry: cantilever constrained by inter-plate spacing	Independent wheel carriers dual-pivot mounting; different types of wheels can be easily changed by adapter
Poor maintainability: full disassembly to access one leg	Modular design: each leg, linkage, and bearing independently accessible and removable
Excessive weight: >5 kg, dominated by thick acrylic plates	Four 8 mm acrylic plates → two 2.5 mm aluminum plates; structural strength was improved efficiently

3.7.3 Directions for Future Improvement

The V2 prototype demonstrated functional feasibility across all specifications. However, substantial engineering work separates it from a field-deployable tool.

The most immediate gap is hardware maturity. The current control circuit uses breadboard-mounted A4988 drivers and jumper wires, which are vulnerable to vibration and other interference in practice. A custom PCB integrating the microcontroller, drivers, and connectors would improve reliability and enable a sealed enclosure. At the controller level, the Arduino Mega was selected for accessibility rather than industrial robustness; a migration to an STM32-based platform with CAN bus communication between distributed driver nodes would provide higher performance and reliability. Environmental hardening is equally critical: the carbon steel fasteners have already begun to corrode under Hong Kong's humidity. Field operation inside pipelines carrying condensed water and corrosive agents would require stainless fasteners, sealed bearings, and coated electronics.

A second gap is test coverage. Validation was conducted in a single 1-metre straight pipe under clean, dry, room-temperature conditions. Real pipeline networks contain bends, diameter transitions, and sections with internal debris. The robot's passive concentricity under non-ideal wall geometry and the universal joint's behavior in curves remain untested.

Finally, the three-legged concentric architecture was selected over alternatives in the concept exploration phase, and V2 validates that choice for straight pipes. However, a wall-adherent platform, rejected primarily on cost grounds in late 2025, deserves re-examination with the benefit of two prototype cycles of experience. If its adhesion challenges were solved, it would offer independence from pipe centerlines and natural negotiation of vertical and curved sections. The concentric architecture remains correct for the project's scope, but extending capability to real pipeline networks may demand a different mobility paradigm.

3.8 Summary of Achievements

This section documented my major achievements and contributions including the complete design, fabrication, validation and integration of the front module of robot V1 and V2, covering three interdependent engineering layers.

1. Designed and manufactured V1 and V2 mechanical structure.
2. Built an embedded control platform.
3. Developed a complete C++ software architecture for robot control.
4. Diagnosed several engineering challenges and resolved them.
5. Implemented a Python serial data pipeline to deliver data to the downstream.
6. Validated the integrated automation system, design specifications substantially met.
7. Identified directions for advancing from a laboratory prototype to a field-deployable tool.

3.9 Appendix

Table B.1. Complete pin mapping for the hardware system

Arduino Mega Pin	Signal	Connected To	Notes
D2	STEP_X	A4988 (Axis X)	Motor 0: Radial extension arm
D5	DIR_X	A4988 (Axis X)	
D3	STEP_Y	A4988 (Axis Y)	Motor 1: Disk rotation
D6	DIR_Y	A4988 (Axis Y)	
D4	STEP_Z	A4988 (Axis Z)	Motor 2: self-centering screw
D7	DIR_Z	A4988 (Axis Z)	
D12	STEP_A	A4988 (Axis A)	Motor 3: Drive wheels (×3, parallel)
D13	DIR_A	A4988 (Axis A)	
D8	EN	CNC Shield (all axes)	LOW = enabled;
D22	DAT (MISO)	PS2 receiver	
D24	CMD (MOSI)	PS2 receiver	
D26	SEL (CS)	PS2 receiver	
D28	CLK (SCK)	PS2 receiver	
D20	SDA	SSD1306 OLED (128×64)	I2C address 0x3C
D21	SCL	SSD1306 OLED (128×64)	
D20	SDA	MPU6050 IMU ×2 (deprecated)	I2C addresses 0x68, 0x69
D21	SCL	MPU6050 IMU ×2 (deprecated)	
Relay			
D52	RELAY_CTL	Relay module (Active LOW)	LOW = relay closed
D0 (RX)	UART RX	Host PC (USB)	115200 baud
D1 (TX)	UART TX	Host PC (USB)	115200 baud
Power			
USB-B	5 V IN	5 V power bank	Powers Arduino Mega (MCU + I/O)
External	12 V	Li-Po 3S battery	→ Relay → A4988 VMOT (motor power)
External	12 V→5 V	DC-DC buck converter	→ A4988 VDD (driver logic supply)

Table B.2 PS2 controller button-to-function mapping

Button	Function	Detection	Relay Protected
▲	Current motor: continuous forward	pressed	Yes
×	Current motor: continuous reverse	pressed	Yes
R1	Current motor: jog forward (+N steps)	justPressed	Yes
R2	Current motor: jog reverse (−N steps)	justPressed	Yes
■	Start automation sequence	justPressed	Yes
↑	Execute single sequence step	justPressed	Yes
START	Toggle 12 V relay (motor power)	justPressed	No
○	Reset all motor positions to zero	justPressed	No
L1	Select previous motor	justPressed	No
L2	Select next motor	justPressed	No
SELECT	Print motor states to serial	justPressed	No

MECH 4900 Final Year Design Project Final Report 2025/2026

4 Individual Section C: Robot Driving Module, Signal Processing and Visualization

By XU You Qi

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4.1 Introduction and Motivation

4.1.1 Introduction

The objective of this final-year project was to design and develop a remotely operated pipeline inspection robot capable of traversing gas pipelines to detect structural anomalies, specifically cracks. The initial phase of the project involved a collaborative mechanical design and manufacturing effort. My primary responsibility encompassed the driving module of the robot, which included the propulsion system, electronics housing, and associated hardware. Following the midterm review, bottlenecks in the data analysis phase necessitated a shift in focus. Consequently, my duties transitioned from mechanical design iteration to the development and implementation of a robust data processing pipeline for sensor acquisition and defect classification.

4.1.2 Motivation for Robot Drive Module Structure

Conventional wheeled rovers present significant limitations in subterranean pipe inspection. A standard rover requires a complex, helical navigational path to ensure comprehensive surface scanning, which complicates precise spatial localization. Furthermore, maintaining wall adhesion against gravitational forces during inverted traversal poses a substantial mechanical challenge. After evaluating various architectural configurations, a concentric design topology was selected. Electromagnetic Acoustic Transducer (EMAT) and ultrasonic sensors exhibit high sensitivity to variations in contact angle and applied coupling force. A mechanism that pivots around the central longitudinal axis of the pipe ensures uniform sensor-to-surface engagement. Post-midterm, the integration of sensing hardware trailed the mechanical development of the robot, prompting my transition toward signal processing tasks to maintain project momentum.

4.1.3 Motivation for Signal Processing and Machine Learning

Both EMAT and ultrasonic sensors are precision instruments that require meticulous geometric positioning, and ultrasonic sensing further demands consistent coupling pressure. These sensor modalities are susceptible to signal degradation stemming from variations in stand-off distance, non-uniform couplant distribution, and inconsistent application force. To optimize defect detection accuracy and minimize both Type I (false positive) and Type II (false negative) errors, a comprehensive data processing pipeline was engineered. This pipeline systematically processes raw sensor data to extract discriminative features that isolate true defect responses from mechanical noise. These features are then used to train standardized machine learning models to enhance the overall reliability of the classification system [36].

4.1.4 Motivation for Data Visualization

To translate this conceptual prototype into a practical industrial tool, efforts were directed toward integrating the mechanical, sensory, and computational subsystems into a unified platform suitable for end-users, such as field inspectors and site engineers. To facilitate intuitive data interpretation, an interactive 3D point cloud visualization framework was developed. This framework computationally maps the processed sensory data to exact spatial coordinates, enabling inspectors to visually assess the geometric profile and precise location of identified structural flaws.

4.2 Personal Objectives, Design Ideas and Methodology

4.2.1 Driving Module Design Ideas

During the conceptual design phase, three primary architectural paradigms were evaluated. The first concept involved a surface-adhering rover capable of traversing the inner pipe wall. However, this required complex adhesion mechanisms (e.g., pneumatic suction or electromagnetic arrays), which severely restricted payload capacity and potentially necessitated an auxiliary tethered cart for additional hardware.

Following a feasibility review, this design was discarded. The second concept proposed a base cart traversing the pipe invert, utilizing a telescoping manipulator arm to sweep the internal circumference. The primary limitation of this design lay in the kinematics of the sensor arm. Given the sensitivity of EMAT and ultrasonic sensors to operational variables, maintaining constant contact parameters throughout a 360-degree sweep would require continuous, multi-axis joint actuation. The resultant mechanical complexity and compounding tolerance stack-up rendered this approach dynamically unfavourable.

Ultimately, the team selected a concentric architecture as the core design focus. To optimize system modularity, the sensing and driving modules were mechanically decoupled, allowing for independent design approaches. While the forward sensing module required strict concentricity, the aft driving module prioritized mechanical simplicity, traction, and maintainability. Rather than synthesizing a complex synchronous expansion mechanism, I engineered an independent, three-legged adjustable suspension system. This configuration provided sufficient relative concentricity while maintaining optimal radial pressure to ensure consistent traction for the drive wheels.

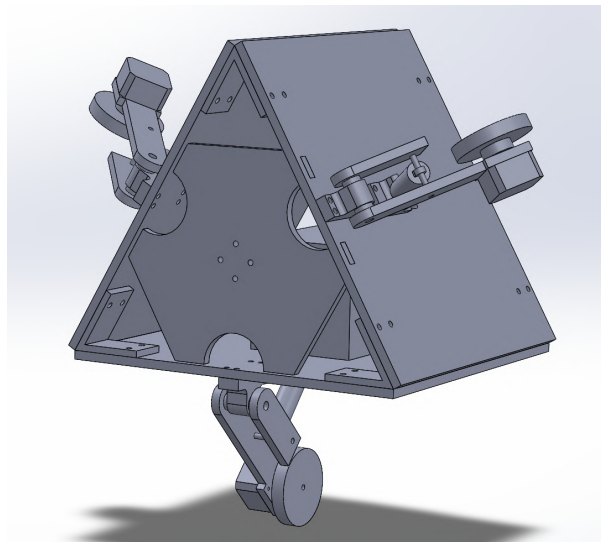


Fig. 4.1 Overall CAD assembly

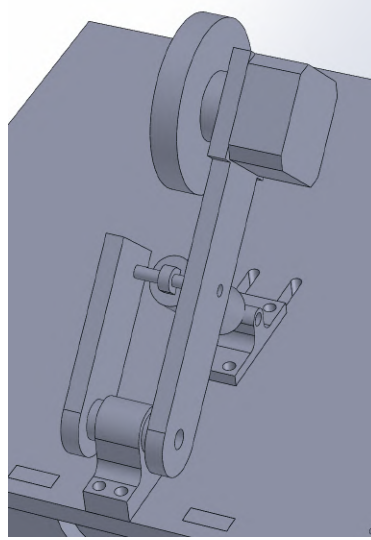


Fig. 4.2 CAD of suspension arm

By adjusting the mount position on the acrylic plate the default extension length of the leg can be tuned. By adjusting the tension of the suspension the spring force can be tuned.

4.2.2 Signal Processing & Feature Engineering Methodology

The signal processing methodology began with empirical observation. Visual inspection of consolidated acoustic waveforms differentiated standard structural profiles from crack-induced anomalies based on echo localization, amplitude attenuation, and wave morphology. Subsequently, analytical mathematical models were developed in Python to systematically quantify these heuristic observations. 25 features were successfully extracted, including peak reflection amplitudes, energy distributions, and spectral centroid shifts. Computational analysis was implemented using Python libraries such as NumPy, SciPy, and Matplotlib. To evaluate the statistical significance and discriminative power of each extracted feature, Cohen's d was employed as the primary metric for calculating the effect size between normal and defective samples [37].

Statistical Significance Metric: Cohen's d

To quantitatively evaluate the discriminative power of each extracted feature, Cohen's d was employed to measure the effect size — specifically, the standardized difference between two means (normal pipe surface readings versus crack anomaly readings). The metric is defined as:

$$d = \frac{\mu_{normal} - \mu_{defect}}{s_{pooled}} \quad (4.1)$$

where μ_{normal} and μ_{defect} represent the mean values of the feature for the respective normal and defective classes, and s_{pooled} is the pooled standard deviation, defined as:

$$s_{pooled} = \sqrt{\frac{(n_{normal} - 1)s_{normal}^2 + (n_{defect} - 1)s_{defect}^2}{n_{normal} + n_{defect} - 2}} \quad (4.2)$$

where:

- s_{normal} = standard deviation of the feature for the normal class
- s_{defect} = standard deviation of the feature for the defective class
- n_{normal} = number of normal samples
- n_{defect} = number of defective samples

By utilizing Cohen's d, the feature selection process was free of heuristic bias. Features that yielded a large effect size ($|d| > 0.8$) were mathematically proven to have high separability and were therefore prioritized for the subsequent machine learning pipeline.

4.2.3 Machine Learning Methodology

The machine learning pipeline was constructed using the Python scikit-learn library. Given the constraints of a limited sample size and the priority of producing a stable proof-of-concept, the classification methodology relied on established, foundational machine learning algorithms [36]. The optimization objective was to maximize the Macro F1-score — ensuring balanced accuracy across both nominal and defective classes — while mitigating the risk of model overfitting or instability inherent to small datasets. Robust computational approaches were integrated, notably Leave-One-Out Cross-Validation (LOOCV) and 10 Fold CV to maximize data utility during training, and exhaustive Grid Search algorithms for rigorous hyperparameter tuning. Principal Component Analysis (PCA) was also utilized to reduce feature dimensionality.

Performance Metric: Macro F1-Score

In pipeline inspection, the dataset is inherently imbalanced: normal structural readings vastly outnumber defect anomalies. Relying strictly on overall accuracy would yield misleading results, as a model could simply predict "normal" universally and still achieve high accuracy [38]. To counter this, the Macro F1-score was selected as the primary optimization metric. It calculates the unweighted mean of the F1-scores for each class, defined as:

$$Macro\ F1 = \frac{1}{C} \sum_{i=1}^C F1_i \quad (4.3)$$

where C is the number of classes and $F1_i$ is the F1-score for class i :

$$F1_i = 2 \cdot \frac{Precision_i \cdot Recall_i}{Precision_i + Recall_i} \quad (4.4)$$

where precision and recall for class i are defined as:

$$Precision_i = \frac{TP_i}{TP_i + FP_i} \quad (4.5)$$

$$Recall_i = \frac{TP_i}{TP_i + FN_i} \quad (4.6)$$

with:

- TP_i = True Positives for class i
- FP_i = False Positives for class i
- FN_i = False Negatives for class i

By treating all classes equally regardless of their support size, the Macro F1-score penalizes models that perform poorly on the minority class, ensuring that the critical objective of the project — identifying structural cracks — is strictly mathematically enforced.

Cross-Validation Strategy: LOOCV, 10 Fold CV

Analysis was performed on a subset of our collected data:

Table 4.1 Sensor types analysed

Sensor Type	Sample Size n
Ultrasound (straight beam transverse wave)	31
Ultrasound (angle beam transverse wave)	69

To conservatively utilize this data without introducing extreme bias, Leave-One-Out Cross-Validation (LOOCV) was implemented for straight beam and Repeated Stratified 10-Fold CV was implemented for angle beam. To prevent data leakage during grid search, this was nested, where the inner loop tuned hyperparameters and the outer loop evaluated generalization performance [39]. In LOOCV, the model is trained on $N-1$ samples and validated on the single remaining sample, iterating N times. While computationally expensive, LOOCV maximizes the amount of training data in each fold, and provides a nearly unbiased evaluation of how the algorithm will generalize to unseen field data [40]. However, because the training sets in each fold overlap almost completely, LOOCV suffers from high estimation variance and fully dependent predictions across iterations. It is therefore adopted here only for the very small straight-beam subset ($N < 40$)[40].

Hyperparameter Tuning: Grid Search

To optimize the predictive performance of the chosen machine learning algorithms (e.g., Support Vector Machines, Random Forests), Grid Search was utilized in tandem with cross validation. Rather than manually tuning algorithm parameters, an exhaustive search was programmed to iterate through a predefined hyperparameter space —varying kernel coefficients or regularization parameters. The pipeline autonomously evaluated discrete combinations and selected the configuration that yielded the maximum cross-validated Macro F1-score, ensuring an optimal and reproducible model architecture.

Dimensionality Reduction: Principal Component Analysis (PCA) and univariate t-test implementation

The signal processing pipeline yielded over 20 time-domain and frequency-domain features. Applying this many dimensions to a geographically constrained dataset drastically increases the risk of overfitting (the "curse of dimensionality"). To mitigate this, other than reducing features, Principal Component Analysis (PCA) and univariate t-test were incorporated into the pipeline prior to model training [41]. PCA applies an orthogonal linear transformation to project the high-dimensional feature space into a lower-dimensional set of uncorrelated variables (principal components) that capture the maximum variance of the data. This mathematically condensed the dataset, discarding redundant acoustic noise and collinear

features, which improved both the computational efficiency and the classification boundary stability of the selected models. T-test keeps the features but selects the top 2 to 5 most discriminative features to pass along to the classification models.

4.3 Project Schedule, Deliverables and Milestones

Table 4.2 — Personal Timeline and Milestones

Period	Phase	Key Activities	Deliverables
2025.10	Concept Exploration	Conducted a literature review on pipeline inspection robotics; evaluated and rejected Concept and Concept 2; selected Concept 3 (three-legged concentric architecture).	Concept sketches, literature review
2025.11	Driving Module Detailed Design	Developed full SolidWorks CAD assemblies of the driving module, featuring three independent suspension legs with adjustable tension and extension mechanisms.	CAD model
2025.12	Driving Module Prototyping & Testing	Manufactured, assembled, and tested the prototype inside a test pipe. Verified all functions and successful locomotion.	Prototype, mid-term progress report
2026.01	Data Pipeline Research	Investigated industry standards and academic literature concerning classification and regression methodologies; established metrics for evaluating feature significance.	N/A
2026.02	Feature Development	Conducted preliminary exploratory data analysis and synthesized custom feature extraction algorithms.	N/A
2026.03	Signal Processing Iteration 1 (Visual/Basic Features)	Deployed a data processing pipeline for raw telemetry ingestion. Calculated Cohen's d metrics and generated statistical distribution graphs comparing nominal and defective acoustic samples.	Python data processing pipeline
2026.03	Signal Processing Iteration 2 (Frequency Domain Features)	Integrated FFT for frequency domain analysis and engineered 10 advanced complex features to optimize Cohen's d effect sizes.	Pipeline upgrade
2026.04	Signal Processing Iteration 3 (Machine Learning)	Implemented a comprehensive machine learning pipeline utilizing scikit-learn, incorporating LOOCV, hyperparameter grid search, and PCA. Evaluated comparative performance across standardized classification models.	Pipeline upgrade
2026.04	Data Visualization	Developed a Python rendering script that translates polar coordinate telemetry and classified acoustic data into an interactive 3D point cloud interface for defect spatial mapping.	Pipeline upgrade

4.4 Results and Discussion

4.4.1 Robot Driving Module Results and Adjustments

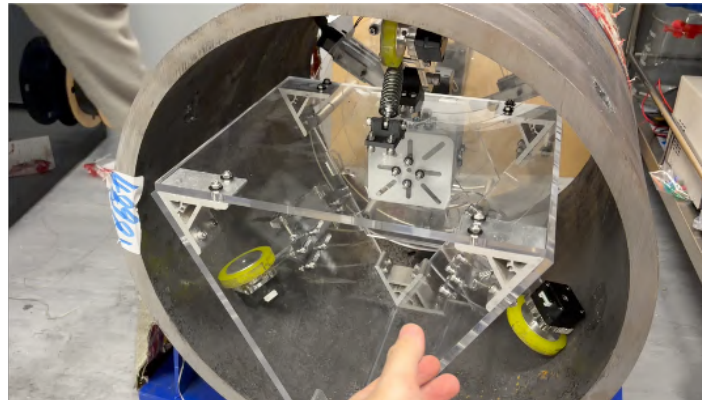


Fig. 4.3 Driving module inside test pipe

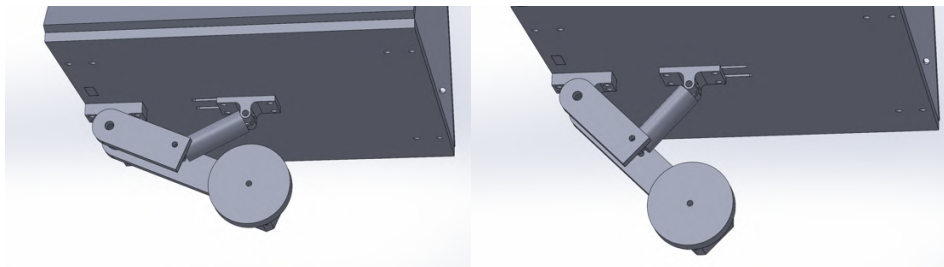


Fig. 4.4 Leg Adjustments

The final physical prototype successfully achieved locomotion using the three-axis suspension system. The goals of being modular, affordable, easy to repair and adaptive to a range of pipe sizes were achieved.

The legs can be adjusted to accommodate pipes with diameters between **420-500 mm**, and adjusting spring force between **10-35 kg** ensures relative concentricity to the pipe. Until today no parts have malfunctioned.

4.4.2 Signal Processing Iteration 1 (Visual/Basic Features)

The first signal processing script was developed to quantify the visual differences observed when manually comparing crack and normal ultrasonic readings. The diagnostic pipeline partitions the raw time-series data into two temporal envelopes: Window 1 (W1) and Window 2 (W2). Window 1 captures the first back-wall reflection. Window 2 captures delayed, secondary phenomena induced by interaction with a physical void or structural defect.

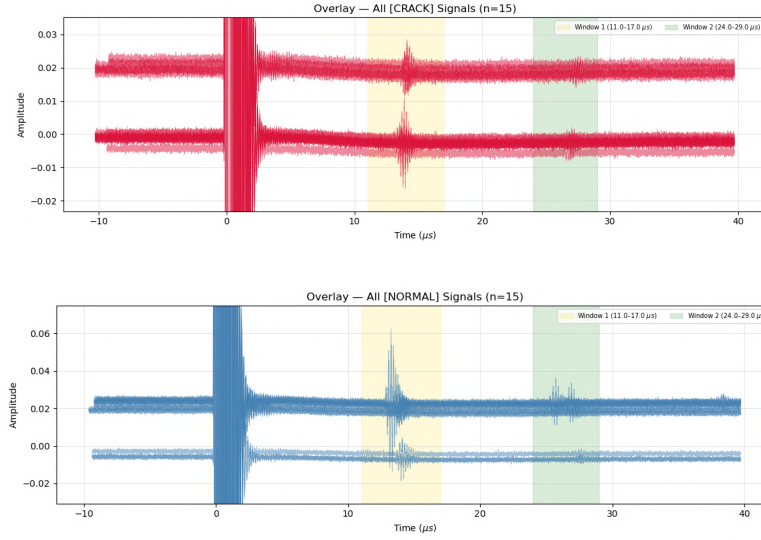


Fig. 4.5 Straight beam signals overlaid

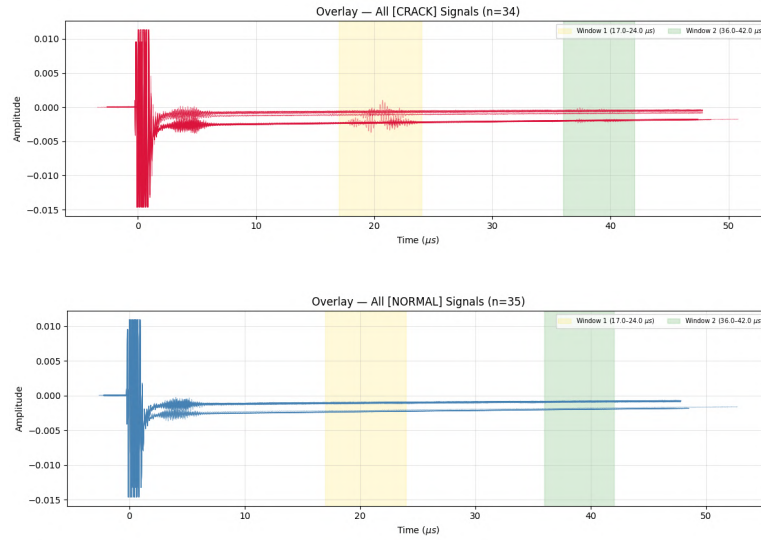


Fig. 4.6 Angle beam signals overlaid

Prior to computational analysis, both temporal windows undergo a rebasing protocol to subtract localized DC offsets, bringing the mean amplitude to zero.

To distinguish structurally sound specimens from those containing cracks, several time-domain features were calculated. The physical principles and mathematical justifications for these analytical features are detailed below.

4.4.2.1 Extracted Features: First Peak Magnitude, Second Peak Magnitude, Peak Ratio, and Peak-to-Peak

Ultrasonic wave propagation is governed by the material's acoustic impedance, defined as [1]:

$$Z = \rho \cdot c \quad (4.7)$$

where ρ is density and c is the speed of sound. When an acoustic pulse propagates through a pristine internal structure, its amplitude decays predictably due to natural material damping. However, a crack introduces a severe localised mismatch in acoustic impedance (typically a solid-to-air boundary). At this boundary, a significant portion of the wave is blocked, reflected, or scattered rather than transmitted.

The transmission coefficient T across a boundary between two media (1 and 2) is given by:

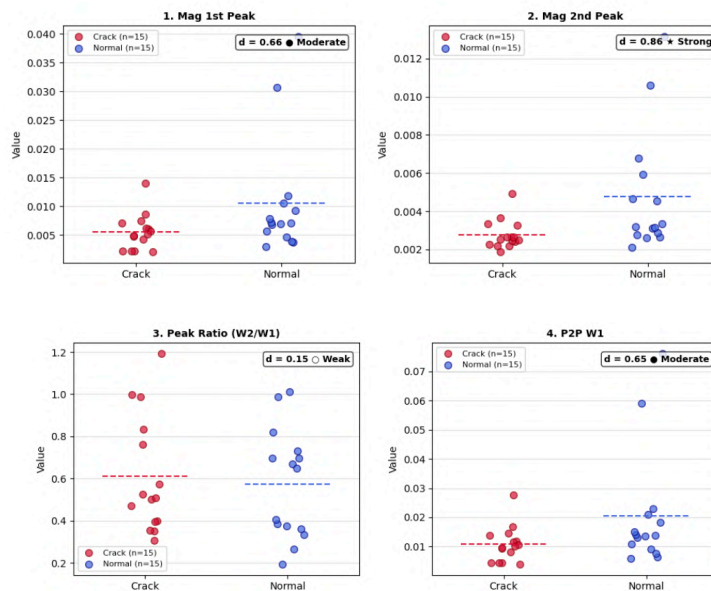
$$T = \frac{2Z_2}{Z_1 + Z_2} \quad (4.8)$$

Because the impedance of air (Z_{air}) in a crack is drastically lower than that of the solid metal (Z_{metal}), T approaches zero, causing a significant drop in transmitted wave amplitude. By measuring the maximum absolute amplitude in both windows (mag_first_peak and mag_second_peak), and computing the proportional drop:

$$\text{peak_ratio} = \frac{\text{mag_second_peak}}{\text{mag_first_peak}} \quad (4.9)$$

A highly attenuated secondary peak relative to the primary peak acts as a strong indicator of interfacial scattering caused by a crack.

The peak-to-peak amplitude ($\text{max} - \text{min}$) captures the full bipolar swing of the ultrasonic oscillation. Whereas the peak magnitude measures only the dominant half-cycle, peak-to-peak reflects the total dynamic range. The same impedance-mismatch mechanism that attenuates the peak magnitude also compresses the peak-to-peak swing. Both $W1$ and $W2$ peak-to-peak values are extracted and $W2$ peak-to-peak is expected to decrease relative to $W1$ in the presence of a crack.



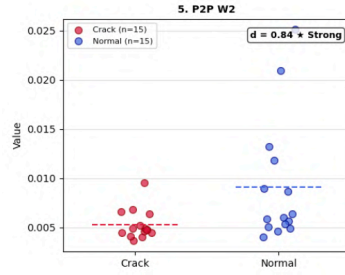


Fig. 4.7 Peak magnitude and peak-to-peak features (straight beam)

For straight beam ultrasound, the second peak magnitude shows a much more significant Cohen's d effect, likely because of compound scattering and attenuation. By the second signal, the wave has interacted with the crack region multiple times, resulting in greater energy loss.

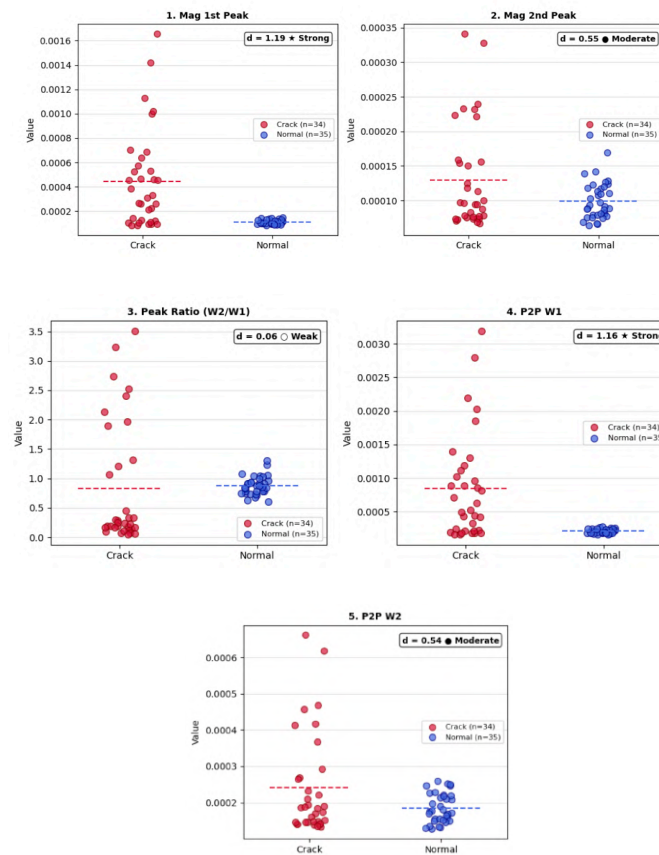


Fig. 4.8 Peak magnitude and peak-to-peak features (angle beam)

For angle beam, the sensor is angled such that normal surfaces deflect the signal away without reflecting back to the receiver. Consequently, the detection of a substantial first peak itself strongly suggests the presence of a defect or crack in the signal path, explaining the strong Cohen's d results for W1 peak magnitude and W1 peak-to-peak. The second window is overall less effective for transverse ultrasound, since the first window receives the clearest and most direct signal.

4.4.2.2 Extracted Features: Energy Ratio and Area Ratio

While peak magnitudes capture instantaneous acoustic reflections, anomalies in internal microstructures (such as multi-branching cracks and micro-voids) cause localized wave scattering that suppresses the total energy of the wave packet. In mechanical waves, acoustic intensity I is proportional to the square of the pressure wave amplitude A :

$$I \propto A^2 \quad (4.10)$$

The pipeline computes a surrogate for relative enclosed energy by taking the sum of the squared discrete amplitudes within each target window:

$$E_{W1} = \sum_{i \in W1} A_i^2, E_{W2} = \sum_{i \in W2} A_i^2 \quad (4.11)$$

The **Energy Ratio** R_E normalises the secondary wave energy against the incident wave energy:

$$R_E = \frac{E_{W2}}{E_{W1}} \quad (4.12)$$

The **Area Ratio** R_A follows the same formulation but uses the sum of absolute amplitudes rather than squared amplitudes, providing a complementary measure less sensitive to outlier spikes:

$$R_A = \frac{\sum_{i \in W2} |A_i|}{\sum_{i \in W1} |A_i|} \quad (4.13)$$

Both metrics are robust because they account for the spatial scattering of the entire secondary wave packet rather than relying solely on a single multipath peak. A substantial drop in either ratio indicates massive internal scattering characteristic of defective specimens.

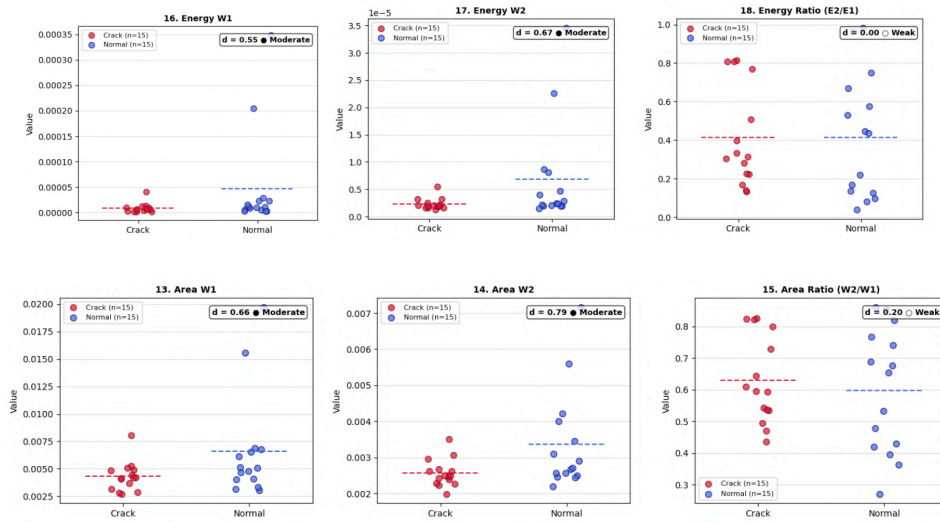


Fig. 4.9 Energy and Area Features (straight beam)

For straight beam testing, secondary window (W2) signals are more discriminative than the first. The Area metric performs better than Energy, suggesting that squaring the signal amplifies outlier noise. Both ratio metrics fail to provide explanatory power, largely because

the crack scatters the wave rather than absorbing it, meaning the total energy trapped within the measurement window is mostly conserved.

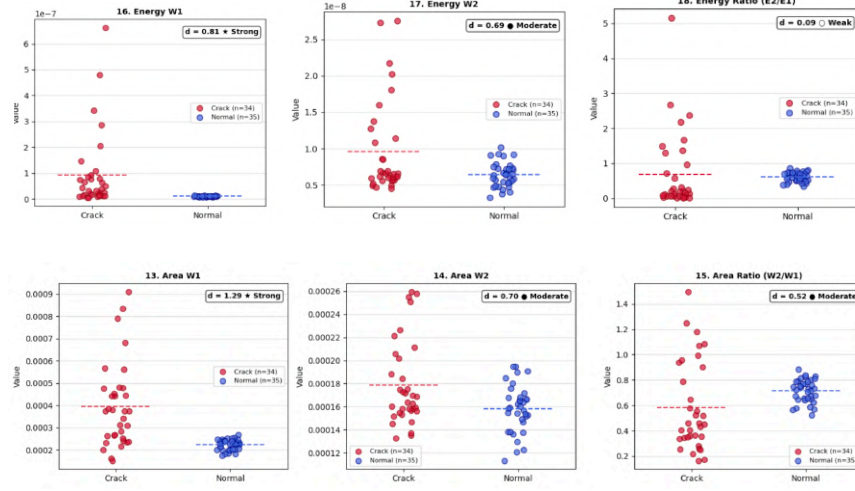


Fig. 4.10 energy and area features (angle beam)

Angle beam results align with theoretical predictions: normal surfaces deflect the signal away, meaning a strong returned wave directly indicates a crack or defect. This direct reflection explains the immense discriminative power observed in the first window (W1). Furthermore, just as seen in straight beam testing, the Area metric significantly outperforms Energy (yielding an exceptional Cohen's d of 1.29 vs 0.81 in W1). This confirms that squaring the signal amplifies random background noise, whereas Area safely ignores it.

4.4.2.3 Extracted Feature: Time Difference (Δt)

In pristine materials, secondary wave reflections typically travel back to the transducer across a known line-of-sight geometric path. In the presence of a macroscopic crack, the direct path is obstructed. According to Huygens' Principle, the propagating wave diffracts (bends) around the crack tips [1]. Because the signal is forced to travel around the extremities of the crack, the physical path length L is elongated. Assuming the material's transverse wave velocity v_L remains constant, this elongated path translates directly into a temporal delay:

$$\Delta t = \frac{\Delta L}{v_L} \quad (4.14)$$

where the transverse wave velocity is given by [1]:

$$c_s = \sqrt{\frac{G}{\rho}} \quad (4.15)$$

with G being the Shear Modulus and ρ the material density. The pipeline isolates the time coordinates of the maximum peak in Window 1 and Window 2 to extract the physical time-of-flight difference (Δt). Larger values correlate stringently to diffracted, non-uniform wave propagation around structural voids.

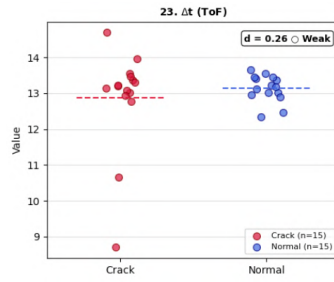


Fig. 4.11 Time difference (straight beam)

Straight beam results are weak. This is likely because the cracks have small cross-sectional area relative to the total width of the ultrasonic beam. Therefore, a majority of acoustic energy bypasses the defect completely.

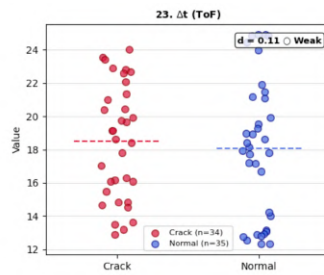


Fig. 4.12 Time difference (angle beam)

Angle beam results are also noisy with almost no discriminative power. This is likely because normal samples are simply picking out a random peak from background noise, while the cracked sample times are highly distributed. For cracked specimens, the return time depends entirely on the depth and location of the flaw along the angled wave path. Since this physical distance can vary greatly, even with slight probe movements when measuring the same crack, the signals return at widely varying times, destroying the consistency of the feature.

4.4.2.4 Iteration 1 Outcome

Overall, Cohen's d performance for these basic time-domain features was acceptable as a proof of concept, but there was clear room for improvement. Several features were heavily correlated with one another, introducing redundancy without yielding additional discriminative information. Others proved too susceptible to baseline ultrasonic noise and variance in defect geometry, limiting their reliability for practical classification. Further refinement through more sophisticated feature engineering was clearly needed.

A comparison of Cohen's d values across the sensor configurations revealed that the angle beam was substantially more potent than the straight beam for defect detection. Consequently, the focus of Iterations 2 and 3 shifted toward deepening the understanding and exploitation of angle beam signals.

4.4.3 Signal Processing Iteration 2 (Frequency Domain & Complex Features)

The second iteration focused on improving Cohen's d by investigating more complex features. Frequency-domain analysis was added, and 25 features were developed in total. Additional

features were explored but ultimately removed to avoid black-box results and overfitting on noise.

The complex features developed in this iteration fall into two categories: time-domain statistical features (Crest Factor, Zero Crossing Rate, Envelope Width, and RMS Ratio) and frequency-domain spectral features (Dominant Frequency Drop, Spectral Centroid Shift, and High-Frequency Ratios). All frequency-domain features share a common preprocessing pipeline, which is described first.

4.4.3.1 FFT and Hanning Window Preprocessing

To move beyond bulk energy losses, the pipeline transfers the analysis into the frequency domain. Because acoustic attenuation is tightly coupled to frequency, establishing the spectral profile of both windows is critical for detecting defect-induced mechanical filtering.

Before converting the signal, it is necessary to address the fact that the raw time windows are truncated abruptly. Applying a raw Fast Fourier Transform to an abruptly truncated signal causes spectral leakage — where energy bleeds into adjacent frequency bins due to artificial discontinuities at the edges of the window.

To force the signal to taper smoothly to zero at its boundaries, the pipeline multiplies the discrete acoustic time-series data $x[n]$ by a Hanning window $w[n]$, defined as:

$$w[n] = 0.5 \left[1 - \cos \left(\frac{2\pi n}{N-1} \right) \right] \quad (4.16)$$

where N is the total number of samples in the un-padded window.

Once smoothed, the pipeline applies a zero-padded, discrete Fast Fourier Transform (FFT) to map the temporal wave to the frequency domain:

$$X[k] = \sum_{n=0}^{N_{\text{FFT}}-1} x[n] \cdot w[n] \cdot e^{-j2\pi \frac{kn}{N_{\text{FFT}}}} \quad (4.17)$$

where:

- $X[k]$ = complex frequency-domain representation at frequency bin k
- $x[n]$ = discrete time-domain signal
- $w[n]$ = discrete Hanning window function (padded with zeros for $n \geq N$)
- N_{FFT} = total FFT size (includes zero-padding)

Zero-padding the sequence artificially interpolates the frequency bins, providing a higher-resolution frequency spectrum to precisely track shifts in the acoustic bandwidth.

4.4.3.2 Extracted Feature: Crest Factor

The crest factor is the ratio of the maximum absolute peak amplitude to the Root Mean Square (RMS) amplitude:

$$CF = \frac{\max(|x[n]|)}{x_{\text{rms}}} \quad (4.18)$$

where the discrete RMS amplitude is defined over the N samples of the window as:

$$x_{\text{rms}} = \sqrt{\frac{1}{N} \sum_{n=0}^{N-1} (x[n])^2} \quad (4.19)$$

A crack echo is impulsive—a sharp spike emerging from a low-energy scattered floor—producing a high peak but low RMS, and therefore an elevated crest factor. A clean structural reflection is more sustained, yielding a lower crest factor. In essence, the crest factor quantifies the "spikiness" of the echo. Window 1 crest factor serves as the baseline (determined by the transducer pulse shape), while Window 2 crest factor captures how the defect interaction reshapes this balance.

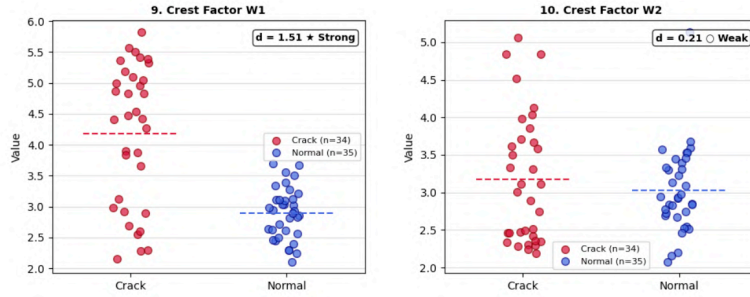


Fig. 4.13 Crest factor (angle beam)

The Crest Factor results for the angle beam configuration strongly validate the physical theory of defect 'spikiness' ($d = 1.51$ for W1). As established, an angle beam in defective material produces a massive, direct specular reflection off the crack face. This direct bounce returns as a sharp, impulsive spike towering over a relatively low-energy background floor, causing a high Crest Factor. Conversely, the normal specimens return only baseline background noise with no dominant peak, resulting in a much lower and tightly clustered Crest Factor. Signal to Noise ratio worsens in the second window.

4.4.3.3 Extracted Feature: Zero Crossing Rate

The zero-crossing rate (ZCR) counts the number of sign changes per unit time, serving as a time-domain proxy for instantaneous frequency: a higher ZCR indicates faster oscillation. When the incident transverse wave interacts with a crack, high frequency components are scattered and dispersed. This mechanical low pass filtering should reduce the overall frequency of the return wave.

$$ZCR = \frac{N_{\text{zero-crossings}}}{T_{\text{window}}} \quad (4.20)$$

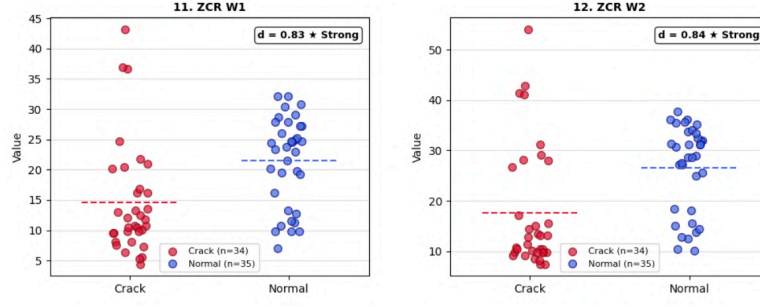


Fig. 4.14 Zero crossing rate (angle beam)

Results from the angle beam demonstrate ZCR has strong discriminative power in both window 1 and window 2. This supports the theory that the crack acts as a low pass filter that scatters high frequency waves.

4.4.3.4 Extracted Feature: Envelope Width (6 dB)

$$\text{Width}_{-6\text{dB}} = \left(\max_i \{ i: \text{Env}_i \geq \frac{\text{Env}_{\max}}{2} \} - \min_i \{ i: \text{Env}_i \geq \frac{\text{Env}_{\max}}{2} \} \right) \cdot \Delta t \quad (4.21)$$

The envelope width measures the temporal duration where the signal envelope (extracted through the Hilbert transform) exceeds half its maximum amplitude. Because the normal signal for an angle beam consists of uniform background noise, the envelope width naturally spans the entire duration of the analyzed window. Conversely, when a highly concentrated amplitude spike is reflected by a crack, the -6 dB threshold is raised significantly, causing the measured width to drop to just the duration of that brief spike.

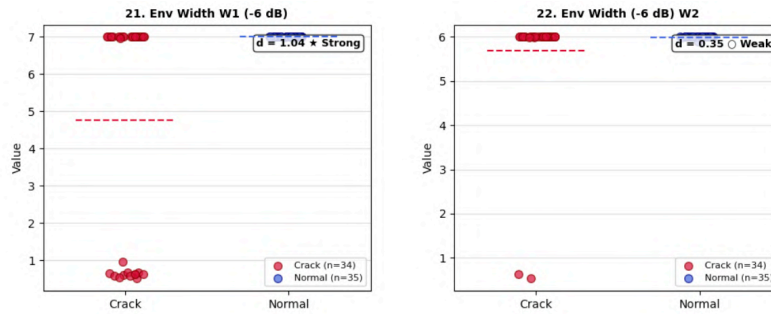


Fig. 4.15 Envelope width (angle beam)

The Envelope Width results match the physical theory, demonstrating strong discriminative capability in the primary window ($d = 1.04$). Pristine samples register the maximum window width as expected, while cracked samples form a distinct cluster of narrow widths around ~ 0.5 units due to their sharp reflections. However, this discriminative power is lost in Window 2 ($d = 0.35$); as the crack reflection scatters over a longer propagation path, the signal degrades back into uniform noise.

4.4.3.5 Extracted Feature: RMS Ratio

Ultrasonic wave transmission is governed by acoustic impedance:

$$Z = \rho \cdot c \quad (4.22)$$

An internal crack constitutes a severe impedance mismatch (metal-to-air), drastically reducing the transmission coefficient T and forcing wave reflection or scattering. To quantify how the defect redistributes this acoustic energy across the temporal windows, the pipeline computes the Root Mean Square (RMS) amplitudes:

$$A_{\text{RMS}, W1} = \sqrt{\frac{1}{N_1} \sum_{i \in W1} A_i^2} \quad (4.23)$$

$$A_{\text{RMS}, W2} = \sqrt{\frac{1}{N_2} \sum_{i \in W2} A_i^2} \quad (4.24)$$

$$\text{RMS Ratio} = \frac{A_{\text{RMS}, W2}}{A_{\text{RMS}, W1}} \quad (4.25)$$

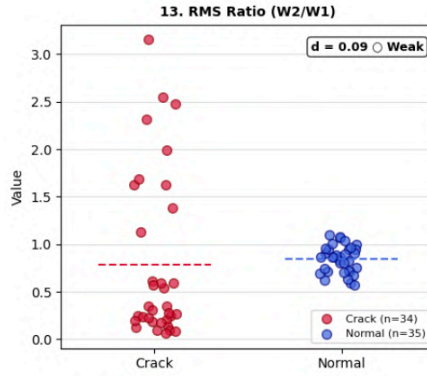


Fig. 4.16 RMS ratio (angle beam)

For normal readings, no clear signal is detected at window 1 and 2, hence the results are relatively clustered around 1. For crack readings, we can see most are focused near 0, which suggests the RMS in window 1 measures a large signal. However, there are also outliers where window 2 RMS is much higher, suggesting variables such as angle and location of the beam can lead to late responses. Ultimately, because this geometric variance scatters the cracked data so widely, its average overlaps with the normal baseline, eliminating its discriminative power ($d = 0.09$).

4.4.3.6 Extracted Feature: Dominant Frequency Drop

A severe crack acts as a spatial, mechanical low-pass filter. According to the physics of Rayleigh scattering, for obstacles smaller than or roughly equal to the acoustic wavelength λ , the scattering cross-section σ scales with the fourth power of frequency [1]:

$$\sigma \propto f^4 \quad (4.26)$$

High-frequency components of the incident wave possess shorter wavelengths, meaning they interact aggressively with the micro-topography of the crack and are scattered destructively. Lower frequencies, possessing longer wavelengths, diffract more easily past the crack tip. The pipeline isolates the peak frequency bin in Window 1 and Window 2 to calculate the Dominant Frequency Drop:

$$\Delta f_{\text{dominant}} = f_{\text{peak}}(W1) - f_{\text{peak}}(W2) \quad (4.27)$$

where:

- $f_{\text{peak}}(W1)$ = frequency corresponding to maximum magnitude in Window 1 spectrum
- $f_{\text{peak}}(W2)$ = frequency corresponding to maximum magnitude in Window 2 spectrum

A large positive $\Delta f_{\text{dominant}}$ firmly indicates Rayleigh scattering caused by an internal structural boundary.

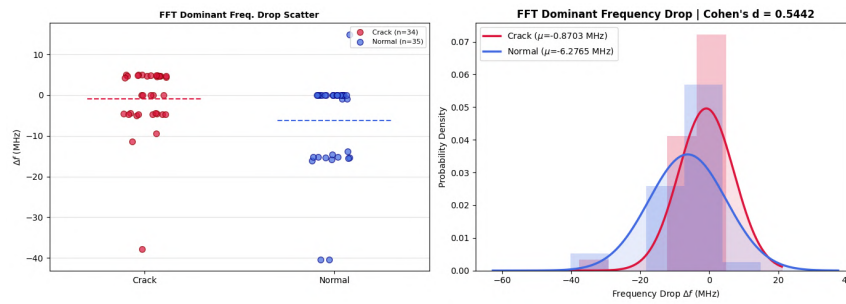


Fig. 4.17 Dominant frequency drop (angle beam)

For angle beam setups, this progressive effect is not obvious. When the initial wave strikes the crack, Rayleigh scattering strips away the high frequencies. Because both windows are detecting the same already-filtered wave response, there is no significant frequency drop between them. Ultimately, because this calculation compares a scattered wave to delayed copies of itself—rather than to the pristine initial wave—the effect of Rayleigh scattering is not visually pronounced in this specific metric.

4.4.3.7 Extracted Features: Spectral Centroid Shift, W1 HF Ratio, and W2 HF Ratio

Beyond simply dropping the peak frequency, a crack redistributes the entire mass of the acoustic spectrum. The pipeline calculates the Spectral Centroid f_{centroid} , the center of mass of the frequency spectrum, using the normalized amplitude array A_{norm} and its corresponding frequency vector f :

$$f_{\text{centroid}} = \frac{\sum f \cdot A_{\text{norm}}}{\sum A_{\text{norm}}} \quad (4.28)$$

The centroid shift is defined as:

$$\text{Centroid Shift} = f_{\text{centroid}}(W1) - f_{\text{centroid}}(W2) \quad (4.29)$$

When the incident transverse wave impacts the jagged geometry of a defect, massive mode conversion occurs, generating a complex mix of secondary waves (including longitudinal and

Rayleigh surface waves) [1]. This chaotic geometric scattering preferentially destroys short-wavelength energy. physical conversion forces the spectral centroid to migrate lower (producing a positive centroid shift) and severely depletes the ratio of energy residing in the upper frequency bands. The high-frequency ratios for each window are therefore defined as the ratio of spectral magnitude above the median frequency, the frequency below which 50% of the cumulative spectral magnitude resides, to the total spectral magnitude within the window:

$$\text{W1 HF Ratio} = \frac{\sum_{f > f_{\text{median}}} |X_{W1}(f)|}{\sum |X_{W1}(f)|} \quad (4.30)$$

$$\text{W2 HF Ratio} = \frac{\sum_{f > f_{\text{median}}} |X_{W2}(f)|}{\sum |X_{W2}(f)|} \quad (4.31)$$

All three metrics serve as indicators of mode conversion and high-frequency scattering by defects.

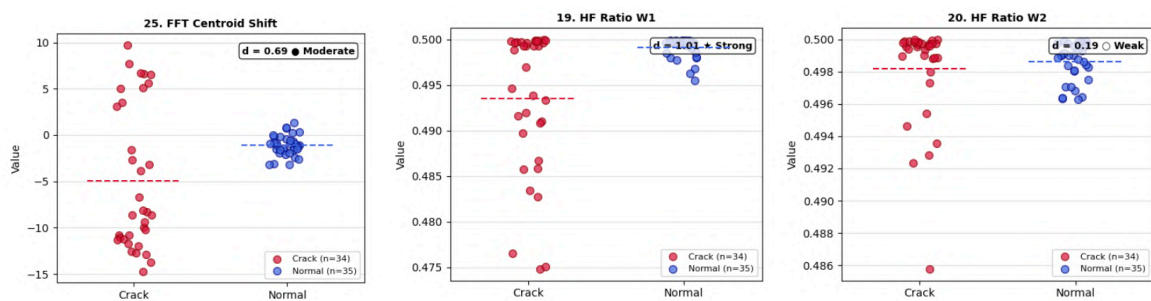


Fig. 4.18 Spectral centroid shift and HF ratios (angle beam)

The Spectral Centroid Shift yields moderate discriminative power ($d = 0.69$). This is expected, as it suffers from the same physical limitations as the Dominant Frequency Drop; comparing the spectral center of a direct W1 reflection against chaotic W2 geometric noise introduces high variance.

Conversely, the High-Frequency (HF) Ratio in Window 1 performs well ($d = 1.01$). This result aligns with physical theory: as the angle beam reflects off the crack, Rayleigh scattering preferentially destroys the short-wavelength, high-frequency components.

4.4.3.8 Iteration 2 Outcome

The integration of frequency-domain and complex features yielded significant improvement over Iteration 1, with Cohen's d reaching up to 1.5 for the strongest features using transverse ultrasound. However, strict threshold-based classification still resulted in unacceptably high error rates, motivating the transition to a machine-learning-based approach in Iteration 3.

This iteration also confirmed the clear physical differences between compression and transverse wave signals. Direct compression wave signals are affected by cracks primarily through diffraction of clean signals and gradual energy decay. In contrast, transverse wave signals exhibit fundamentally different behavior: the angled sensor receives meaningful reflections only when a crack or defect is present, making the transverse modality inherently

more discriminative. This reinforced the decision to prioritize transverse ultrasound in subsequent analysis.

4.4.4 Signal Processing Iteration 3 (Machine Learning)

Strict-threshold classification based on individual features yielded unacceptably high error rates when processing acoustic scattering data. Consequently, the pipeline transitioned to a machine learning (ML) framework built on scikit-learn. The final pipeline accommodates the severely constrained sample pool using robust modelling architectures designed to eliminate data leakage and overfitting. All results in this section are reported for the transverse ultrasound sensor, which demonstrated the strongest discriminative performance. Results for the straight beam sensor are provided in the appendix.

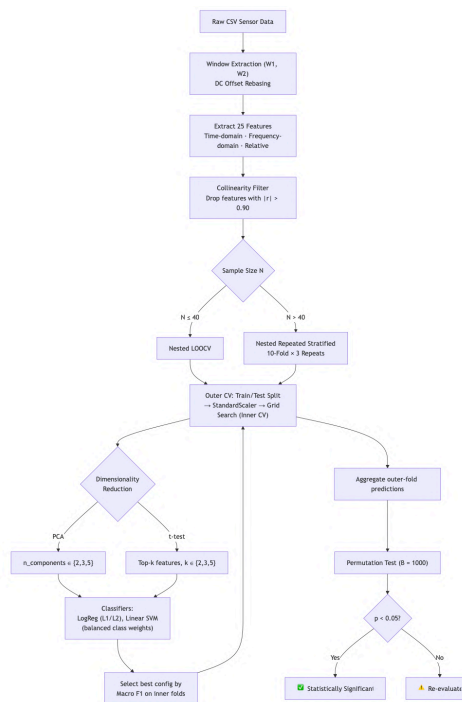


Fig. 4.19 Pipeline flowchart

4.4.4.1 Data Preprocessing and Feature Filtration

All missing values were imputed using a mean-based SimpleImputer, and all features were standardized using StandardScaler to ensure zero mean and unit variance. To counteract the curse of dimensionality, several filtration steps were applied to the 25 extracted features before model training.

First, an absolute Pearson correlation filter ($|r| > 0.90$) was applied to strip collinear and mathematically redundant variables. This eliminated features that carried no additional information beyond what was already captured by their correlated counterparts.

4.4.4.2 Cross-Validation Strategy

Given the limited dataset (69 transverse ultrasound samples), the cross-validation strategy was chosen to maximize data utility while maintaining statistical rigor. The infrastructure automatically toggles its complexity based on sample pool size N :

- $N < 40$: Strict Nested Leave-One-Out Cross-Validation (LOOCV), where the model trains on $N-1$ samples and validates on the single remaining sample, iterating N times.
- $N \geq 40$: Nested Repeated Stratified K-Folds (10 folds, repeated 3 times), which isolates the testing fold from the training and validation folds across multiple random partitions.

The nested structure is as follows: an inner cross-validation loop tunes hyperparameters, while an outer loop evaluates generalization performance. This separation prevents the hyperparameter search from peeking at the test data — a common source of overoptimistic results in small-sample studies.

4.4.4.3 Classification Models and Hyperparameter Tuning

Three classification models were evaluated:

1. Logistic Regression (L2 penalty) — a linear model with ridge regularization.
2. Logistic Regression (L1 penalty) — a linear model with lasso regularization, capable of implicit feature selection.
3. Linear Support Vector Machine (SVM) — a maximum-margin classifier.

All models were configured with "balanced" class weights to directly penalize false negatives for the minority crack class, ensuring that the models did not simply learn to predict "normal" for every sample.

Hyperparameter tuning was performed using exhaustive Grid Search within the inner cross-validation loop. The search spaces were:

- Logistic Regression: $C \in \{0.01, 0.1, 1.0, 10.0\}$
- Linear SVM: $C \in \{0.1, 0.5, 1.0, 5.0\}$

The optimization objective for all grid searches was the Macro F1-score, ensuring balanced performance across both classes regardless of their relative frequencies.

4.4.4.4 Dimensionality Reduction Comparison

Rather than assuming a single dimensionality reduction technique, the pipeline compared two distinct methodologies within the nested cross-validation folds:

PCA Pipeline: Principal Component Analysis was applied to project the high-dimensional feature space into 2, 3, or 5 uncorrelated principal components capturing the maximum data variance. The optimal number of components was selected by grid search within each inner fold.

Univariate t-test Pipeline: An independent-samples t-test (Welch's t-test) was performed on each feature to assess its statistical separation between crack and normal classes. The top 2, 3, or 5 features ranked by p-value were retained. The optimal number was again selected by grid search.

Both pipelines fed their reduced feature sets into each of the three classifiers, and the grid search selected the combination of dimensionality, classifier, and hyperparameters that maximized the cross-validated Macro F1-score.

4.4.4.5 Model Evaluation Results

Strategy	Model	Best Params	Recall (Crack)	Spec (Normal)	Macro F1	PR-AUC	ROC-AUC	TP	FN	TN	FP
PCA	SVM (Linear)	C=1.0, n_components=5	94.117647	97.142857	95.648518	97.291329	96.058420	32	2	34	1
PCA	LogisticReg (L1)	C=10.0, n_components=3	91.176471	94.285714	92.747538	96.685409	94.453782	31	3	33	2
PCA	LogisticReg (L2)	C=1.0, n_components=3	88.235294	97.142857	92.729189	97.726072	96.302521	30	4	34	1
t-test	LogisticReg (L2)	C=10.0, n_select=5	76.470588	97.142857	86.778795	91.607147	86.974790	26	8	34	1
t-test	LogisticReg (L1)	C=10.0, n_select=5	70.588235	97.142857	83.715941	91.151098	86.554622	24	10	34	1
t-test	SVM (Linear)	C=5.0, n_select=5	64.705882	100.000000	81.968641	90.933069	85.714286	22	12	35	0
single-feature	Single-Feature [area_first_window]	-	61.764706	100.000000	80.350493	89.982945	85.842017	21	13	35	0

Fig. 4.20 Overall performance comparison of PCA, t-test, and single-feature approaches (angle beam)

The linear SVM with PCA dimensionality reduction achieved the highest Macro F1-score of **95.6%**, outperforming all t-test configurations and single-feature baselines. Overall, PCA strategies consistently outperformed t-test strategies, suggesting that the principal components successfully captured variance from multiple weakly discriminative features that would have been discarded by univariate thresholding, and that the orthogonalization of input features improved the stability of the linear decision boundary for SVM.

For reference, the best single-feature classifier (logistic regression using the top individual feature alone) achieved a markedly lower Macro F1-score of 80%, confirming that the multi-feature ML approach provided a substantial improvement over the threshold-based methods of Iterations 1 and 2.

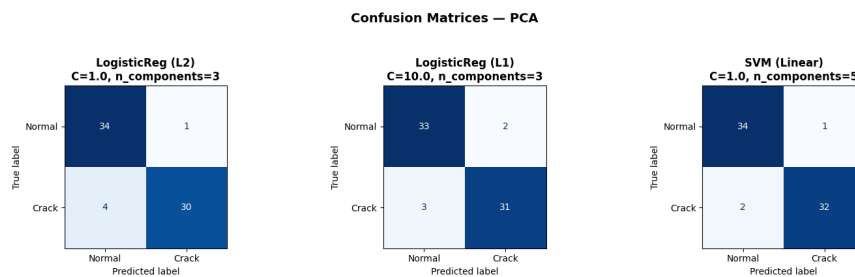


Fig. 4.21 Confusion matrices of PCA approach models

The winning SVM model classified 2 false negatives and 1 false positive.

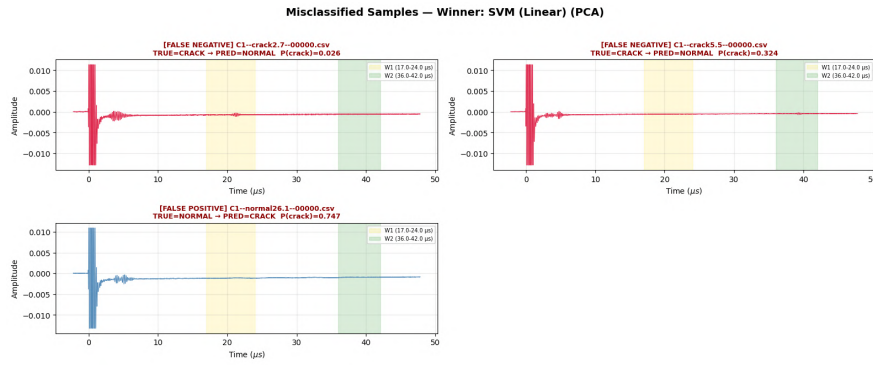


Fig. 4.22 Misclassified samples of winning SVM model

4.4.4.6 Statistical Significance: Permutation Test

To verify that the 95.6% Macro F1-score was not attributable to random chance given the small sample size, a permutation test was conducted with 1,000 iterations. In each iteration, the ground-truth labels were randomly shuffled, and the pipeline was re-executed using the winning model and hyperparameter configuration (SVM + PCA). The proportion of permuted runs achieving a Macro F1-score equal to or exceeding the observed score yielded a p-value of 0.001, well below 0.05, confirming that the model's discriminative performance is statistically significant and not an artefact of the limited dataset.

4.4.4.7 Performance on Trial Operation

We performed a fully automated trial operation test where the robot automatically collected samples in an area grid encompassing a crack under realistic inspection conditions. All normal and crack measurements were classified correctly.

Saved 25 predictions to production_predictions.csv

	File	Prediction	Confidence
0	C1--crack17--00000.csv	CRACK	0.544669
1	C1--crack18--00000.csv	CRACK	0.704336
2	C1--crack19--00000.csv	CRACK	0.968927
3	C1--normal1--00000.csv	NORMAL	0.857553
4	C1--normal10--00000.csv	NORMAL	0.841540
5	C1--normal11--00000.csv	NORMAL	0.877812
6	C1--normal12--00000.csv	NORMAL	0.850839
7	C1--normal13--00000.csv	NORMAL	0.828796
8	C1--normal14--00000.csv	NORMAL	0.843557
9	C1--normal15--00000.csv	NORMAL	0.830880
10	C1--normal16--00000.csv	NORMAL	0.892743
11	C1--normal2--00000.csv	NORMAL	0.891496
12	C1--normal20--00000.csv	NORMAL	0.814695
13	C1--normal21--00000.csv	NORMAL	0.876279
14	C1--normal22--00000.csv	NORMAL	0.767154
15	C1--normal23--00000.csv	NORMAL	0.911966
16	C1--normal24--00000.csv	NORMAL	0.924433
17	C1--normal25--00000.csv	NORMAL	0.906459
18	C1--normal3--00000.csv	NORMAL	0.773910
19	C1--normal4--00000.csv	NORMAL	0.875622
20	C1--normal5--00000.csv	NORMAL	0.841469
21	C1--normal6--00000.csv	NORMAL	0.898013
22	C1--normal7--00000.csv	NORMAL	0.884010
23	C1--normal8--00000.csv	NORMAL	0.873508
24	C1--normal9--00000.csv	NORMAL	0.884754

Fig. 4.23 Trial operation classification results

4.4.4.8 Iteration 3 Outcome

The transition to machine learning yielded a decisive improvement over the threshold-based approaches of Iterations 1 and 2. The combination of nested cross-validation, collinearity

filtering, PCA dimensionality reduction, and linear SVM classification achieved a Macro F1-score of 95.6% with statistical significance confirmed by permutation testing ($p < 0.05$). The winning pipeline — SVM with PCA ($n_components = 5$, $C = 1$) — demonstrated strong generalization to unseen trial data. This performance establishes the pipeline as a functional proof-of-concept for automated crack detection using angle beam ultrasonic sensing.

4.4.5 Data Visualization

To bridge the gap between abstract algorithmic inference and practical application for field inspectors, the machine learning predictions are computationally mapped to an intuitive visualization platform. Using the Python Plotly framework, processed pipeline telemetry is dynamically rendered into a scalable, interactive 3D point cloud. This allows inspectors to visually assess the geometric profile and precise spatial location of identified structural flaws without interpreting raw classification probabilities.

4.4.6 Spatial Coordinate Mapping

The robot's stepper motor controller records the angular position, axial distance, and sensor reading at each measurement point throughout the inspection sweep. These data points are recorded in a polar coordinate system (angle, radius, axial offset) aligned with the pipe's central longitudinal axis.

To render them in a 3D Cartesian visualization, each point is transformed using the standard polar-to-Cartesian conversion:

$$x = r \cos(\theta), y = r \sin(\theta), z = z_{\text{axial}} \quad (4.32)$$

where r is the pipe radius, θ is the angular position of the sensor head, and z_{axial} is the axial displacement along the pipe.

Each rendered point is color-coded according to the classification result from the machine learning pipeline:

Table 4.3 Color Codes

Color	Meaning
Green	Normal — structurally sound
Red	Crack — defect detected
Grey	Not tested — no sensor coverage at this coordinate

The interactive Plotly renderer allows the user to rotate, pan, and zoom into the 3D point cloud, enabling detailed inspection of crack clusters, their spatial extent, and their exact location relative to known pipe features.

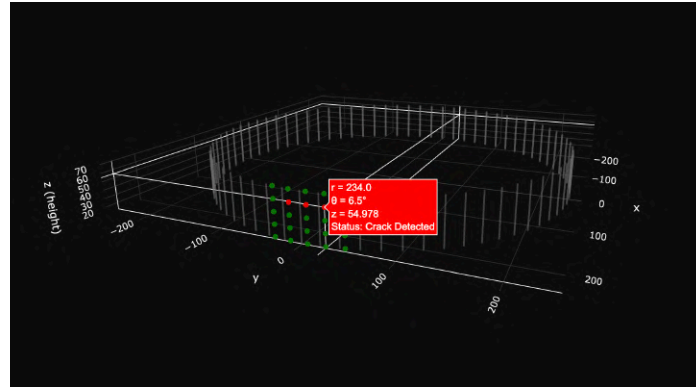


Fig. 4.24 3D point cloud with cursor over crack data point

4.4.7 Practical Value

This visualization closes the loop between signal processing and field deployment. Rather than requiring inspectors to interpret numerical ML outputs or raw waveforms, the color-mapped 3D model presents an immediate, spatial answer: where the defect is, how large the affected region appears, and whether adjacent areas are clean. This format is directly compatible with the workflow of site engineers who are accustomed to interpreting structural drawings and spatial scans.

4.4.8 Limitations

A foundational challenge of this prototype was the data acquisition bottleneck. Collecting high-fidelity NDT telemetry is extremely time-intensive, and after exploring various sensors and wave types the final angle beam dataset was limited to approximately 70 samples. While this small subset constrained the deployment of deep learning tools such as convolutional neural networks, the implementation of rigorous cross-validation, dimensionality reduction, and physics-informed acoustic feature engineering nonetheless produced a functional and statistically significant inference model.

Several specific limitations were identified:

Data Representativeness. The training data was collected from artificially machined cracks, whose dimensions and acoustic signatures differ from naturally occurring weld cracks. The model's generalization to real-world defects remains unverified. Retraining on field-collected data from actual in-service pipelines would be necessary before production deployment to ensure the model can identify the subtler, more irregular crack morphologies encountered in practice.

Window Configuration. The temporal windows used for feature extraction are tuned to the specific pipe geometry and sensor configuration of this prototype. A more robust, adaptive windowing approach (potentially using automated onset detection or pipe-geometry-aware gating) should be explored to improve generalization across varying pipe diameters and wall thicknesses.

Sample Size and Model Robustness. Future work could investigate synthetic data generation, such as physics-based simulation of ultrasonic wave propagation through cracked media, to augment the training set and improve model robustness against edge cases. This was scoped

out of the current prototype due to timeline constraints but represents a natural next step for transitioning the proof-of-concept into a field-ready tool.

4.5 Summary of Achievements

- Conceptualized, simulated, and built a three-legged adjustable suspension drive-base to maintain strict concentricity inside pipes.
- Quickly adapted to project needs by learning and implementing a full signal processing pipeline mid-year.
- Extracted and evaluated complex time-domain and frequency-domain features from EMAT and ultrasonic sensors using Cohen's d.
- Successfully trained an ML classification model utilizing nested cross-validation, collinearity filtering, and dimensionality reduction via PCA were implemented to overcome the severe sample-size limitations inherent to prototype NDT data collection.
- A linear support vector machine achieved a cross-validated Macro F1-score of 95.6%, with statistical significance confirmed by permutation testing ($p < 0.05$).
- Generated an interactive, interpretable 3D point-cloud map for end-user crack visualization.

4.6 Appendix

4.6.1 Driving Module Data

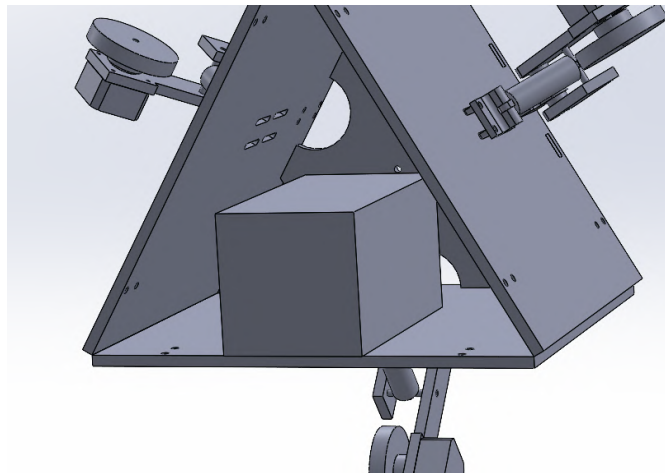


Fig. C.1 CAD of Driving Module (Back)

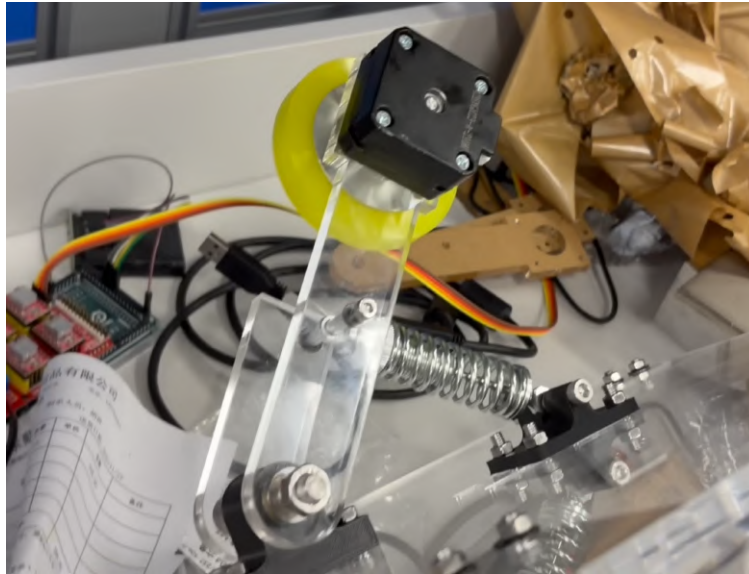


Fig. C.2 Suspension Leg Closeup

4.6.2 Straight Beam Data

Raw Signal Pre-Viewer (Y-Axis scaled to ignore first 5.0 μs)

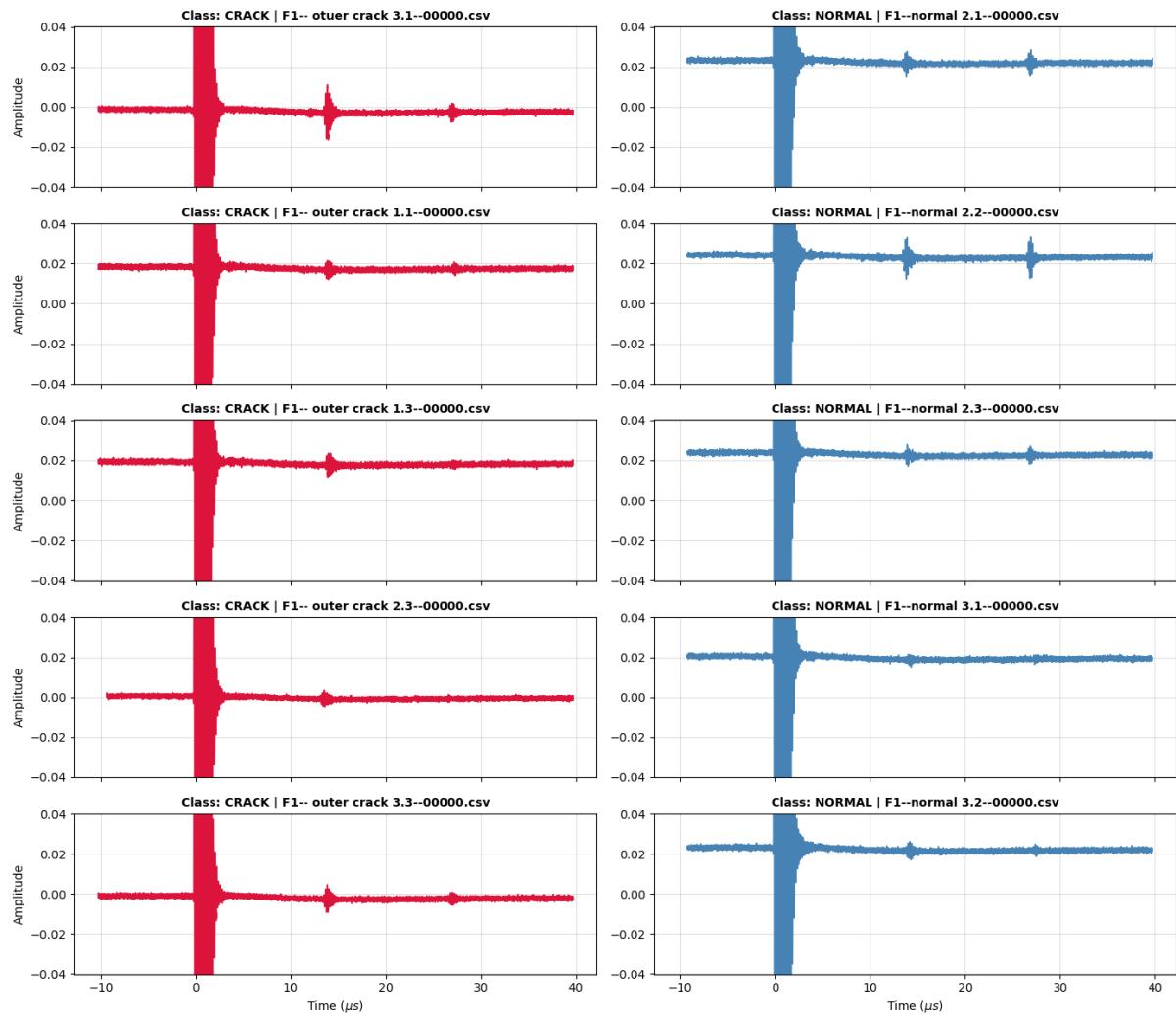


Fig. C.3 10 Straight Beam Ultrasound Raw Signals

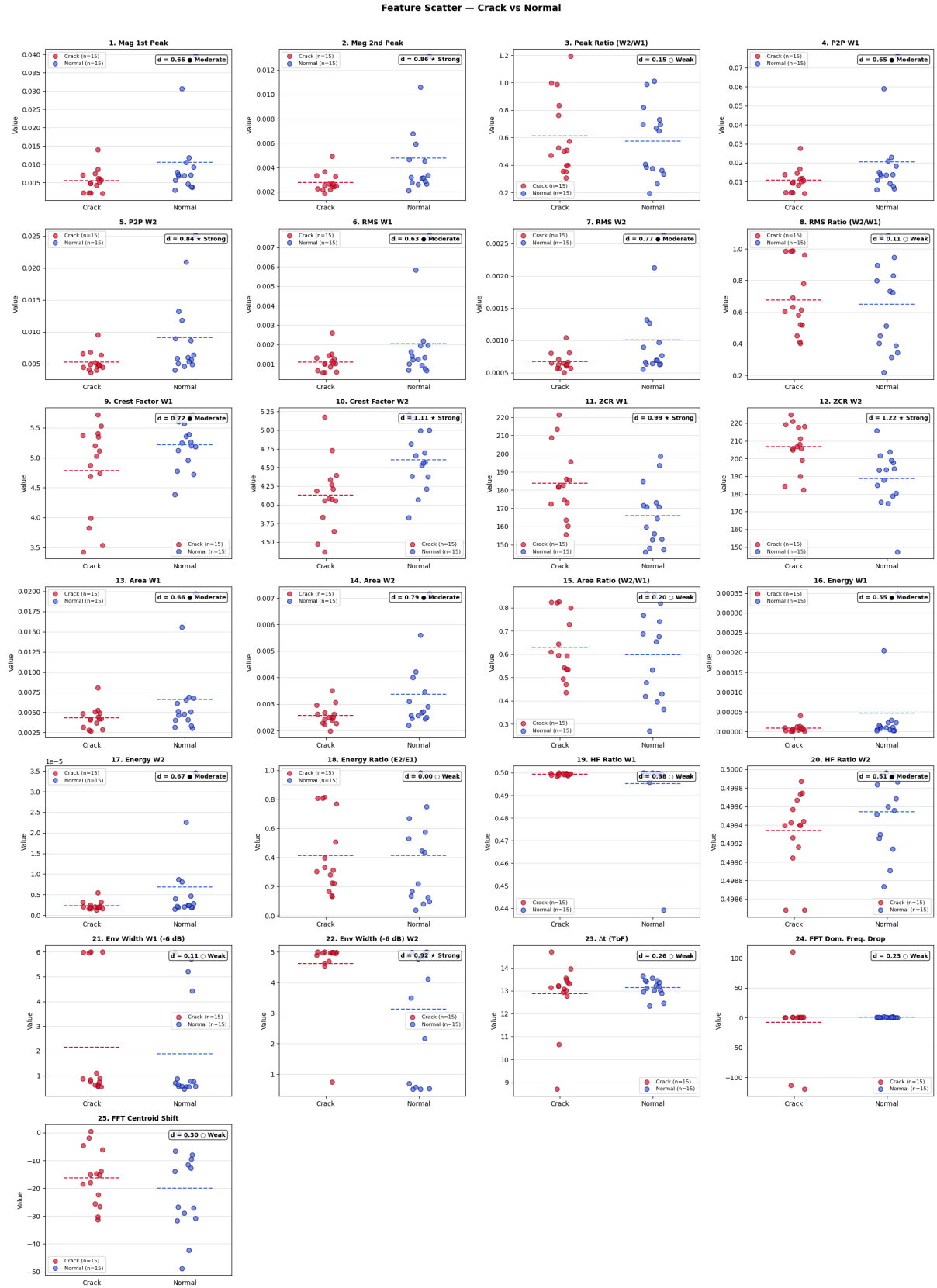


Fig. C.4 Feature Scatter Plots (Straight Beam)

Feature Distributions — Crack vs Normal

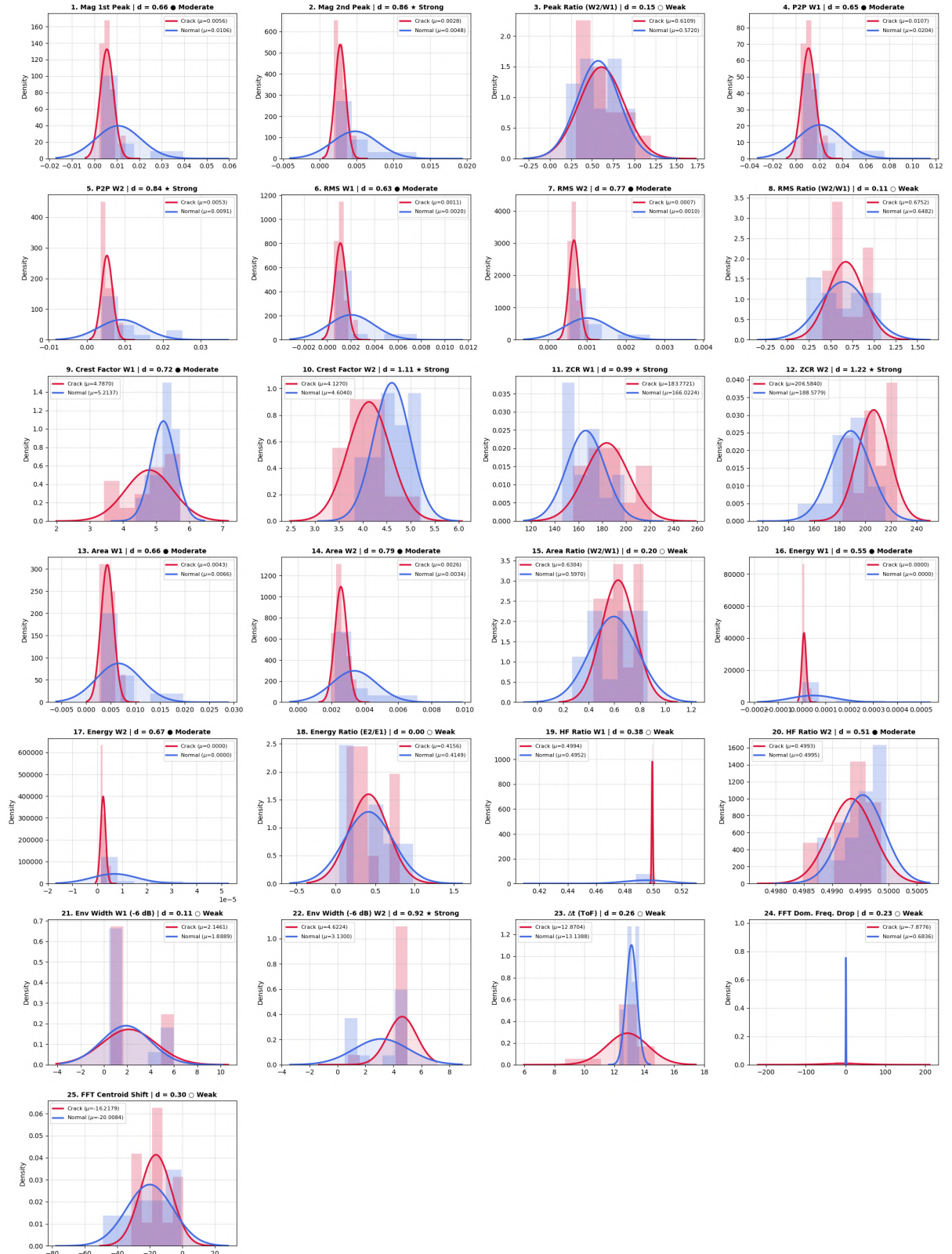


Fig. C.5 Feature Distributions (Straight Beam)

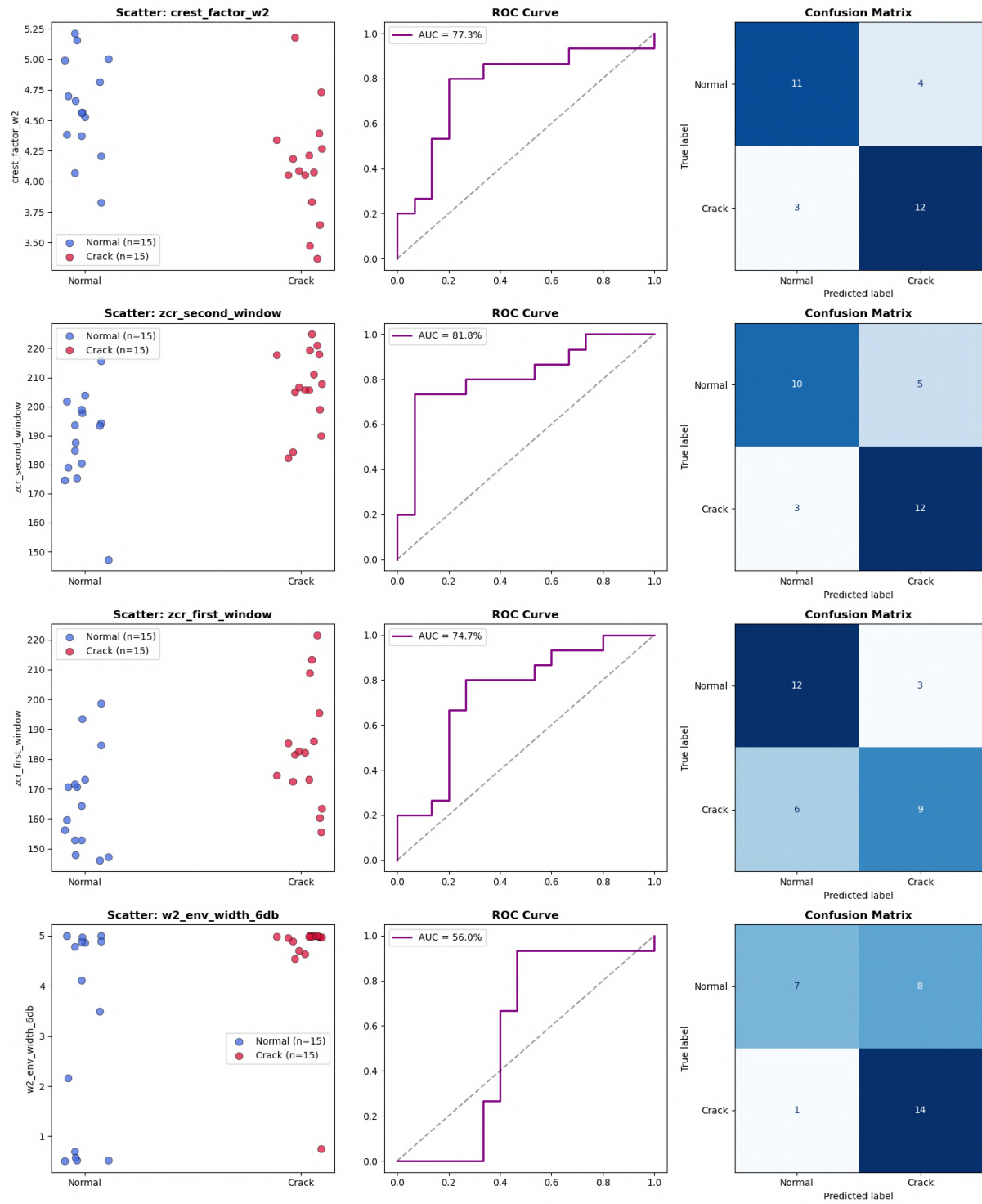


Fig. C.6 Single Feature Classification Results (Straight Beam)

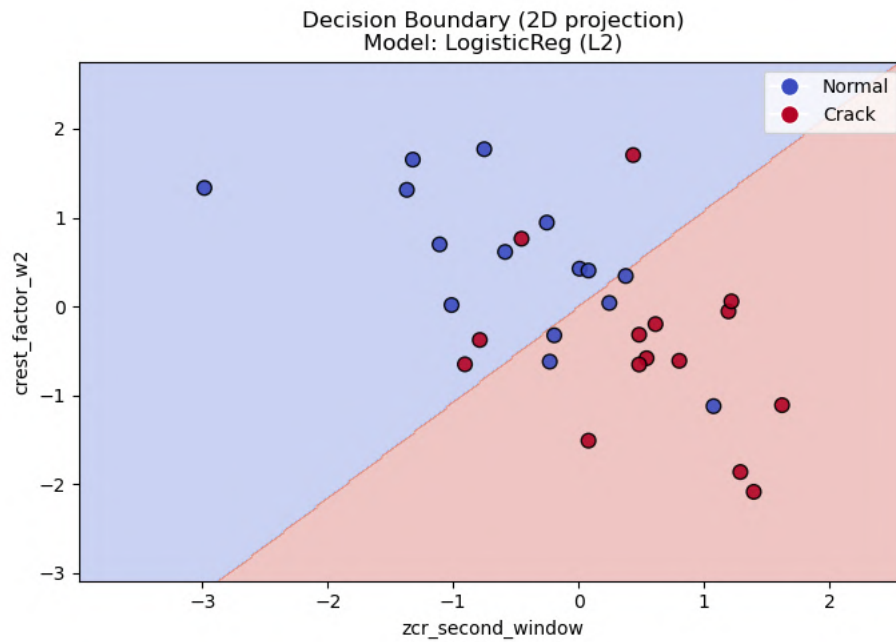


Fig. C.7 Logistic Regression Boundary (Straight Beam, Features)

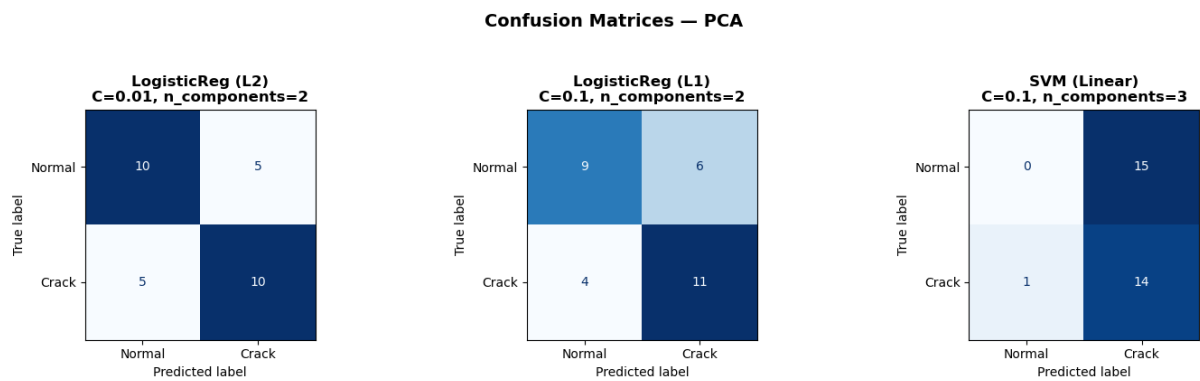


Fig. C.8 Confusion Matrices (Straight Beam, PCA)

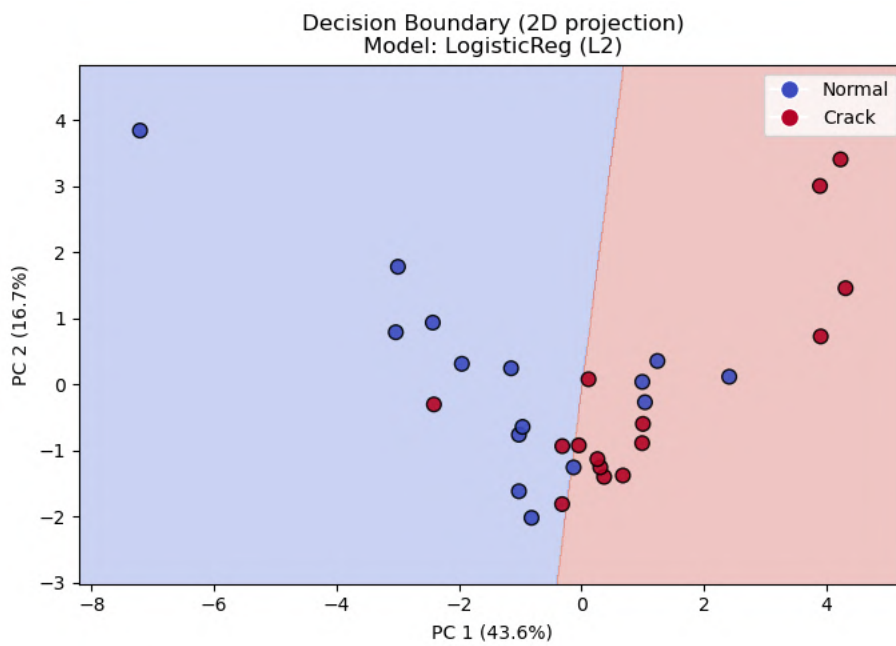


Fig. C.9 Logistic Regression Boundary (Straight Beam, PCA)

Strategy	Model	Best Params	Recall (Crack)	Spec (Normal)	Macro F1	PR-AUC	ROC-AUC	TP	FN	TN	FP	
single-feature	Single-Feature	[crest_factor_w2]	-	80.000000	73.333333	76.640712	78.350427	77.333333	12	3	11	4
	PCA	LogisticReg (L2)	C=0.01, n_components=2	66.666667	66.666667	66.666667	85.886859	83.111111	10	5	10	5
t-test	LogisticReg (L1)	C=1.0, n_select=3	73.333333	60.000000	66.517857	78.100376	73.333333	11	4	9	6	
t-test	LogisticReg (L2)	C=0.01, n_select=3	73.333333	60.000000	66.517857	74.477060	67.555556	11	4	9	6	
PCA	LogisticReg (L1)	C=0.1, n_components=2	73.333333	60.000000	66.517857	65.537744	67.111111	11	4	9	6	
t-test	SVM (Linear)	C=0.1, n_select=3	73.333333	46.666667	59.276018	59.811500	64.000000	11	4	7	8	
PCA	SVM (Linear)	C=0.1, n_components=3	93.333333	0.000000	31.818182	38.545628	22.666667	14	1	0	15	

Fig. C.10 Performance Comparison (Straight Beam)

4.6.3 Angle Beam Data

Raw Signal Pre-Viewer (Y-Axis scaled to ignore first 5.0 μ s)

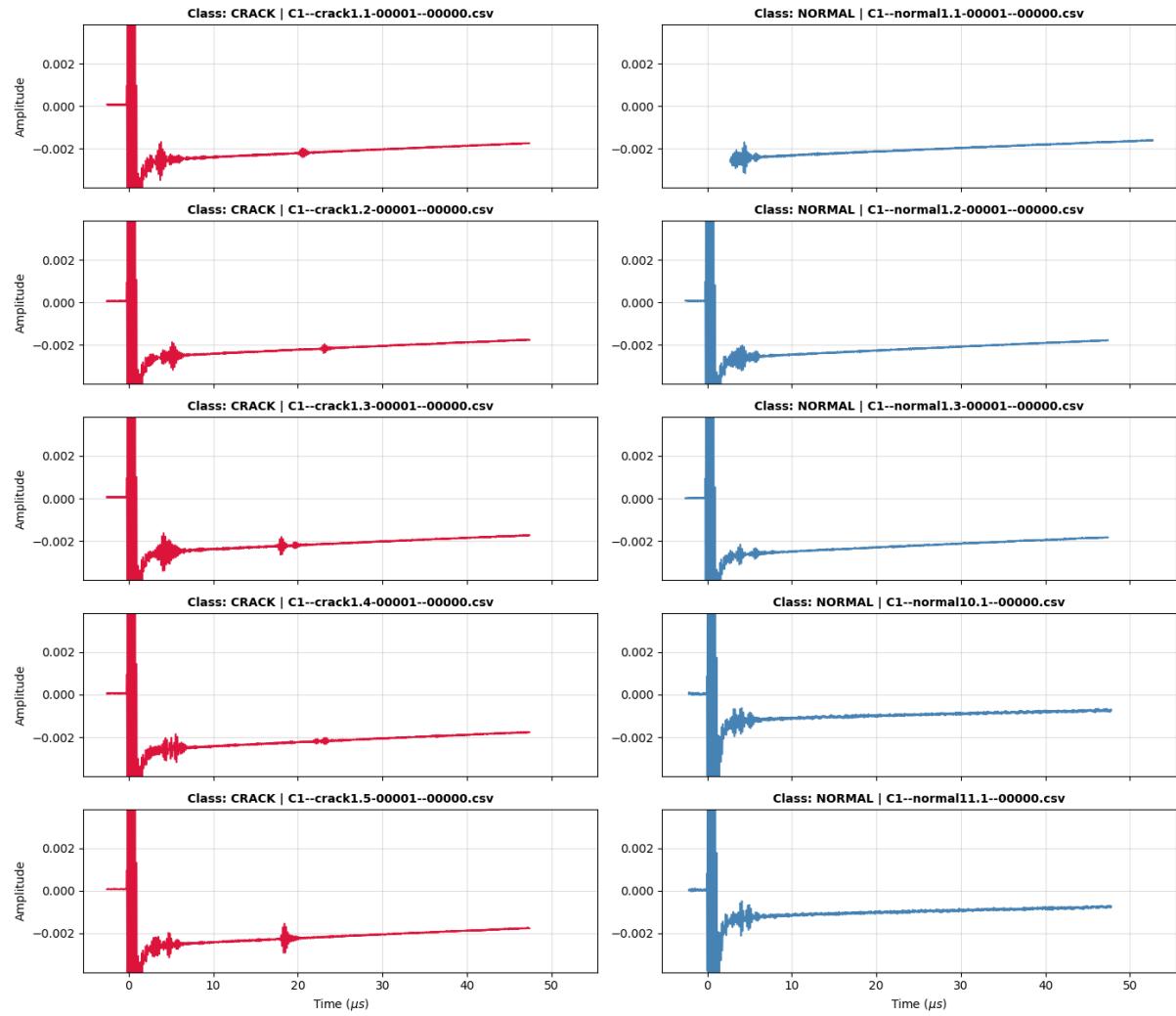


Fig. C.11 Angle Beam Ultrasound Raw Signals

Feature Scatter — Crack vs Normal

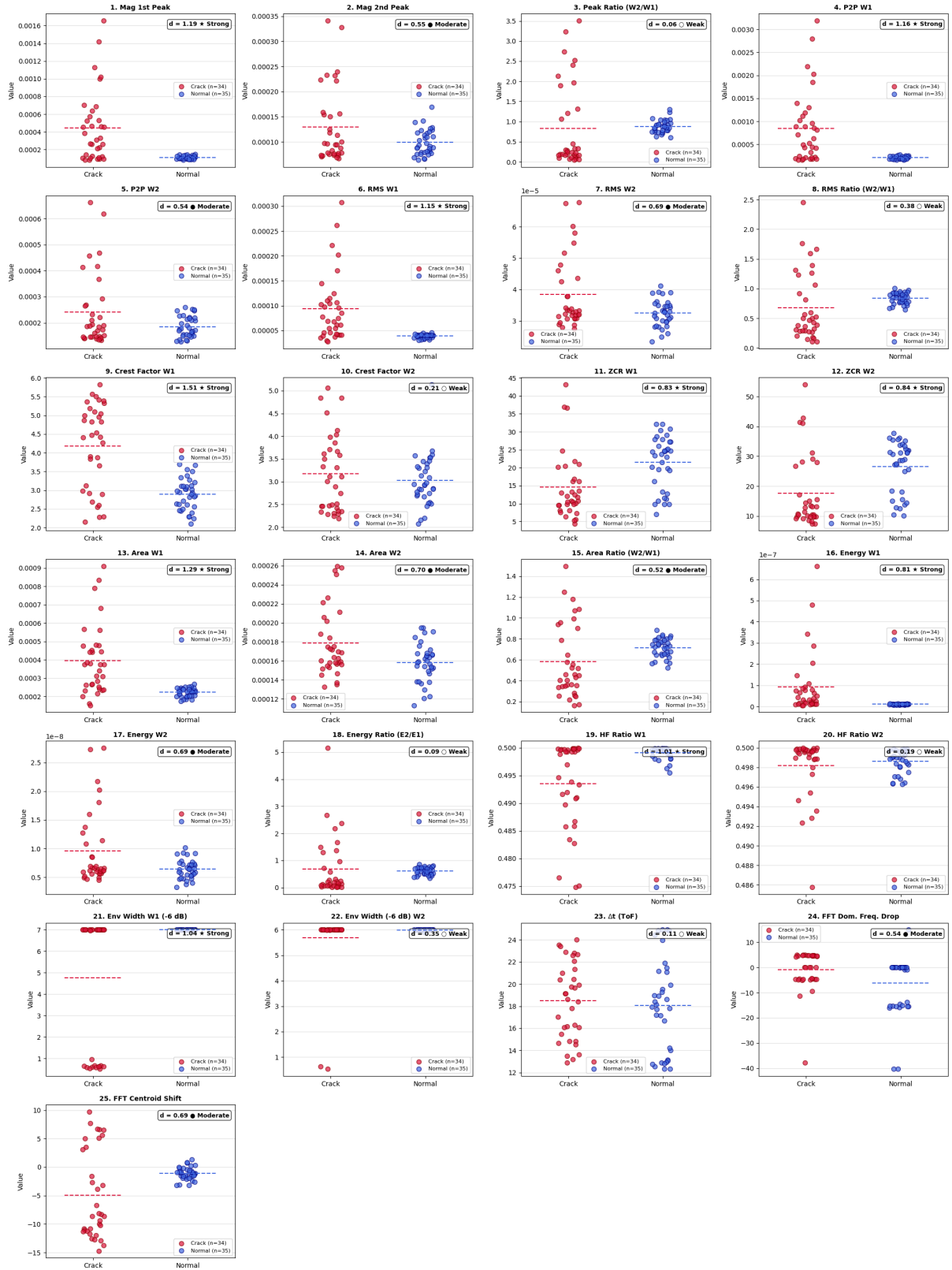


Fig. C.12 Feature Scatter Plots (Angle Beam)

Feature Distributions — Crack vs Normal

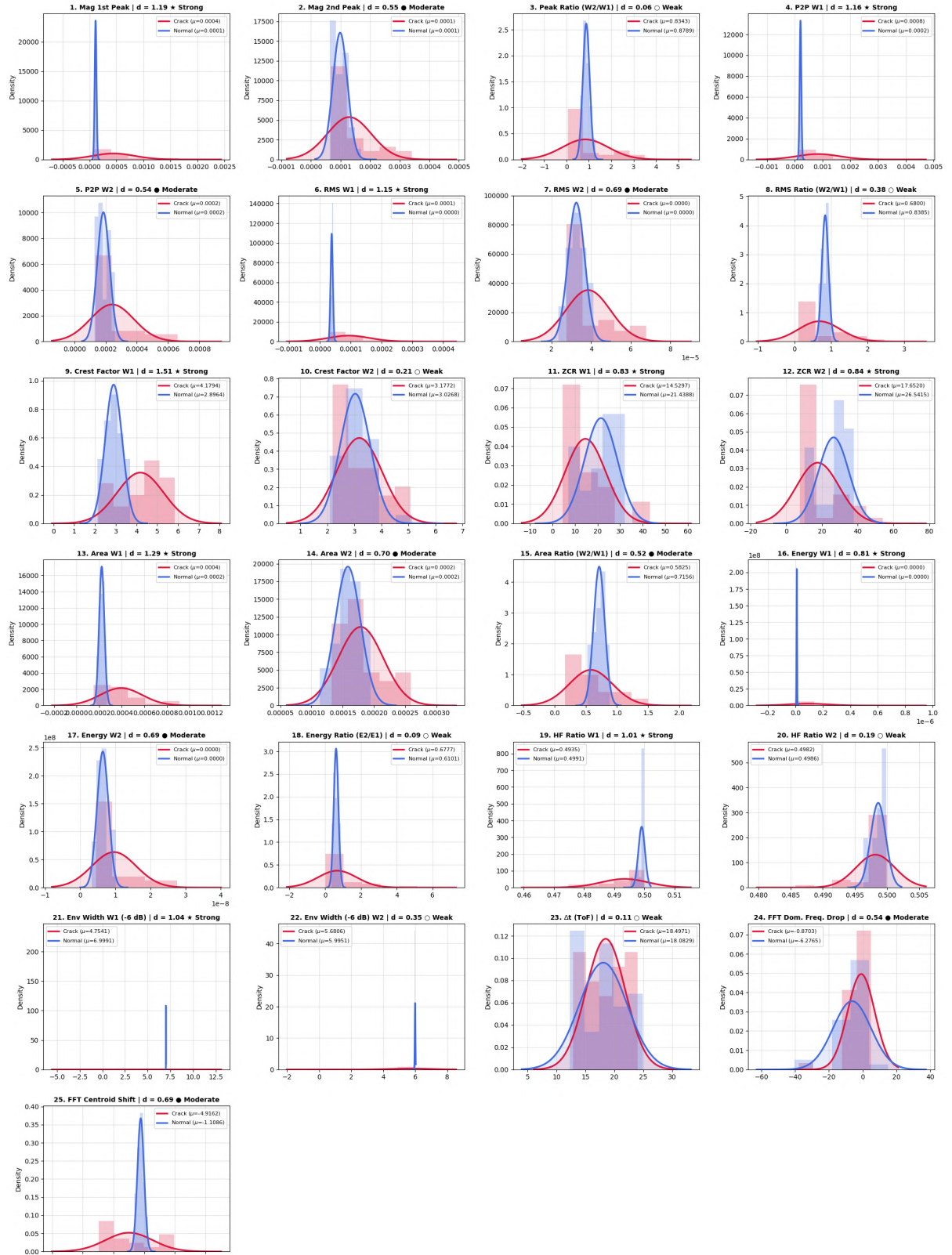


Fig. C.13 Feature Distributions (Angle Beam)

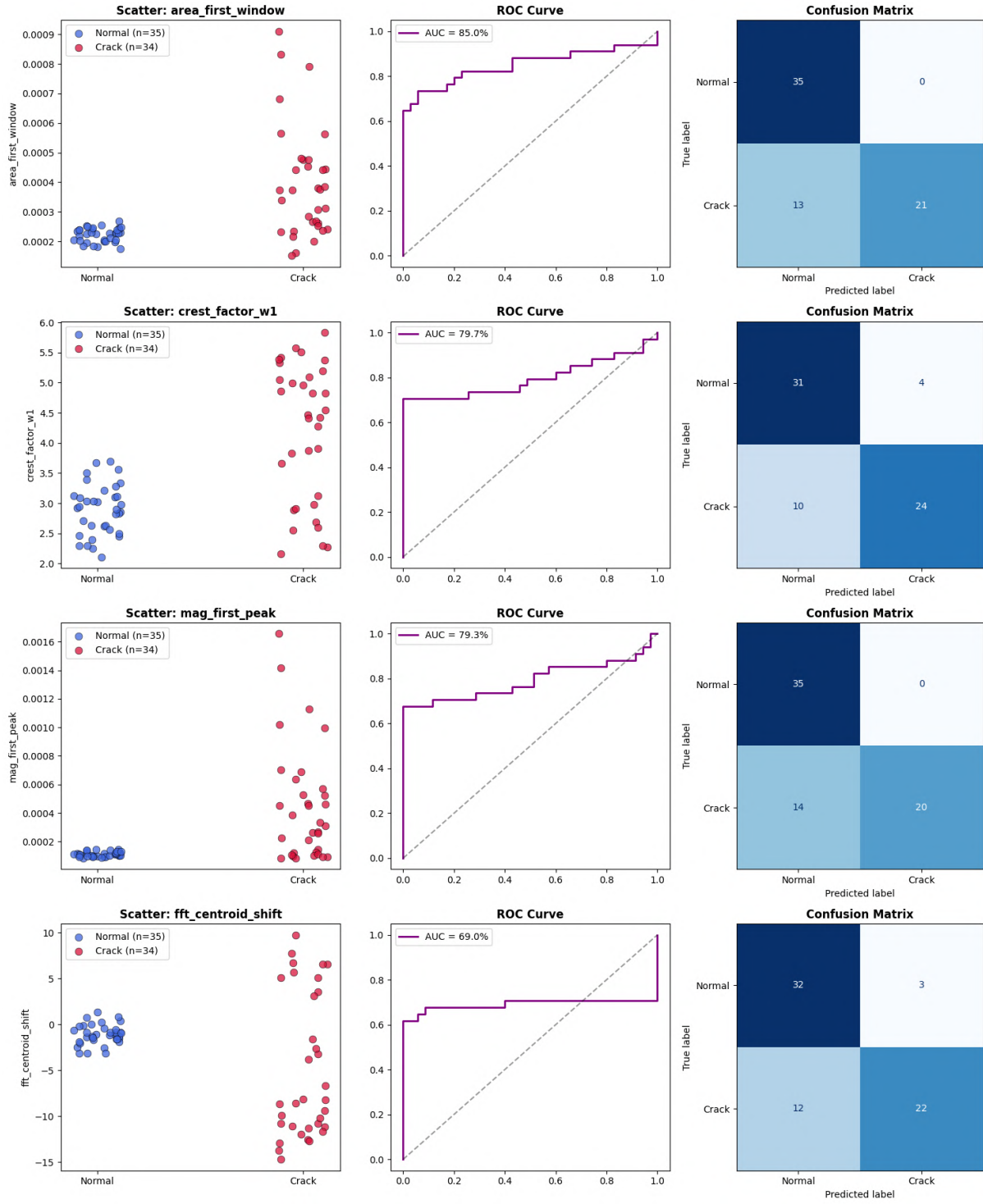


Fig. C.14 Single Feature Classification Results (Angle Beam)

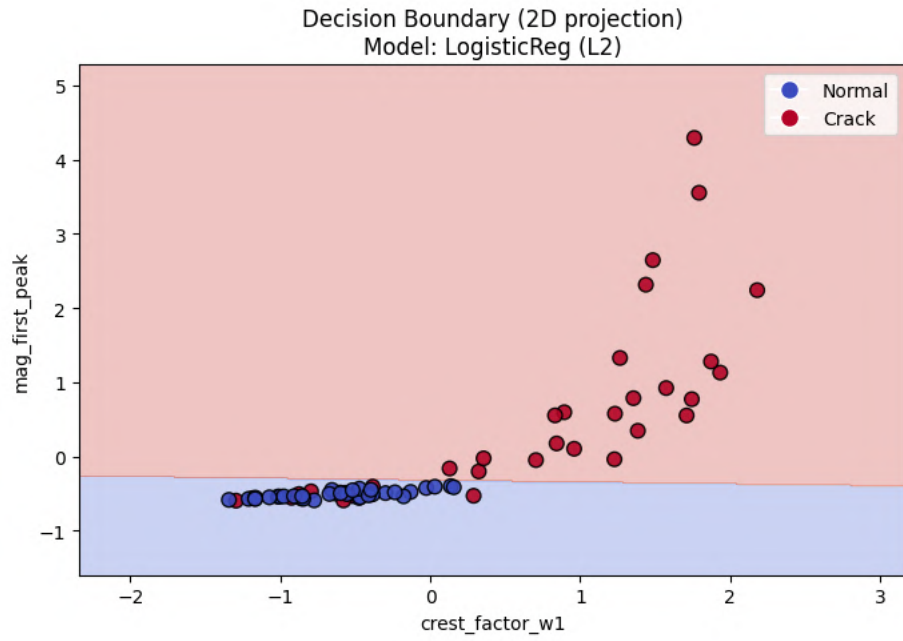


Fig. C.15 Logistic Regression Boundary (Angle Beam, Features)

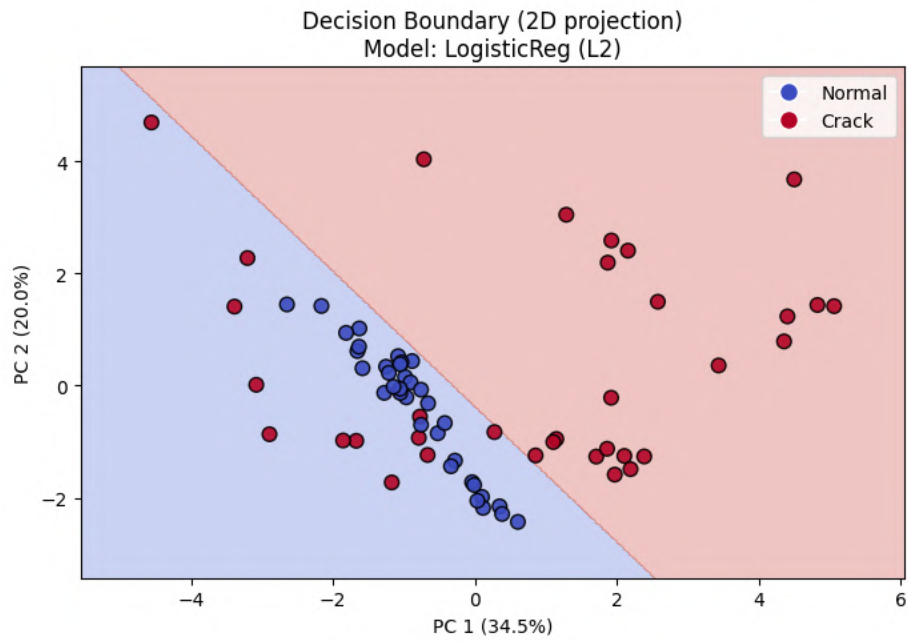


Fig. C.16 Logistic Regression Boundary (Angle Beam, PCA)

Strategy	Model	Best Params	Recall (Crack)	Spec (Normal)	Macro F1	PR-AUC	ROC-AUC	TP	FN	TN	FP
PCA	SVM (Linear)	C=1.0, n_components=5	94.117647	97.142857	95.648518	97.291329	96.050420	32	2	34	1
PCA	LogisticReg (L1)	C=10.0, n_components=3	91.176471	94.285714	92.747530	96.605409	94.453782	31	3	33	2
PCA	LogisticReg (L2)	C=1.0, n_components=3	88.235294	97.142857	92.729189	97.726072	96.302521	30	4	34	1
t-test	LogisticReg (L2)	C=10.0, n_select=5	76.470588	97.142857	86.778795	91.607147	86.974790	26	8	34	1
t-test	LogisticReg (L1)	C=10.0, n_select=5	70.588235	97.142857	83.715941	91.151098	86.554622	24	10	34	1
t-test	SVM (Linear)	C=5.0, n_select=5	64.705882	100.000000	81.968641	90.933069	85.714286	22	12	35	0
single-feature	Single-Feature [area_first_window]	-	61.764706	100.000000	80.350493	89.982945	85.042017	21	13	35	0

Fig. C.17 Performance Comparison (Angle Beam)

4.6.4 Code Repository

<https://github.com/jonybarr/Feature-Analysis-and-Classification-Pipeline.git>

MECH 4900 Final Year Design Project Final Report 2025/2026

5 Individual Section D: Robot Assembly and System Integration

By LAU Tsz Kit

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5.1 Introduction
5.2 Personal Objective
5.3 Project Schedule and Milestones
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5.1 Introduction

Converting a validated ultrasonic sensing technology to a fully functional field of deployable inspection system requires careful assembly and integration of all mechanical, electrical and software components into a single robotic platform. Following the prior comparative study which determined the ultrasonic angle beam probe transducer to be more ideal sensors for the detection of pipeline cracks, the next crucial step in the project was to physically construct the robot and integrate seamlessly its actuation, control, and sensing subsystems.

This phase is motivated by the concrete reality that even the most sensitive ultrasonic sensor is ineffective without reliable positioning against the pipe wall, the robot's ability to navigate the pipeline, and the correct synchronization of the control sequence for the various motors. To achieve a working prototype, there are numerous engineering challenges that need to be overcome, namely mechanical misalignments, inadequate torque, gear vibration, and wheel traction problems. Furthermore, mechanical validation was also executed via examining various motors. This report describes the complete assembly process, challenges encountered and their solutions, automation sequencing integration, field demonstration results, and limitations identified with the proposed improvements for future iterations.

The outline of this section is as follows. Section 5.2 covers personal objectives for the robot assembly and integration phase. Section 5.3 presents the project schedule, key milestones, and the tangible deliverables achieved. Section 5.4 illustrates the systematic assembly methodology including step-by-step construction from the backroad to connecting with the second module of the robot and indicates encountered assembly challenges and their feasible solutions. Section 5.5 details the mechanical and motion validation, covering clamping and concentricity, disk rotation, radial extension, and forward/backward motion tests. Section 5.6 explains the integration of the transducer and the automation system. Section 5.7 reveals the experimental results, current potential issues of the robot, and proposed advancements. Section 5.8 summarizes the personal achievements of this phase.

5.2 Personal Objective

The primary objective of this phase was to create a fully operational robotic platform that could perform autonomous movement within a specific pipeline, transporting the transducers, and executing a predetermined sequence of motor actions for automated crack detection. Specifically, the goals included:

- To systematically construct the first part of the robot, and successfully connect to its second part, ensuring each subassembly was verified before full integration.
- To identify and resolve key mechanical challenges, involving non-concentric center lead screw, poor gear engagement, enormous friction from the steel ball wheels, and stepper motor torque or directional failures.
- To perform mechanical and motion validation tests, encompassing clamping concentricity via the self-centering four-bar linkage, full-range disk rotation utilizing the 11.2:1 gear reduction, radial extension of the scanning arm along linear guides, and forward/backward locomotion following the substitution of steel ball wheels with rubber wheels.
- To analyze an automation sequence in C++ that controls four types of motors: transducer lift (motor 0), rocker arm rotation (motor 1), leg expansion (motor 2), and forward/backward movement (motor 3).
- To conduct practical demonstrations inside a test pipeline, inspecting multiple points across a known crack location.

- To critically evaluate the structure of the robot and the performance of the integrated system, document existing problems, and propose concrete improvements for future design revisions.

5.3 Project Schedule and Milestones

Table 5.1. Personal timeline and milestones

Period	Phase	Key activities	Deliverables
2025.9	Concept exploration	Literature reviews on cutting-edge pipeline inspection robots and different types of ultrasonic transducers' principles	Literature review
2025.10	Future investigations direction planning	Understanding the operation of oscilloscope and signal generator; planning the experiments of P-wave contact transducers and air-coupled transducers	N/A
2025.11	Experiments of contact and air-coupled transducers	2 types of transducers' experiments conducted; Their data signals analyzed	P-wave contact and air-coupled transducers data signal collection and analysis
2025.12	Alternative investigations direction, pipeline processing	Understanding EMAT principles and its operation devices; 5 more artificial, realistic cracks on the inner and outer surfaces of the pipeline	Mid-term report, new cracks producing
2026.01	Experiments of EMAT	The experiment of EMAT conducted; its data signals analyzed	EMAT data signal collection and analysis
2026.02	Experiments of EMAT and T & P wave contact transducers	The experiment of EMAT and T & P wave contact transducers conducted; Recording the required materials for the robot assembly	Data signal collection and analysis
2026.03	Conducting experiment, robot assembly, additional pipeline processing	The experiment of T & P wave contact transducers conducted; Version 2 robot assembly; Making deeper the previous defects	More data signal of T & P wave contact transducers collection, V2 prototype
2026.04 early - mid	Experiment of ultrasonic angle beam probe transducer	Understanding the principle of guide wave ultrasonic testing; The experiment of the ultrasonic angle beam probe transducer conducted;	Data signal of ultrasonic angle beam probe transducer collection
2026.04 mid - late	Pipe validation, system readiness	Testing the feasibility of the robot travelling and operating inside the pipe; Integrating ultrasonic angle beam probe transducer to the robot and recording demo	Automation sequence and defect-detection demo recording

5.4 Robot Assembly

The robot assembly was conducted in a logical procedure, commencing with the bottom mechanical layer and progressing upper and outer parts. Before installing the broader framework, each subassembly would be completed and validated, avoiding inconsistency, misalignment, and collision.

5.4.1 Process of Assembly

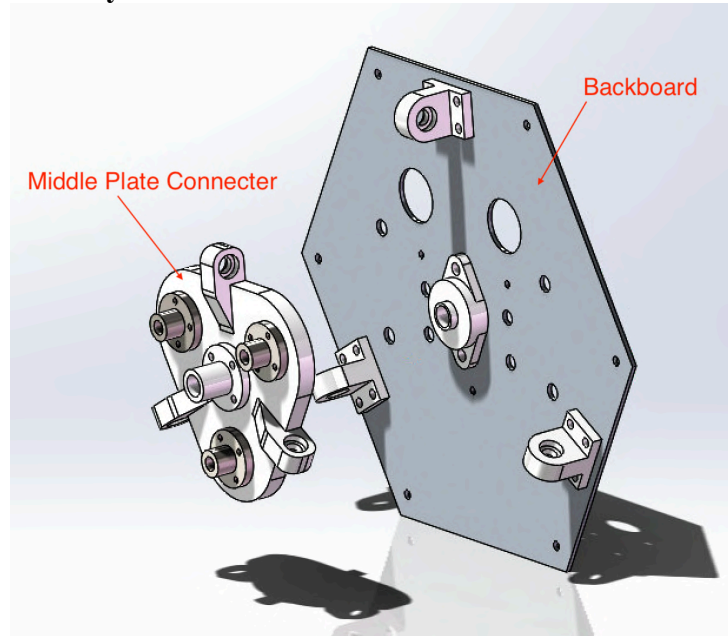


Fig. 5.1 The first step of the robot assembly

The backboard and the middle plate connected to the legs were initially built, as shown in Fig. 5.1. There are several flanges and connections which have to be installed. In addition, a certain amount of space has been reserved at each joint which requires installing bearing, optimizing the movement of the legs.

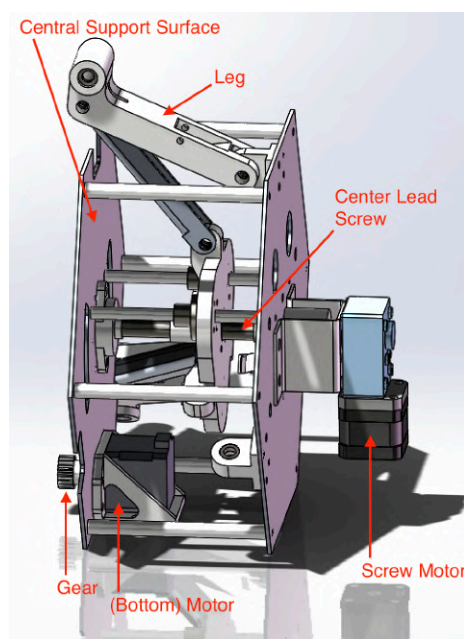


Fig. 5.2 The overall structure of the robot body

Next, as displayed in Fig. 5.2, several shafts, motor at the bottom with gear and the middle plate connector were assembled to connect to the central support surface, forming a general structure and fixing the location. Then, the center lead screw with its motor and 3 legs were installed. It is noted that the center lead screw and legs can be moved freely before tightening all screws.

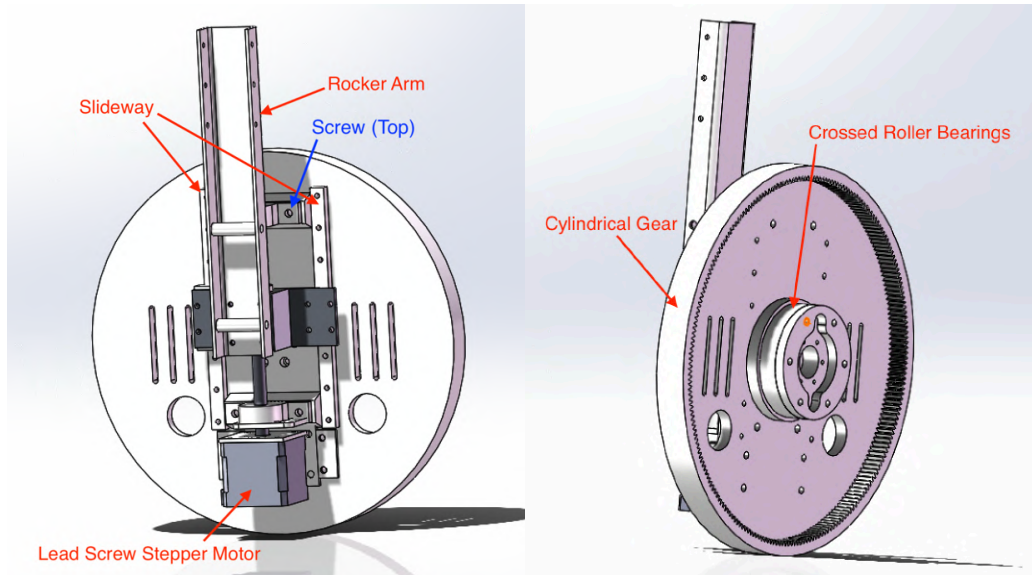


Fig. 5.3 The head part assembly of the robot

After that, Fig. 5.3 shows that the cylindrical gear from the head part was isolated to install each component, including rocker arm, slideway, lead screw stepper motor and crossed roller bearing. Similarly, if the top screws were originally tightened, the lead screw stepper motor might not rotate owing to alignment, severely affecting the performance of the rocker arm. Hence, it is essential to verify the movement of the rocker arm in each step.

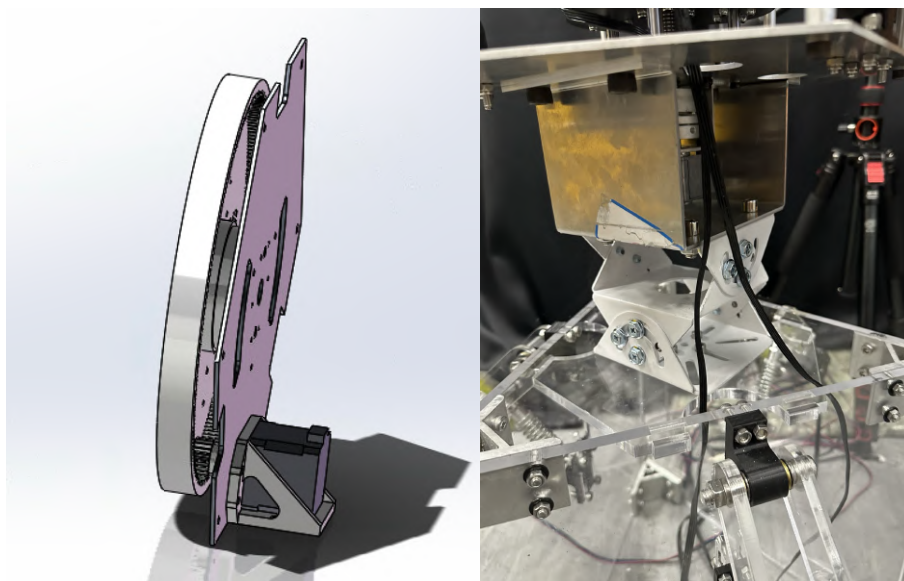


Fig. 5.4 The connection between the head part and the main body of the robot (left) and connection between the main body and the second part of the robot (right)

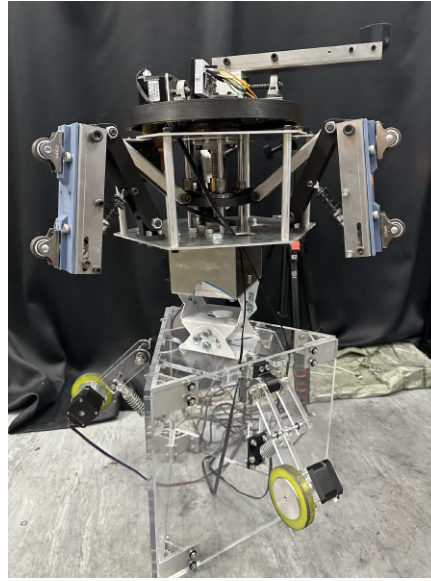


Fig. 5.5 The overview of the finished robot

The final procedure is exhibited in Fig. 5.4 that the cylindrical gear and the central support surface were integrated. The gear at the bottom has to align with the teeth of the head part, otherwise the robot fails to rotate. Additionally, spring, wheel holders and wheels could be assembled, and the second part of the robot could be connected. Fig. 5.5 exhibits an overview of the entire robot. At the end, all motors could be examined to verify whether they are success to rotate all specific parts. And further testing could be conducted in the pipeline.

In this systemic assembly process, not only does the process become simpler, particularly for tightening screws, but the components and connection points are also ensured to be functionable and aligned. Therefore, it is necessary to plan the assembly process before assembling, minimizing issues with the robot's structure.

5.4.2 Challenges and Solutions

Despite careful planning, a number of major difficulties still emerged during the assembly and in-pipeline testing. Each issue was required thorough investigation and a feasible solution.

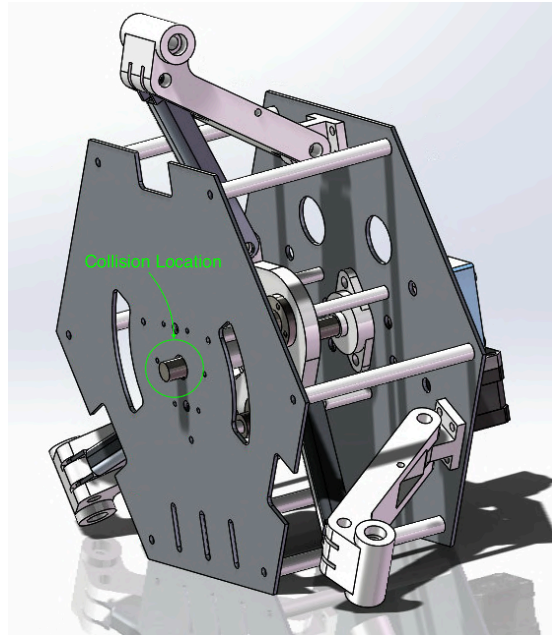


Fig. 5.6 The position of the center lead screw collision

The first problem we encountered is the non-concentric of the middle screw, as shown in Fig. 5.6. When the center lead screw was inserted from the backboard to the central support surface, the edge of the center lead screw seems to scratch the middle hole of the central support surface. Initially, we questioned its manufacturing accuracy, but CNC manufacturing would not contain major deviation. Eventually, we discovered that screws were tightened without proper alignment, so we loosened the screws and realigned them.

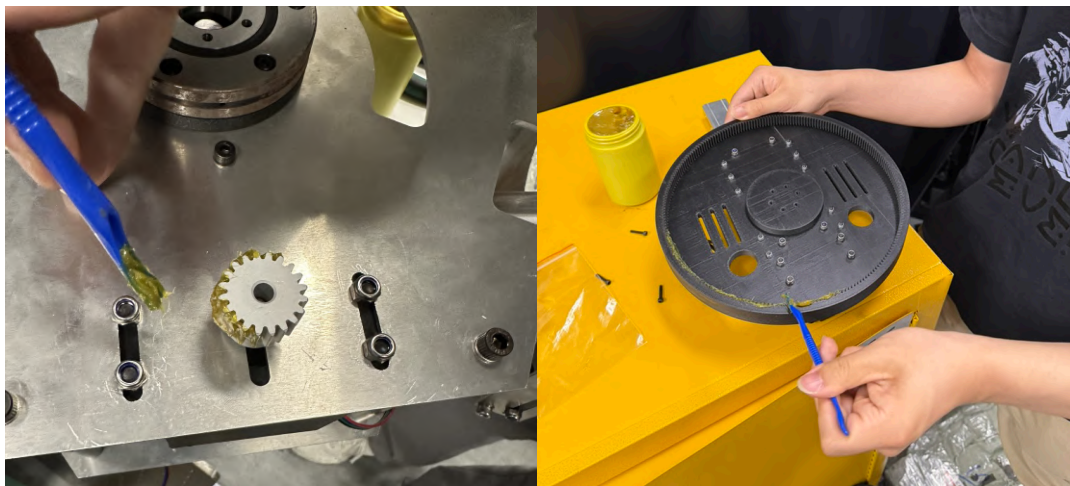


Fig. 5.7 The lubrication of gear (left) and cylindrical gear (right)

After assembling all accessories, all motors were tested once. Their operation was normal, while there was loud noise due to strong vibration during running motor 1. We considered gear engagement that might exhibit poor meshing. To clarify this problem, all gear interfaces were lubricated with a high viscosity grease, as shown in Fig. 5.7. Lubrication was used to reduce friction and to inhibit vibrations from gear engagement. Ultimately, the noise had subsided. Due to the motor frequency, there might still be resonance, resulting in a noticeable noise.

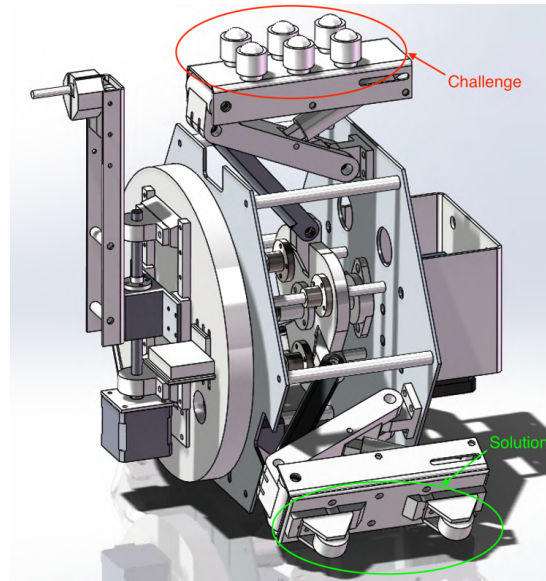


Fig. 5.8 The general picture of the steel ball wheels and the rubber wheels in the robot

More importantly, the biggest challenge is that the robot cannot move inside the pipeline during initial testing. The second-generation robot was driven by steel ball wheels, creating huge friction that the robot unable to move. After that, 21mm stepper motors driving the rear tires were changed to 42mm stepper motors, but the robot was still unable to travel inside the pipe. Consequently, the rubber wheels from version one robot were replaced, as shown in Fig. 5.8, reducing tremendous friction. Thus, free movement was achieved.

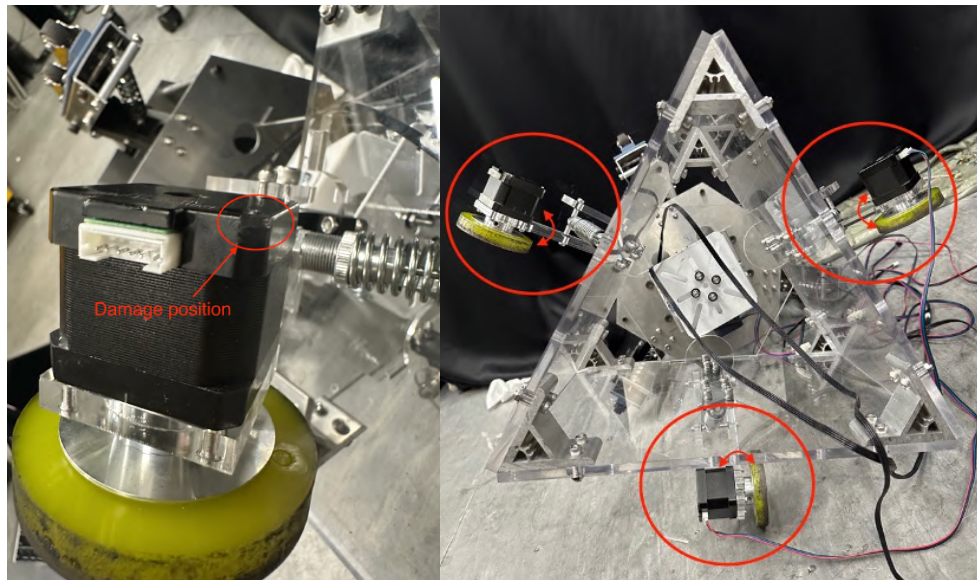


Fig. 5.9 The damage position of the motor (left) and the latest location of the rear tires and the motors after the adjustment (right)

Subsequently, the following problem of changing larger stepper motors is that the motors would contact the inner surface of the pipe, scratching the shell of the motor during testing, as shown in Fig. 5.9. The movement of the entire robot was sometimes influenced. In this regard, as redesigning requires a significant investment of time and time was tight at that moment, the

ultimate solution displayed in Fig. 5.9 is to reverse the motor and wheel positions, creating more space for installing the motors.

In summary, all major mechanical problems had been solved with systematic diagnosis and particular modifications. The solutions included realigning the center lead screw, lubricating gears, replacing steel wheels with rubber wheels, and repositioning motors. After these adjustments, the robot was enabled to achieve stable, controllable movement within the pipeline.

5.5 Mechanical and Motion Validation

Before the integration of the ultrasonic transducer and the full automation sequence, it was necessary to verify that the fundamental mechanical subsystems of the robot performed as expected in real pipeline conditions. This section validates four key motion functions: the clamping and concentricity of the self-centering four-bar linkage, the smooth rotation of the disk in the full 360° range, the radial extension and retraction of the scanning arm along its linear guides, and the forward/backward locomotion after the replacement of the steel ball wheels with rubber wheels. Each test was performed on a section of the test pipe, and performance parameters such as number of steps per revolution, displacement resolution, and travel speed were recorded. The results confirmed that the robot fulfilled the design requirements of clamping stiffness, rotational precision, positional control, and controllable motion, with a robust basis for the subsequent integration of the transducer and the automation system. The detailed procedures and the measured parameters are described in the following subsections.

5.5.1 Clamping and Concentricity

The self-centering four-bar linkage was driven with the PS2 wireless controller located in the test pipe. Through Motor 2, the three-wheel carriers expanded until they made contact with the pipe inner wall. Due to the spring suspension design, the rear spring-mounted wheel contacted the wall first; further extension brought the front pivot wheel (R4 hinge joint) into contact, at which point clamping was complete. The front module stood inside the pipe independently once clamped, without any external support, and the structure exhibited a high rigidity with no detectable looseness. The module center was constantly aligned with the pipe centerline, satisfying the concentricity requirement. In practice, the mechanism was pre-adjusted at the mid-travel before insertion and fine-tuned inside the pipe by manually increasing the speed to approximately 700 μ s pulse interval (~ 1.43 kHz); the entire clamping procedure lasted several minutes.

5.5.2 Head Part Rotation and Radial Extension

When the rocker arm and probe bracket were installed, a complete 360° rotation of Motor 1 (the head part rotation) was evaluated. There was no perceptible backlash or stutter during the rotation. The drive train yields an 11.2:1 reduction ratio by using a 20-tooth steel pinion (module 1) meshing with a 224-tooth internal ring gear that is integrated within the PETG-CF rotating disk. The manual calibration produced 2230 steps per revolution, with a 0.45% divergence from the theoretical angular resolution of 0.16° every complete step and 2240 steps per revolution. The cross-roller bearing carried radial, axial, and overturning-moment loads all

at once; the disk performed better than expected under the overhung arm load and showed no discernible tilt at any angle.

The scanning arm was driven by a Motor 0 radial extension via a range of 0–62 mm along the identical pair of linear sideways. A displacement precision of 0.01 mm per step was offered by the lead screw with a 2 mm pitch, direct drive, and no reducer. The linear speed was 1.67 mm/s at the default blocking speed of 6000 μ s per step, which was slow enough to visually assess contact with the pipe wall without incurring the danger of collision. There was no stalling or missed steps across the whole travel range of extensions and retractions. Touch-wall detection relied on operator observation; neither a proximity sensor nor a limit switch was installed.

5.5.3 Forward and Backward Motion

The entire robot was propelled forward and backward inside the pipe by Motor 3, which was set up with 1/8 micro stepping (1600 microsteps per revolution) and drove all three wheels using paralleled A4988 drivers. Following the replacement of the steel ball wheels with rubber wheels (two for each leg, for a total of six), mobility was smooth and controlled, with no discernible difference between forward and backward speed. The average speed was approximately 2-3 cm/s (22.9 mm/s in blocking mode at intervals of 6000 μ s and 34.3 mm/s in continuous mode at intervals of 4000 μ s). A few microsteps, or sub-millimeters, were the minimum controlled displacement.

5.5.4 Motion Parameter Summary

Table 5.2 Four-axis motion parameters

Motor	Function	Resolution	Speed (at default)	Travel / Range	Reduction
Motor 0	Radial arm	0.01 mm/step	1.67 mm/s	0–62 mm	Direct drive (2 mm lead)
Motor 1	Head rotation	0.16°/step	—	360° continuous	11.2:1 (20T / 224T)
Motor 2	Linkage	0.0001 mm/step	0.018 mm/s	0–50 mm (screw stroke)	100:1 gearbox + 2 mm lead
Motor 3	Drive wheels	0.137 mm/step	~20–30 mm/s	Continuous	1/8 microstep (1600 Step/rev), 70 mm wheel

5.6 Automation System Integration

Upon the resolution of the robot's issues and the validation of its mechanism and motion, integrated automated testing was conducted. The automation mainly focuses on rocker arm extension (motor 0), rocker arm rotation (motor 1), and robot movement (motor 3), while the extension of the legs (motor 2) was pre-expanded to provide solid support inside the pipe. Its leg extension was slightly larger than the inner diameter of the tube due to the elasticity of the rubber wheels.

The automation program for motor control is managed by C++. During preliminary test, it was discovered that there are effective means to control the actions of the robot:

- For motor 0 (transducer lift), the extension step was set to be 0 when the transducer contacted the surface. Then, the data was altered to -1000 to leave the surface.
- For motor 1 (rocker arm rotation), the rotation step of the rocker arm rotating 180 degrees clockwise is +1115 (-1115 rotation steps for 180 degrees counterclockwise), so the other angles can be roughly estimated.
- For motor 3 (forward movement), the distance of 100 forward steps of the robot is 13.7mm.

Based on these calibrated parameters, automation sequences were designed for practical demonstrations to exhibit the robot's performance and feasibility.

5.6.1 Automation Sequencing without Transducer

To assess the robot's mechanical performance independently of the ultrasonic sensing technology, an automation sequence was originally conducted without utilizing the transducer. This test focused solely on the repeatability and controllability of motor-controlled motions. The rocker arm was initially positioned vertically, which served as the zero angle reference.

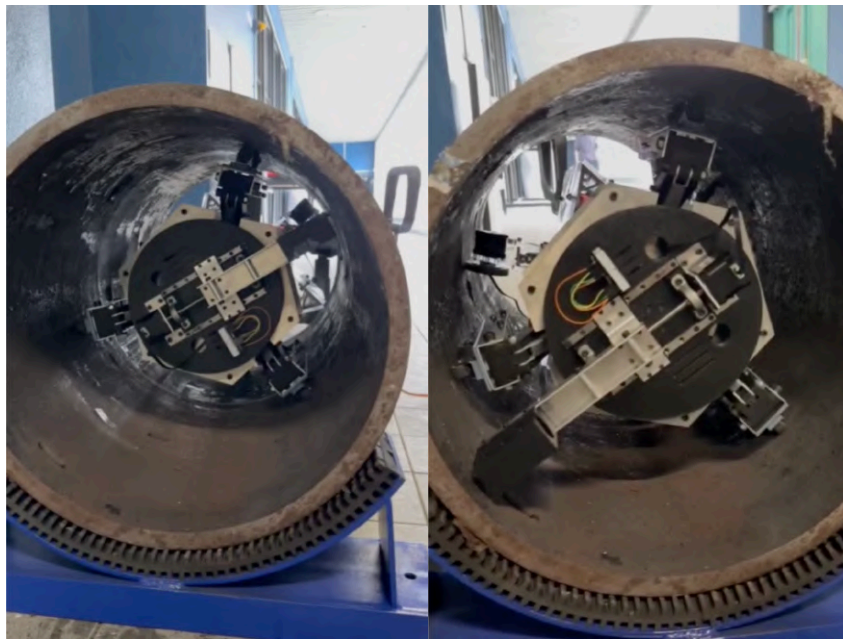


Fig. 5.10 The scenario diagrams of automated testing without transducer

Fig. 5.10 briefly presents the situations and process of the automated robot test. The complete sequence consisted of the following steps:

1. The robot moved forward by setting motor 3 to +500 steps (68.5mm).
2. Motor 0 was commanded to extend (position 0) and then retract (position -1000) to simulate the lifting action without actual surface contact.
3. Motor 1 was then rotated clockwise by 60 degrees, which corresponded to a step value of +372 in the control script.
4. The lifting action of motor 0 (step 2) was repeated.

5. Motor 1 was rotated at a further 60 degrees clockwise, reaching a total clockwise rotation of 120 degrees (another +372 steps).
6. The lifting action of motor 0 was repeated.
7. Motor 1 was rotated to another 60 degrees clockwise, reaching a total clockwise rotation of 180 degrees (another +372 steps).
8. The lifting action of motor 0 was repeated.
9. Motor 1 was then rotated anticlockwise by 240 degrees, which brought the rocker arm to a net angle of -60 degrees (anticlockwise 60 degrees from vertical). This required a step change of -1487.
10. The lifting action of motor 0 was repeated.
11. Motor 1 was rotated anticlockwise by a further 60 degrees (step -+72), reaching a net angle of -120 degrees.
12. The lifting action of motor 0 was repeated.
13. Motor 1 was rotated anticlockwise by another 60 degrees (step -372), reaching a net angle of -180 degrees.
14. The lifting action of motor 0 was repeated.
15. Finally, motor 1 was rotated clockwise by 180 degrees (step +1115) to return the rocker arm to its original vertical position (net angle 0 degrees).
16. The entire sequence from step 1 to step 15 was repeated four additional times, yielding a total of five complete cycles.

The robot was able to execute all of the commands during this test without experiencing any motor failure or loss of positional control. The robot's range of motion was broad and completely functional, but it is important to note that the greatest possible rotation on each direction was restricted to around 180 degrees due to the wiring that connected the transducer and its related electronics. Excessive rotation would cause the wires to become tighter and encounter the danger of being damaged or disconnected. Nevertheless, its ability to forward stepping motion and consistently sweeping between -180 and +180 degrees verify that the mechanical and control systems are suitable for examining a large circumferential region inside the pipeline. For the intended use, the observed performance was judged to be satisfactory.

5.6.2 Automation Sequencing with Transducer

For the robot demonstration integrated with the ultrasonic angle beam probe transducer, a grid of 25 inspection spots (5 rows $\times$ 5 columns) was examined at and near the artificial crack position, as shown in Fig. 5.11. The rocker arm was initially vertical, with motor 1 set to 0. The motor 0 was also set to -1000, providing a safe standoff distance for the transducer to avoid scraping.



Fig. 5.11 The target locations detection for automated testing with transducer

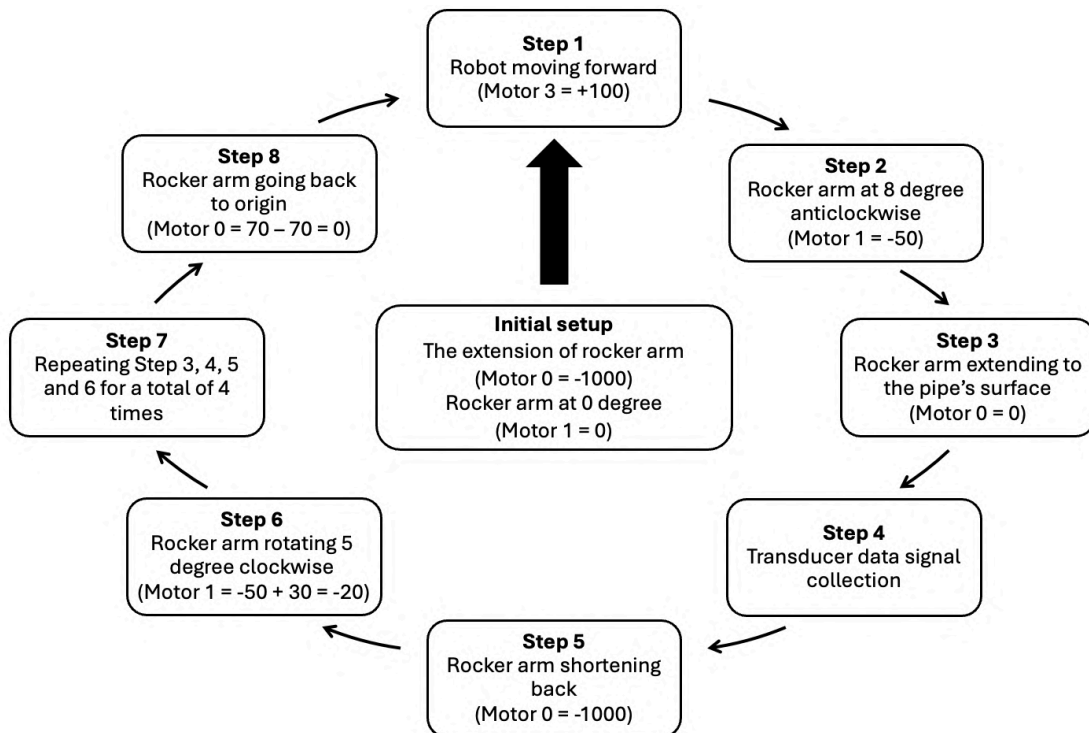


Fig. 5.12 The flow chart of the automation sequencing

The outline of the experimental procedure is displayed in Fig. 5.12, and its details are as follows:

1. The robot moved forward by setting motor 3 to +100 steps.
2. Motor 1 was then set to -50 (approximately 8 degrees counterclockwise) to angle the rocker arm.

3. Motor 0 was changed to 0, raising the rocker's arm until the transducer contacted the inner pipe surface.
4. Once contact was established, the transducer began acquiring ultrasonic signals. If the received data performed by the oscilloscope were effective and stable (consistent normal surfaces or crack signal amplitude and no significant fluctuations), the contact condition was reliable and successful, and the data could be recorded.
5. After the data collection, motor 0 was set back to -1000 , retracting the rocker arm to its original lifted position.
6. Motor 1 was then increased by $+30$ (approximately 5 degrees clockwise) to rotate the rocker arm to the next inspection angle.
7. Steps 3–5 were repeated until five angular positions had been inspected, resulting in a total clockwise rotation of approximately 25 degrees.
8. The rotation step of motor 1 was then decreased to 0 that the rocker arm was going back to the origin.
9. The robot then moved forward again by another $+100$ steps (motor 3), and the entire angular scanning sequence was repeated.
10. This process was iterated five times, covering a total of 25 distinct inspection points.

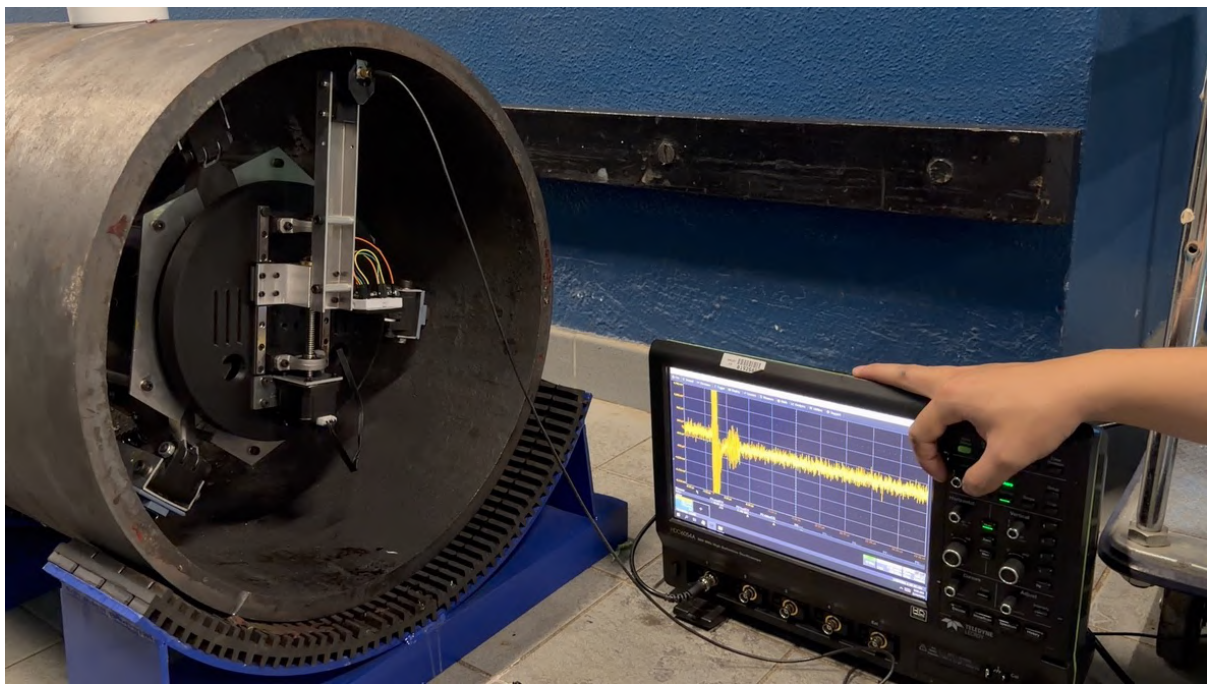


Fig. 5.13 The actual situation of the demonstration

Throughout the demonstration, the robot operated smoothly and successfully inside the pipeline. The entire operation is displayed in Fig. 5.13. On top of that, the ultrasonic angle beam probe transducer recorded effective ultrasonic signals constantly. The recorded ultrasonic signals could be obviously distinguished from the echo of defective and normal surfaces via the oscilloscope. Therefore, the feasibility of the entire integration was verified.

5.7 Result and Discussion

In a trial pipeline with an artificial fracture, the integrated robotic system was effectively validated, and all of the primary objectives were successfully fulfilled. As it executed the preprogrammed automation sequence, the robot could travel inside the pipeline. The ultrasonic angle beam probe transducer obtained signal data by making contact at each of the 25 grid points under the control of motors 0, 1 and 3. The recorded signals evidently displayed a visible fault echo at the specified crack site, demonstrating the functionality of the whole system, from assembly and integration to automation. Additionally, all motors did not exhibit any abnormalities when performing repetitive automatic sequences. There was no motor stalling or loss of traction, and the gear noise was within acceptable ranges. In this regard, the success of this final inspection result is attributed to assembling systemically, tackling a series of challenges of robot assembly and testing, verifying its mechanism and motion, and accomplishing the automation sequences.

5.7.1 Current Potential Problems

Despite the successful demonstration, a variety of limitations were still found which currently constrain the system's dependability and long-term durability.

1. Non-concentricity of the robot within the pipe — The robot may not maintain perfect concentricity while traveling realistic pipes due to the non-uniformity of the interior diameter and the wheel mounting tolerances permitting minor lateral displacements. This can lead to uneven contact pressure of the transducer and even put additional load on the rocker arm and its motor.

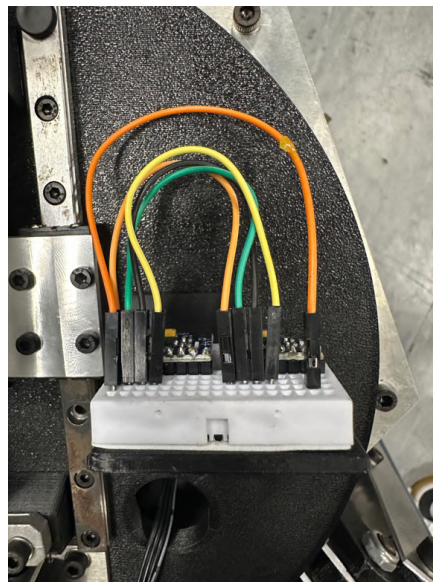


Fig. 5.14 The picture of the dual IMU

2. Unaware of the precise angle of the transducer — the number of rotation steps sent to motor 1 is only known. The exact angular position of the rocker arm (as well as transducer orientation) is not directly recognized. The dual IMU is displayed in Fig. 5.14 which had been tried to measure the angle of the transducer, while the motors could not operate normally when the imu was connected during the start of the automation experiment. We suspected that the power supply was not enough to motivate both motors and imu. On the

other hand, the possibility of software or hardware issues might be included as well. Consequently, the IMU was not considered to be utilized. Hence, the accuracy of multi-angle inspections is still restricted by the absence of feedback systems.

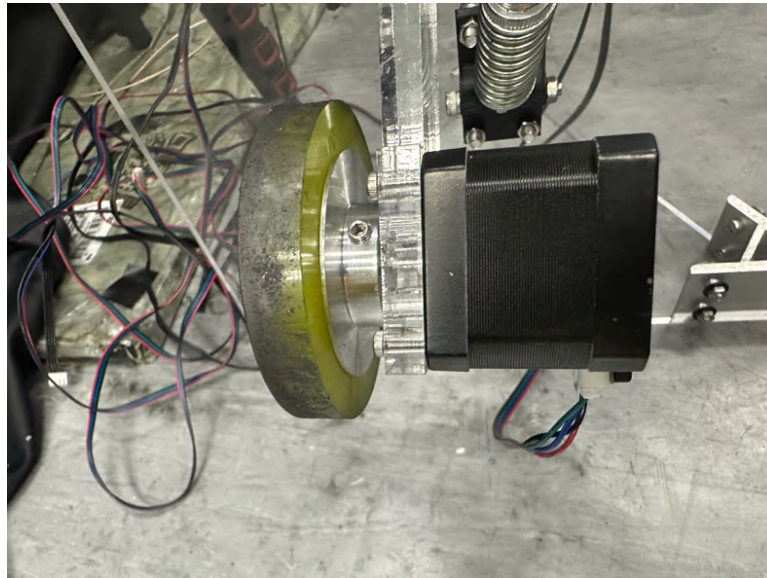


Fig. 5.15 The level of contamination of the rear wheel

3. Possibility of slipping for the rear wheels — Fig. 5.15 reveals that the rear rubber wheels may slip sometimes due to the adhesion of steel powder, especially when the robot is moving on uneven, oily or dusty surfaces. This rubber compound fails to offer enough friction under all circumstances.

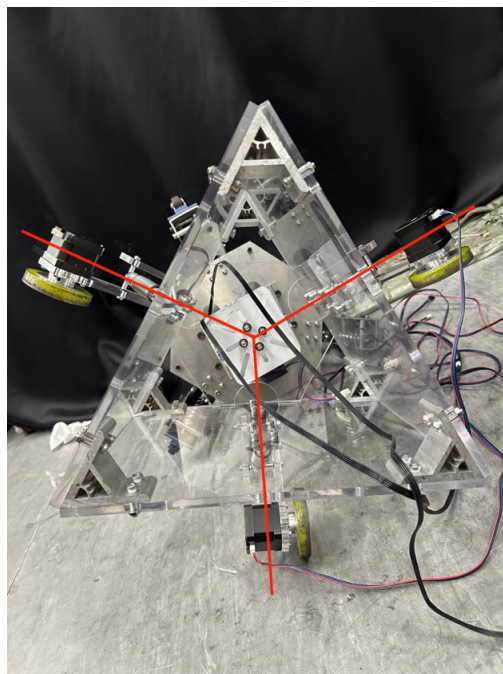


Fig. 5.16 The condition of rear tires asymmetry

4. Uneven load distribution on the rear wheels due to asymmetry — Fig. 5.16 exhibits that the asymmetric condition causes the non-concentricity of the second part of the robot, resulting in an imbalanced load on the rear wheels. Consequently, the rubber tires

accelerate the rate of wear and involve an irregular degree of abrasion, decreasing their service lifespan.

5. Insufficient efficiency due to manual programming and current sampling methods — The automation sequence was successfully implemented, but the system currently is missing true autonomous adaptability. All motion parameters (step counts for the forward movement, angular positions for the rocker arm, extension/retraction of the transducer) are manually pre-programmed and hard-coded. Any change in pipe diameter, surface condition or inspection requirement requires an operator to reprogram and recalibrate the system. This manual intervention limits the inspection speed and makes the system impractical for fast, field-deployable scenarios where multiple pipes with different geometries must be inspected in succession. Additionally, the discrete point was only inspected by the ultrasonic angle beam probe transducer, and the recorded data still required further analysis, which are major obstacles for enhancing efficiency as well.

5.7.2 Proposed Improvement

In light of the identified problems, the following improvements are recommended for future iterations of the robotic platform.

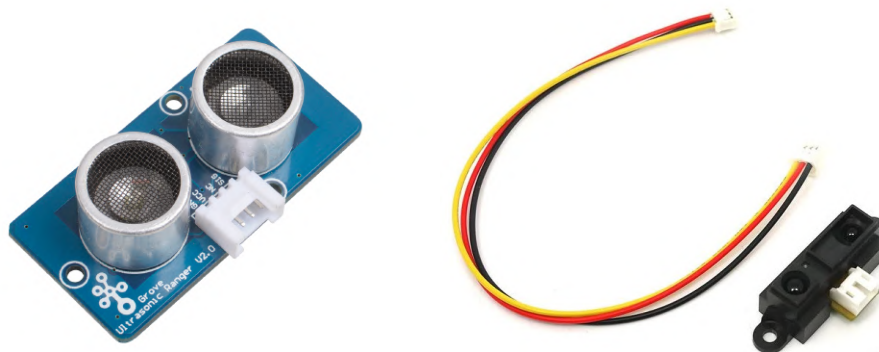


Fig. 5.17 The product of ultrasonic rangefinder (left) and infrared rangefinder (right) [42]

1. Adding a distance sensor to avoid collision and maintain consistent standoff — To mitigate the non-concentricity issue and prevent transducer collision, a non-contact distance sensor, such as an infrared or ultrasonic rangefinder shown in Fig. 5.17, should be installed adjacent to the transducer. The distance between the transducer face and the inner wall of the pipe would be continually measured by this sensor. Regardless of small changes in pipe diameter or robot misalignment, the system can automatically adjust the extension and maintain a constant, safe contact pressure by feeding the measured distance back to motor 0 (transducer lift) in real time. This closed-loop distance control would enhance signal consistency and drastically lower the chance of collision damage.

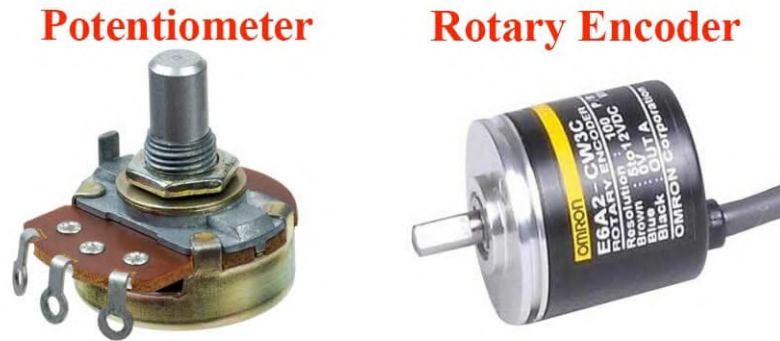


Fig. 5.18 The product of potentiometer (left) and rotary encoder (right) [43]

2. Installing a new piece of hardware to measure the rocker arm angle — A potentiometer or rotary encoder shown in Fig. 5.18 can be added near the rocker arm which would offer real-time angular feedback. This would enable accurate angle mapping of defects identified and increase inspection accuracy.
3. Replacing the latter rubber tires — To improve traction on inner surfaces, the current rubber tires with polyurethane are changed to wheels with grooves to prevent skidding.
4. Redesigning the second part of the robot to avoid interference between the motors and the surface — The motor housing is scratching and worn to the inner wall of the pipe because of the present configuration. A redesign can improve clearance between the motor and the inner wall, repositioning the motors and the tires and replacing more stepper motors. Furthermore, the front rubber wheels can also try to convert to the steel ball wheels to observe its feasibility after changing motors.
5. Developing an online real-time fully automated integration software pipeline — The most advanced enhancement would be to develop a complete software pipeline to automate all the steps from the robot locomotion to ultrasonic inspection, data collection, defect diagnosis and visualization. Such a pipeline would consist of the following components:
 - Autonomous path planning and locomotion: The robot would use onboard distance sensors and encoder feedback to navigate the pipe without pre-programmed step counts, modifying its speed and direction based on real-time conditions.
 - Automated inspection sequencing: The system would automatically position the transducer based on closed-loop feedback from the distance sensor and the angular encoder, without the need for manual calibration.
 - Real-time data acquisition and processing: The ultrasonic signals would be sent to an on-board or remote computer where signal processing algorithms (e.g., envelope detection, noise filtering, thresholding) would be run in real time.
 - Defect diagnosis and classification: Machine learning or rule-based algorithms would analyze the processed signals in order to detect, locate and classify cracks (e.g. inner vs outer wall) without human intervention.

- Visualization and reporting: A graphical user interface would display a real-time map of the pipe with detected flaws, generate inspection reports, and highlight areas requiring maintenance.
6. Replacing the contact transducer with an EMAT for future inspection — The existing ultrasonic angle beam probe transducer requires physical contact and precise normal alignment which complicates automation and risks of wear with extended use. However, the Electromagnetic Acoustic Transducer (EMAT) has been widely used in the commercial field due to their suitability for measuring surface roughness/coated metal parts [44]. It offers non-contact operation, improved lift-off tolerance, and higher Signal-to-Noise Ratio for inspection of metallic pipeline. The use of an EMAT instead of a contact probe would simplify the control sequence (motor 0 may be used only for coarse positioning) and improve inspection speed and reliability.

5.8 Summary of my own Achievement

In this phase of the project, my teammates and I successfully assembled a complete robotic platform and integrated automation and the ultrasonic angle beam probe transducer for pipeline inspection. From the backboard to the last connection with the second module, I methodically recorded the assembly process, giving concise justifications for the selected assembly sequence and verification procedures.

Moreover, several significant engineering challenges were identified and resolved during assembly, for instance non-concentric screw alignment, excessive gear vibration reduced by high viscosity grease, the critical substitution of steel wheels with rubber wheels to achieve movement, and the motor contact issue solved by reversing motor and wheel positions. In addition, a reliable automation sequence that accurately controlled transducer lift (motor 0), rocker arm rotation (motor 1), leg expansion (motor 2), and locomotion (motor 3) was successfully developed using C++ after several tests.

The mechanical and motion aspects were comprehensively validated. The self-centering four-bar linkage exhibited good clamping and concentricity within the pipe; the front module was free-standing, and no looseness was observed. The disk rotation test confirmed the smooth 360° rotation with 2230 steps per revolution measured (0.45% deviation from theoretical) and no visible tilt under the load of the overhung arm. The radial extension test confirmed the smooth travel of the scanning arm over the entire range of 0-62 mm with a 0.01 mm resolution. After replacing the steel ball wheels with rubber wheels, forward and backward motion was smooth and controllable, with speeds of 2–3 cm/s and a maximum tested travel of 50 cm.

At the end of the integration experiment, the ultrasonic angle beam probe transducer correctly identified an artificial flaw via the robot and the entire system, and a complete 25-point grid inspection was successfully demonstrated inside a test pipeline.

Last but not least, I conducted a critical evaluation of the current system, recognizing five major issues: concentricity, restricted wheel degrees of freedom, imbalanced loading, wheel slip, and lack of angular feedback. And four specific enhancements for further work were also suggested.

These accomplishments indicate that an automated, the ultrasonic angle beam probe transducer-based robotic inspection system for pipeline infrastructure is feasible and establish a strong foundation for future development toward a field-deployable solution.

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